QUARK MASSES

J. GASSER and H. LEUTWYLER

Institut für Theoretische Physik, Universität Bern, Sidlerstrasse 5, CH-3012 Bern, Switzerland

NORTH-HOLLAND PUBLISHING COMPANY – AMSTERDAM

PHYSICS REPORTS (Review Section of Physics Letters) 87, No. 3 (1982) 77–169. North-Holland Publishing Company

QUARK MASSES*

J. GASSER and H. LEUTWYLER

Institut für Theoretische Physik, Universität Bern, Sidlerstrasse 5, CH-3012 Bern, Switzerland

Received 12 March 1982

Contents:

Abstract:

We review the current information about the eigenvalues of the quark mass matrix. The theoretical problems involved in a determination of the running masses m_u, m_d, m_s, m_c and m_b from experiment are discussed with the aim of getting reliable numerical values equipped with error bars that represent a conservative estimate of the remaining uncertainties.

* Work supported in part by Schweizerischer Nationalfonds.

Single orders for this issue

PHYSICS REPORTS (Review Section of Physics Letters) 87, No. 3 (1982) 77–169.

Copies of this issue may be obtained at the price given below. All orders should be sent directly to the Publisher. Orders must be accompanied by check.

Single issue price Dfl. 48,00, postage included.

0 370-1573/82/0000–0000/$23.25

1. Introduction – historical prelude

The earliest information about quark masses derived from current algebra, although in this framework the quark mass problem showed up only implicitly in the symmetry properties of the Hamiltonian, i.e. in the commutation rules involving the currents and the energy-momentum tensor. Partial conservation of vector and axial currents implies that the piece of the Hamiltonian that is not symmetric under $SU(3)\times SU(3)$ is small. The breaking of SU(3) was identified as an octet term in the Hamiltonian very early [1]. The structure of chiral symmetry breaking could however only be uncovered once it was recognized that the symmetry generated by the axial currents is not realized as a normal symmetry of the states, but is spontaneously broken by an asymmetric vacuum [2–6]. A further important step that had to be taken to understand the structure of the symmetry breaking terms concerns isospin symmetry. For a long time the strong interaction was supposed to be isospin symmetric; the observed isospin breakings were attributed to the electromagnetic interaction. Several puzzles however occurred in this picture: the sign of the proton–neutron and K^+-K^0 mass differences, the decay $\eta\rightarrow 3\pi$ and the phenomenon of $\rho-\omega$ mixing. A first step towards a solution of these puzzles was the tadpole mechanism [7], which associated the bulk of the electromagnetic self-energies with an octet operator. The origin of the tadpole remained mysterious for quite some time. Within the quark model an operator of this type is generated by the electromagnetic self-energy diagrams of the u and d quarks. The quark self-energies are however proportional to e^2m_u and e^2m_d – too small to explain the size of the tadpole term. The principal source of isospin breaking finally turned out not to be the photon cloud that surrounds the particles but the fact that the u and d quarks have very different masses even if electromagnetism is switched off. (The smallness of these quark masses makes it likely that they are entirely due to an electroweak radiative correction determined by a larger fermion mass scale – the masses m_u and m_d may well turn out to be of order e^2 in a more complete theory. This does not concern us here; the point we are discussing is that other degrees of freedom than light quarks and photons must be involved if one is to understand the size of m_u-m_d.)

The role of symmetries and their breaking through mass terms was clarified by Wilson [8] and it gradually emerged that the part of the strong interaction Hamiltonian that breaks the symmetries of current algebra has all the symmetry properties of a quark mass term.

In QCD this structure of the Hamiltonian is built in: the only renormalizable symmetry breaking term that this theory admits is a quark mass term. The parameters that break $SU(N_f)\times SU(N_f)$ are the eigenvalues of the quark mass matrix: m_u, m_d, m_s, ... The strength of isospin breaking is measured by m_u-m_d, the strength of SU(3) breaking by $m_s-\frac{1}{2}(m_u+m_d)$; chiral $SU(2)\times SU(2)$ is broken by m_u and m_d etc.

On the basis of current algebra symmetries alone it is not possible to determine the absolute size of the light quark masses, but it is possible to determine the two ratios $m_u:m_d:m_s$. We will discuss the present status of the subject in sections 13 and 14.

In some of the early works on the quark model the quarks were treated as very heavy in order to explain on kinematical grounds why the bound states are not ionized in high energy reactions. QCD indicates that this picture is wrong, that confinement of colour is a dynamical phenomenon that persists even if the quark masses are put equal to zero. In order for a massless theory to make sense and to represent a good approximation to this world it is crucial that this theory breaks conformal invariance [9, 10]. It was realized very early that the trace of the energy-momentum tensor (which measures the breaking of scale invariance) contains a large piece that is symmetric under $SU(3)\times SU(3)$. The origin of this term however remained mysterious [see e.g. 11]. In QCD the trace of the energy-momentum tensor contains an anomaly [12–16] proportional to the square of the gluon field strength:

$$\theta^{\mu}_{\mu} = \frac{\beta(g)}{2g^3} F^a_{\mu\nu} F^{\mu\nu a} + \{1 + \gamma(g)\} \{m_u \bar{u}u + m_d \bar{d}d + \cdots\} . \tag{1.1}$$

The trace anomaly is a consequence of the fact that even though the coupling constant g is dimensionless, a change in scale requires a change in the value of g, determined by the β-function. Conformal invariance is broken ab initio by the renormalization group invariant scale Λ (dimensional transmutation).

Nambu and Jona-Lasinio [3] had pointed out that in a theory based on spontaneously broken chiral symmetry the bare fermion masses m_0 needed to produce the proper pion mass must be surprisingly small, a rough estimate indicating $m_0 \lesssim 5$ MeV. Various estimates of the size of these masses were given in early papers on the subject [4, 17, 18]. Okubo [19] analyzed sum rules for the two point functions involving scalar, vector, tensor, axial and pseudoscalar densities, saturated them with the lowest lying physical states and obtained

$$m_u \simeq m_d \simeq 7\,\text{MeV}, \qquad m_s \simeq 156\,\text{MeV}. \tag{1.2}$$

He also observed that these values are consistent with the assumption that the mass difference between baryons is given by the difference of the quark masses they contain (additivity rule). Within QCD the sum rules used by Okubo are divergent and saturation by lowest intermediate states does not make sense [20]. One has to work harder along the lines suggested by Vainshtein et al. [21] to estimate quark masses from QCD sum rules.

A different framework that allows one to determine the light quark masses was proposed by one of us [22]. Using SU(6)-symmetry to relate the matrix elements of the vector current to those of the pseudoscalar density

$$\langle 0|\bar{u}\gamma_\mu d|\rho^-\rangle = f_\rho M_\rho \varepsilon_\mu \qquad f_\rho = (204 \pm 11)\,\text{MeV}$$

$$\langle 0|\bar{u}\,\mathrm{i}\gamma_5 d|\pi^-\rangle = \frac{f_\pi M_\pi^2}{m_u + m_d} \qquad f_\pi = (131.9 \pm 0.1)\,\text{MeV}$$

we obtained the relation

$$3\hat{m}(M_\rho - \hat{m}) f_\rho = (M_\pi^2 - \hat{m}^2) f_\pi \tag{1.3}$$

which gives

$$\hat{m} \equiv \tfrac{1}{2}(m_u + m_d) = 5.4\,\text{MeV} \tag{1.4}$$

for the mean mass of u and d. Using SU(3)-symmetry for the operator matrix elements the model furthermore implies the relations

$$\frac{M_\pi^2 - 4\hat{m}^2}{2\hat{m}} = \frac{M_K^2 - (m_s + \hat{m})^2}{m_s + \hat{m}}$$

$$M_\varphi - M_\rho = 2(M_{K^*} - M_\rho) = (m_s - \hat{m})(1 + f_\pi/f_\rho) \tag{1.5}$$

which allow one to extract a value for the strange quark mass in two independent ways, with the result

$$m_s = 125\,\text{MeV} \qquad \text{and} \qquad m_s = 153\,\text{MeV}$$

respectively. The first relation is a slightly modified version of the Gell-Mann–Oakes–Renner–Glashow–Weinberg formula expressing the ratio $M_\pi^2 : M_K^2$ in terms of SU(2) × SU(2) to SU(3) × SU(3) symmetry breaking parameters. The second relation is a modified version of the additivity rule according to which the masses of the bound states ρ, K*, φ only differ through the masses of the quarks they contain: $M_\varphi - M_\rho = 2(m_s - \hat{m})$. The reason for this additivity rule to emerge within a symmetry scheme is the fact that the symmetry relations are consistent both with the chiral limit $m_{\text{quark}} \to 0$ and with the nonrelativistic situation for heavy quarks, for which additivity holds trivially: up to corrections of order v^2/c^2 the energy of the bound state is given by the sum of the rest energies of the quarks.

Since SU(6) has not been established as an approximate symmetry property of the bound states in QCD it is not clear that the relations among the matrix elements that follow from this symmetry are correct.

In the early seventies a number of different quark mass estimates were obtained [23–27] on the basis of deep inelastic scattering data, using various hypotheses about the short distance behaviour of the electromagnetic current. These hypotheses turn out not to be satisfied in QCD. It does not seem to be possible to extract more than crude upper bounds on the quark masses from deep inelastic scattering data at the presently available precision.

In 1975 we reanalyzed the Cottingham formula for the proton–neutron mass difference to obtain an estimate for the isospin breaking due to the electromagnetic interaction. Using the SU(6)-values for the mean mass of u and d we then obtained the estimates [28]

$$m_u \simeq 4\,\text{MeV}\,, \qquad m_d \simeq 7\,\text{MeV}\,. \tag{1.6}$$

At first sight this substantial mass difference between the u and d quarks seems to be in conflict with isospin symmetry: why is isospin an almost perfect symmetry of the strong interactions if these masses differ by almost a factor two? First, the presence of a scale in the strong interaction implies that the size of the quark mass terms is not to be measured in comparison to zero, but in comparison to typical eigenvalues of the chiral Hamiltonian – free u and d quarks would have very different energies at rest, interacting quarks always have energies of the order of the strong interaction scale and it only makes a small difference whether their masses are zero or are of the order 4 or 7 MeV. The same is true of the huge difference between the masses of u, d and s quarks indicated in (1.2): as long as the typical energy of an s quark is large in comparison to 150 MeV, this difference will not produce large flavour asymmetries. There is however one place where the large difference between m_u, m_d and m_s is not shielded by the strong interaction. The mass of the Goldstone bosons would vanish if the quark masses were zero; the ratio $M_\pi^2 : M_K^2 : M_\eta^2$ is a direct measure for the ratio $\hat{m} : m_s$ of quark masses (see section 8). This relation poses the second puzzle: why does the ratio $m_u : m_d \simeq 4:7$ not lead to a large isospin violation of the Goldstone boson masses? The reason why the strong interaction hides the SU(2) asymmetry in the quark mass term even better than it hides the SU(3) asymmetry is the fact that only three of the four states that consist mainly of u and d quarks ($\bar{u}u$, $\bar{u}d$, $\bar{d}u$, $\bar{d}d$) become massless in the limit $m_u = m_d = 0$. The fourth state remains heavy because of the anomaly in the axial current whose presence is vital for an understanding of the meson spectrum (if one e.g. ignores the anomaly in the context of the SU(6) symmetry-scheme mentioned above this model goes to pieces). The isospin violating part of the quark mass term, $\frac{1}{2}(m_u - m_d)(\bar{u}u - \bar{d}d)$ does not have any matrix elements between the three $I = 1$ Goldstone modes, it only produces transitions from these modes to the $I = 0$ combination $\bar{u}u + \bar{d}d$. These transitions are however inhibited by the large mass difference produced by the anomaly. [The presence of s quarks allows the combination $\bar{u}u + \bar{d}d$ to go partly into the η-meson and thus to remain at a relatively low mass. The mixing $\pi^0 - \eta$ does produce a tiny second order effect in $M_{\pi^+} - M_{\pi^0}$ proportional to $(m_u - m_d)^2/(m_s - \hat{m})$ (see e.g. [29]) which is however difficult to separate from genuine second order perturbations (section 12).] Nevertheless there are sizeable isospin asymmetries e.g. in the σ-terms of πN-scattering [30], but the smallness of these terms makes it again difficult to identify the effect in the data.

Since the quark masses break SU(3) × SU(3) one may attempt to determine their value on the basis of the observed violations of chiral symmetry. Chiral symmetry breaking effects were analyzed by several authors [31–40]. The observed deviations from chiral symmetry are indeed found to be consistent with small quark masses, typical estimates giving $\hat{m} \sim 15$ MeV or $\hat{m} < 40$ MeV. The smallness of these effects of course makes it difficult to obtain accurate values. It is however crucial that one does not find any large violations of chiral symmetry. In the case of SU(2) × SU(2) a deviation of the order of 7% is a very large effect [41]. For a recent discussion of deviations from the Goldberger–Treiman relation see e.g. Dominguez [42] and the references quoted therein. We briefly comment on schemes that try to explain some of the "observed deviations" in terms of large quark masses of order $\hat{m} \sim 40$ MeV in appendix E.

In the meantime the perturbative properties of QCD had been firmly established. Georgi and Politzer [43] pointed out that the ratios of running masses coincide with the symmetry breaking parameters of current algebra at large values of the scale μ. (In the $\overline{\text{MS}}$ scheme the ratios are renormalization invariant so that this connection holds for any value of the running scale.) The scale dependence of the quark mass reflects the fact that a bare quark is surrounded by a cloud of gluons and quark pairs; the energy of the cloud contained in a sphere of radius r increases with r. For sufficiently small radius perturbation theory provides a reliable description of this cloud (asymptotic freedom) and hence a reliable expression for the running mass at large values of $\mu \sim r^{-1}$. Unless the scale μ is changed by orders of magnitude or is shifted into the region where perturbation theory breaks down the variation in the running mass is rather small. [The value of the effective strong coupling constant has recently been measured in several independent ways with the result α_s (20 GeV^2) = 0.18 ± 0.03, a value that is considerably smaller than previously claimed [44–48]. With this value of α_s the perturbative expressions make sense at values of μ of the order of 1 GeV. The change of the running mass from $\mu = 3$ GeV down to $\mu = 1$ GeV e.g. is less than 25% if $\Lambda_{\overline{\text{MS}}}^{N_f=3} \leq 150$ MeV.] *In the following we will always be quoting values for the running mass at the scale* $\mu = 1$ *GeV* (the renormalization group invariant mass is rather sensitive to the value of Λ which is not known to sufficient accuracy).

Bound states of very heavy quarks have a small diameter and it makes sense to use the perturbative separation of self-energy effects from gluon exchange. Within perturbation theory the pole mass M_q defined by the start of the cut in the quark propagator appears to be a meaningful quantity that is relevant for the description of heavy quark bound states (section 16). For light quarks however nonperturbative effects play a crucial role in the bound state motion. Politzer [49] determined the leading nonperturbative contribution to the effective quark mass $M(p^2)$ defined by $S(p) = Z(p^2)[M(p^2) - \not{p}]^{-1}$.

Many attempts were made to establish a connection between the quark mass in the Lagrangian and the *constituent quark mass* parameter used in phenomenological nonrelativistic models. These models do give a semiquantitative account of the bound state properties, but it has not been possible to interpret their parameters in terms of the masses and of the coupling constant in the Lagrangian which is supposed to be the basis of these models. Rules of thumb such as the rule that the constituent mass to be used in a nonrelativistic description is given by the quark mass plus a term that comes from the scale of QCD and is of the order of 300 MeV are consistent with other rules of thumb such as $M_{\text{proton}} \simeq 3m_{\text{const.}}$, $M_\rho \simeq 2m_{\text{const.}}$ or $\mu_{\text{proton}} \simeq e/2m_{\text{const.}}$. We will not discuss any constituent quark mass values but stick to the masses that appear in the Lagrangian.

The first determination of light quark masses within the framework of QCD was given by Vainshtein, Voloshin, Zakharov, Novikov, Okun and Shifman [21] who analyzed the two point function for the axial divergence. We will discuss this subject in detail in section 15.

This concludes our historical survey. The following sections describe the current theoretical status of the quark masses.

2. Origin of quark and lepton masses – QCD as an effective low energy theory

The gauge theory of the electroweak interaction has made it clear that the quark and lepton masses do not have the status of basic constants of nature: the mass of, say, the electron does not occur in the Lagrangian of the theory. (Gauge fields can only couple to conserved currents; the weak V-A currents are conserved only if the fermions are massless.) The electron mass is attributed not to a mass term in the Lagrangian but to the fact that the ground state of the theory happens not to be symmetric with respect to the transformations generated by the weak currents. The electron moves at less than the velocity of light, because it has to move through and interact with a condensate that spontaneously breaks the symmetry of the Lagrangian. In the framework of gauge theory the values of the lepton and quark masses are determined by the properties of the condensate. Many models for the spontaneous breakdown of gauge symmetries have been studied, but we still know very little about the electroweak condensate we actually live in. It does not seem to be possible at this time to calculate the fermion masses, although some of the relationships among these masses that follow from simple assumptions about the symmetry properties of the condensate are consistent with the facts (section 18).

Even if the quark and lepton masses have lost the status of basic constants of nature, the success of QED shows that the electron mass is a crucial physical quantity whose value allows one to account for a large body of facts in a very precise manner. The reason for this success of an effective low energy theory like QED is its renormalizability. The predictions of the theory only depend on the charge and the mass of the particle. The mechanism responsible for the generation of the electron mass is irrelevant as far as low energy matrix elements such as e.g. the magnetic moment are concerned (provided of course that the energy scale involved in that mechanism is large in comparison with the electron mass – the larger the more accurate the effective low energy theory).

At energies small in comparison with M_{W}, M_{Z} the quarks and leptons interact almost exclusively with gluons and with photons. The weak interactions (and all other degrees of freedom relevant for the structure of the symmetry breaking condensate) are frozen in. The degrees of freedom that are not frozen – quarks, leptons, gluons, photon – interact in a renormalizable manner. The corresponding effective low energy theory (QED + QCD) contains two coupling constants e and g and the masses m_e, m_μ, m_τ, m_{u}, m_{d}, m_{s}, m_{c}, $m_{\mathrm{b}}, \ldots$ as free parameters. The theory is expected to account for the low energy phenomena involving these particles to a very high degree of accuracy comparable to the precision of QED in atomic physics: as long as the quark or lepton masses are not too large the direct effects of the weak interactions can safely be ignored at low energies.

Since the charge e is a small parameter electromagnetic effects are well accounted for by the first few terms of an expansion in powers of e^2. The first part of this paper will be devoted to the zero order term in this expansion – electromagnetic effects will be neglected. In this approximation the effective low energy theory mentioned above reduces to QCD:

$$\mathcal{L}_{\mathrm{QCD}} = -\frac{1}{4g^2} F^a_{\mu\nu} F^{\mu\nu a} + \sum_{\mathrm{q}=\mathrm{u},\mathrm{d},\mathrm{s},\ldots} \bar{q}(\mathrm{i}\not{D} - m_{\mathrm{q}})q$$
$$D_\mu = \partial_\mu + \mathrm{i}F^a_\mu \lambda^a/2 . \tag{2.1}$$

The hadronic energy levels are of course only supposed to be reproduced by this Lagrangian up to effects of order e^2. The mass of the proton e.g. is expected to come out too low by about 1 MeV – we will discuss these electromagnetic corrections in sections 11 and 12.

For completeness we mention that (2.1) is not the most general renormalizable Lagrangian involving quarks and gluons. The effective Lagrangian of the strong interactions presumably also contains a small *CP* violating term proportional to $F\tilde{F}$. (Such a term arises e.g. if before diagonalization the quark mass matrix has a complex determinant.) We will not discuss the connection between the quark mass matrix and the weak angles (Cabibbo, Kobayashi–Maskawa) and disregard the small *CP* violating term in the Lagrangian of QCD.

3. Renormalization

The Lagrangian of QCD contains the bare coupling constant g^0, the bare quark masses $m^0_{\mathrm{u}}, m^0_{\mathrm{d}}, \ldots$ and some ultraviolet cutoff λ that regularizes the divergences encountered in the expansion in powers of g^0. The theory is fully specified by the values of the bare constants $g^0, m^0_{\mathrm{u}}, \ldots$ once a suitable regularization procedure is chosen. In principle, the renormalization program is straightforward: one calculates quantities of physical interest (say eigenvalues of the Hamiltonian H_{QCD}) in terms of the bare parameters at a given, large value of λ. Once a sufficient number of physical quantities have been determined as functions of the bare parameters one inverts the result and expresses the bare parameters in terms of physical quantities, always working at some given, large value of λ. Finally, one uses these expressions to eliminate the bare parameters in all other quantities of physical interest. Renormalizability guarantees that this operation at the same time also eliminates the cutoff.

In QED there is a very natural set of physical quantities in terms of which to express the bare constants: the charge and the mass of an isolated electron. This choice is by no means of crucial importance. Even if isolated electrons were not available (i.e. even if electrons occurred only in positronium bound states) the renormalization program would go through just as well: the masses of two such bound state levels could serve as physical parameters in terms of which the bare constants of the theory could be calculated. The crucial point in the renormalization program is that one is able to establish a quantitative connection between the constants in the Lagrangian and some physical quantities. Since one is able to calculate the positronium levels in terms of the bare constants of QED almost as well as the charge and the mass of an isolated electron, the renormalization program would go through almost as well if there were no measurements on isolated electrons.

In QCD confinement prevents us from observing isolated quarks – the colour and the mass of an isolated quark are not available as physical quantities in terms of which to express the bare constants of the theory. There are various observables that one may use instead. The most obvious quantities of physical interest, the masses of the bound states, have recently become accessible to a quantitative comparison with the parameters of the Lagrangian. In the framework of lattice calculations it appears to be possible to calculate the lowest mesonic and baryonic eigenvalues in units of the inverse lattice spacing $\lambda = a^{-1}$ as a function of the bare coupling constant and of the bare quark masses. Inverting these results one obtains values for the bare coupling constant and for the bare quark masses which seem to be perfectly consistent with the values obtained from other methods. If the violations of chiral symmetry produced by the lattice can be brought under control then these methods may well provide us with the most accurate determination of the parameters of QCD.

A different set of observables is related to high energy processes or to the bound state properties of very heavy quarks which play the role of test charges probing the structure of the QCD Lagrangian. For observables of this type the relation to the parameters appearing in the Lagrangian may be established within perturbation theory.

Perturbation theory also fixes the cutoff-dependence of the bare parameters: in order that a change in the value of the cutoff λ does not modify physical quantities such as bound state masses, the bare constants have to be adapted to the change according to the renormalization group [53–56]. As $\lambda \to \infty$ both the bare coupling constant and the bare quark masses tend to zero, roughly $g^0 \sim (\ln \lambda)^{-1}$, $m_i^0 \sim (\ln \lambda)^{-1/2}$. More precisely, the bare coupling constant and the bare quark masses are given by

$$\exp\left\{\frac{16\pi^2}{\beta_0 (g^0)^2}\right\} = \frac{\lambda^2}{\Lambda^2}\left(\ln\frac{\lambda^2}{\Lambda^2}\right)^{\beta_1/(\beta_0)^2}, \qquad m_i^0 = \bar{m}_i\left(\ln\frac{\lambda}{\Lambda}\right)^{-2\gamma_0/\beta_0} \tag{3.1}$$

where Λ and $\bar{m}_i$ are held fixed as $\lambda \to \infty$. The quantity Λ is the renormalization group invariant scale and $\bar{m}_i$ is the renormalization group invariant mass. The constants β_0, β_1 and γ_0 are the leading coefficients of the β- and γ-functions [57–67]:

$$\begin{aligned}
&\beta(g) = -\beta_0 \frac{g^3}{16\pi^2} - \beta_1 \frac{g^5}{(16\pi^2)^2} + O(g^7)\\
&\gamma(g) = \gamma_0 \frac{g^2}{4\pi^2} + \gamma_1 \frac{g^4}{(4\pi^2)^2} + O(g^6)\\
&\beta_0 = 11 - \tfrac{2}{3}N_f\,; \qquad \gamma_0 = 2\\
&\beta_1 = 102 - \tfrac{38}{3}N_f\,; \qquad \gamma_1 = \tfrac{101}{12} - \tfrac{5}{18}N_f
\end{aligned} \tag{3.2}$$

where N_f is the number of quark flavours. In the $\overline{\text{MS}}$-scheme the renormalized coupling constant $g(\mu)$ and the renormalized quark masses $m_i(\mu)$ are determined by the differential equations

$$\begin{aligned}
&\mu \frac{d}{d\mu} g(\mu) = \beta[g(\mu)]\\
&\mu \frac{d}{d\mu} m_i(\mu) = -\gamma[g(\mu)]\, m_i(\mu)
\end{aligned} \tag{3.3}$$

Table 1
Running coupling constant and ratio of running mass to renormalization group invariant mass in the $\overline{\text{MS}}$ scheme with 3 flavours. The last two rows contain the corresponding values of Λ in the $\overline{\text{MS}}$ scheme with 4 and 5 flavours (to two loops, taken from Bernreuther and Wetzel [71]). For details see text.

	$\Lambda_3 = 50$ MeV		$\Lambda_3 = 100$ MeV		$\Lambda_3 = 150$ MeV		$\Lambda_3 = 200$ MeV		$\Lambda_3 = 250$ MeV		$\Lambda_3 = 300$ MeV	
μ (GeV)	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$	$\alpha_s(\mu)$	$\frac{m(\mu)}{\bar{m}}$
0.75	0.19	0.62	0.25	0.72	0.31	0.81	0.37	0.90	0.46	1.01	0.56	1.14
1.0	0.18	0.59	0.22	0.67	0.27	0.74	0.31	0.81	0.36	0.88	0.41	0.95
1.25	0.17	0.57	0.21	0.64	0.24	0.70	0.27	0.75	0.31	0.81	0.35	0.86
1.5	0.16	0.56	0.19	0.62	0.22	0.67	0.25	0.72	0.28	0.76	0.31	0.81
2.0	0.15	0.54	0.18	0.59	0.20	0.63	0.22	0.67	0.24	0.70	0.27	0.74
2.5	0.14	0.52	0.17	0.57	0.19	0.61	0.21	0.64	0.22	0.67	0.24	0.70
3.0	0.14	0.51	0.16	0.56	0.18	0.59	0.19	0.62	0.21	0.65	0.22	0.67
3.5	0.13	0.50	0.15	0.55	0.17	0.58	0.19	0.60	0.20	0.63	0.21	0.65
4.0	0.13	0.50	0.15	0.54	0.16	0.57	0.18	0.59	0.19	0.61	0.20	0.63
4.5	0.13	0.49	0.14	0.53	0.16	0.56	0.17	0.58	0.18	0.60	0.19	0.62
5.0	0.12	0.49	0.14	0.52	0.16	0.55	0.17	0.57	0.18	0.59	0.19	0.61
$\bar{m}/m(1)$	1.69		1.49		1.35		1.24		1.14		1.05	
Λ_4	36 MeV		76 MeV		120 MeV		165 MeV		211 MeV		259 MeV	
Λ_5	21 MeV		47 MeV		78 MeV		111 MeV		146 MeV		184 MeV	

with the boundary condition that $g(\lambda)$, $m_i(\lambda)$ coincide with the bare constants g^0, m_i^0 for sufficiently large λ (more precisely, the difference $g(\lambda)^{-2} - (g^0)^{-2}$ tends to zero, the ratio $m_i(\lambda)/m_i^0$ tends to one).

The solution to these differential equations reads

$$\alpha_s(\mu) = \frac{g(\mu)^2}{4\pi} = \frac{4\pi}{\beta_0 L}\left\{1 - \frac{\beta_1}{\beta_0^2}\frac{\ln L}{L} + \mathrm{O}\left[\frac{(\ln L)^2}{L^2}\right]\right\} \tag{3.4}$$

$$m_i(\mu) = \bar{m}_i(\tfrac{1}{2}L)^{-2\gamma_0/\beta_0}\left\{1 - \frac{2\beta_1\gamma_0}{\beta_0^3}\frac{\ln L + 1}{L} + \frac{8\gamma_1}{\beta_0^2 L} + \mathrm{O}\left[\frac{(\ln L)^2}{L^2}\right]\right\}$$

with $L = \ln(\mu^2/\Lambda^2)$. In the $\overline{\text{MS}}$-scheme the renormalized coupling constant $g(\mu)$ depends only on the ratio μ/Λ and is independent of the quark masses; N_f is the total number of flavours, including heavy quarks. In the low energy region which we will be concerned with the heavy quark degrees of freedom are however frozen. In this region a renormalization prescription that ignores the heavy flavours is more appropriate than the $\overline{\text{MS}}$-scheme: the decoupling theorem [68, 69] asserts that infinitely heavy quarks decouple from all quantities of physical interest. In the following we will therefore use the $\overline{\text{MS}}$-scheme with $N_f = 3$. The behaviour of the effective coupling constant across the thresholds for the heavy flavours was studied by Ovrut and Schnitzer [70] and by Bernreuther and Wetzel [71]. These authors show that the value $\Lambda_3 = 130$ MeV in the $\overline{\text{MS}}$-scheme for $N_f = 3$ is equivalent to the value $\Lambda_4 \simeq 100$ MeV in the $\overline{\text{MS}}$-scheme for $N_f = 4$ and to $\Lambda_5 \simeq 65$ MeV for $N_f = 5$.

The numerical values of the effective coupling constant and of the ratio of the running mass to the renormalization group invariant mass are listed in table 1 for $N_f = 3$ and different values of Λ_3.

4. Chiral limit

To appreciate the significance of the quark masses in QCD it is important to understand the theoretical limit in which these parameters are put equal to zero. In fact, the masses of the light quarks u, d and s are small in a sense to be specified below – to put them equal to zero does not distort the properties of the theory beyond recognition. In section 7 we will discuss how to account for the actual, nonzero values of m_u, m_d and m_s in a perturbation expansion around the chiral limit $m_u = m_d = m_s = 0$.

The remaining quarks c, b, ... are not light: although one may of course study the theoretical limit in which these masses also vanish, it does not seem to be possible to recover the actual mass values by an expansion around that limiting case. At low energies a better approximation is obtained if the quarks c, b, ... are instead treated as infinitely heavy. In this limit the degrees of freedom associated with these quarks freeze and may be ignored in the effective low energy theory. For this reason we will in this section ignore the presence of heavy quarks altogether, and study the limit $m_u = m_d = m_s = 0$. (The discussion would not change in any significant way if we instead considered a different number of flavours provided all quark masses are put equal to zero.)

In the chiral limit QCD contains only one parameter: the renormalization group invariant scale Λ (note that in the chiral limit the presence of a *CP* violating term proportional to $F\tilde{F}$ has no observable consequences). The mass of say the proton is some pure number multiplying Λ and likewise for all other physical states of the theory: all mass values of the spectrum are given by some pure number multiplying the proton mass. The numbers M_ρ/M_p, $M_\Delta/M_p, \ldots$ are determined by the theory in a parameter free manner. In this sense the chiral limit of QCD may be called a theory without any adjustable parameters: QCD is of course unable to predict the value of M_p in GeV units, but it determines all dimensionless hadronic quantities in a parameter free manner. The elastic cross section for pp scattering e.g. is some fixed function of the variables s/M_p^2 and t/M_p^2 multiplying the square of the proton Compton wavelength. It is most remarkable that a theory that does not contain a single dimensionless parameter is expected to provide a good approximation to the hadronic spectrum. To solve this theory and to verify that the solution indeed closely resembles the world we live in presents one of the most fascinating challenges in particle physics.

The Lagrangian of massless QCD has a high symmetry. The nine vector and eight axial currents

$$\begin{aligned} V_\mu^a &= \bar{q}\gamma_\mu \frac{\lambda^a}{2} q\,, \qquad a = 0, 1, \ldots, 8 \\ A_\mu^a &= \bar{q}\gamma_\mu\gamma_5 \frac{\lambda^a}{2} q\,, \qquad a = 1, \ldots, 8 \end{aligned} \tag{4.1}$$

are conserved – the corresponding charges commute with the Hamiltonian. The ninth axial current is not conserved ($\varepsilon_{0123} = 1$):

$$\partial_\mu(\bar{q}\gamma^\mu\gamma_5 q) = \frac{3}{16\pi^2} F^a_{\mu\nu}\tilde{F}^{\mu\nu a}\,. \tag{4.2}$$

[Since $F\tilde{F}$ is itself a divergence, it is possible to write down a ninth axial current that is conserved. The consequences of this conservation law are referred to as the U(1)-problem. The conserved current is not gauge invariant and the associated charge makes sense only if the gauge field strength vanishes at infinity. We assume that the U(1)-problem does find a solution within QCD and we will in the following disregard the ninth axial current. For a review of the U(1)-problem see Crewther [72].]

5. Spontaneous breakdown of chiral symmetry

In the chiral limit the symmetry group of the QCD Hamiltonian is SU(3) × SU(3) × U(1). It is very easy to see that the ground state of the theory cannot possibly be symmetric with respect to this group if massless QCD is to be a good approximation to this world. In fact, consider the vector and axial vector two point functions. If the state $|\phi\rangle$ is symmetric under SU(3) × SU(3) then we must have ($a, b = 1, \ldots, 8$)

$$\langle\phi|A^a_\mu(x)\, A^b_\nu(y)|\phi\rangle = \langle\phi|\, V^a_\mu(x)\, V^b_\nu(y)|\phi\rangle\,. \tag{5.1}$$

[To prove this, observe that under SU(3) × SU(3) the combinations $R = V + A$, $L = V - A$ transform like ($\underset{\sim}{8}$, $\underset{\sim}{1}$) and ($\underset{\sim}{1}$, $\underset{\sim}{8}$) respectively. If $|\phi\rangle$ is invariant then $\langle\phi|RL|\phi\rangle$ must vanish. This implies (5.1).]

The equality (5.1) requires that whenever there is a particle of spin 0^+ or 1^- (contributing to the right hand side of (5.1)) there must be a mass degenerate partner of spin 0^- or 1^+ (contributing with equal strength to the left hand side of (5.1)). There is no trace of such parity doubling in the spectrum of hadrons. If massless QCD is a good approximation to this world then the vacuum cannot be symmetric under SU(3) × SU(3).

Note that the equality (5.1) is true order by order in perturbation theory. There may be a state $|\phi\rangle$ – the dressed Fock vacuum of perturbation theory – which is Poincaré invariant up to a phase and symmetric with respect to SU(3) × SU(3) such that (5.1) holds. This state is not the state of lowest energy, however. The state of lowest energy, the physical vacuum $|0\rangle$, must be asymmetric under SU(3) × SU(3) [73, 74].

A spontaneously broken symmetry calls for Goldstone bosons: if some generator T^a of SU(3) × SU(3) does not leave the vacuum invariant, $T^a|0\rangle \neq 0$ then we must have a physical state $T^a|0\rangle$ with the same energy eigenvalue as $|0\rangle$, because H_{QCD} commutes with T^a. If T^a is a vector charge then the state $T^a|0\rangle$ describes a massless scalar, if T^a is an axial charge then we get a massless pseudoscalar [75].

The eight lightest hadrons (π, K, η) are pseudoscalars. They are the Goldstone particles associated with the axial charges. [That these particles are not exactly massless as it would be appropriate for Goldstone particles is attributed to the fact that the QCD Hamiltonian is not really chirally invariant – the masses of the light quarks break the symmetry SU(3) × SU(3). If we were able to vary the quark mass parameters in the real world then we could see how π, K and η indeed become massless as we let m_{u}, m_{d} and m_{s} tend to zero.]

The vector charges are not broken spontaneously: (i) there is no trace of light scalars in the spectrum of the mesons (the lightest scalars are heavier than the proton), (ii) there is very good evidence for the corresponding symmetry group SU(3) to be a normal symmetry of the spectrum. Hadrons do occur in nearly degenerate multiplets that constitute representations of SU(3).

6. The vacuum expectation value of $\bar{q}q$

The vacuum expectation value of the scalar quark operators $\bar{u}u$, $\bar{d}d$, $\bar{s}s$ was identified as a quantitative measure for the asymmetry of the vacuum long ago [76]. In the chiral limit the operators $\bar{u}_L u_R$, $\bar{u}_L d_R, \ldots$ transform like $(\underset{\sim}{3}^*, \underset{\sim}{3})$ under $SU(3)_L \times SU(3)_R$. Hence their expectation value in a chirally invariant state vanishes. The physical ground state is however not chirally symmetric and there is no reason for the expectation values $\langle 0|\bar{q}_L q_R|0\rangle$ to vanish. As a product of two fields of dimension $\frac{3}{2}$ the operators $\bar{q}q$ are of course only defined up to counter terms of dimension less than or equal to 3. In massless QCD $\bar{q}q$ is defined up to a multiple of the unit operator. The requirement that $\bar{q}_L q_R$ transforms like $(\underset{\sim}{3}^*, \underset{\sim}{3})$ fixes this multiple uniquely. To see this, one may simply define the scalar quark densities by the commutators of the axial charges with the pseudoscalar densities. The commutator of two quark bilinears is however a very singular object at short distances – the commutator in general explodes in the equal time limit. To show that the Gell-Mann–Oakes–Renner relation may nevertheless be derived within QCD we use the language of the operator product expansion which exhibits these singularities explicitly. For definiteness we consider the charged axial current and the corresponding pseudoscalar density

$$A_\mu = \bar{d}\gamma_\mu\gamma_5 u\,, \qquad P = \bar{d}\mathrm{i}\gamma_5 u$$

and write the operator product expansion in the form

$$A_\mu(x)\,P^+(y) = \sum_n C^n_\mu(x-y)\,O^n(y)\,. \tag{6.1}$$

The coefficient of the unit operator, $C^{\mathbb{1}}_\mu(z)$, is Lorentz covariant. Furthermore, in the chiral limit which we are considering here, the axial current is conserved, i.e. $\partial^\mu C^{\mathbb{1}}_\mu(z) = 0$. The general Lorentz covariant solution to this constraint is

$$C^{\mathbb{1}}_\mu = k \cdot z_\mu/(z^2 - \mathrm{i}\varepsilon z^0)^2 \tag{6.2}$$

where k is some constant of dimension 3. Applying the same argument to the coefficients of the other scalar operators $\bar{u}u$, $\bar{d}d$, $\bar{s}s$, $F^2, \ldots$ we conclude that they must also have the form (6.2). In fact the scalars $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$ have the same dimension as P^+. Their expansion coefficients are therefore canonical and may be read off from the equal time commutator in free field theory:

$$C^{\mathrm{u}}_\mu(z) = C^{\mathrm{d}}_\mu(z) = \frac{1}{2\pi^2}\frac{z_\mu}{z^4}\,, \qquad C^{\mathrm{s}}_\mu(z) = 0\,. \tag{6.3}$$

Since QCD is asymptotically free the effects of the interaction have to disappear at short distances. The operators of higher dimension can therefore not be accompanied by singular coefficients of the type (6.2) – the corresponding constants k vanish. Finally we observe that by a suitable redefinition of the quark scalars of the type $\bar{q}q \to \bar{q}q + \lambda\mathbb{1}$ we may absorb the contribution of the unit operator to obtain $C^{\mathbb{1}}_\mu(z) = 0$. This operation unambiguously specifies the quark scalars. With this convention the operator product contains a single scalar contribution:

$$A_\mu(x)\,P^+(y) = \frac{z_\mu}{2\pi^2 z^4}(\bar{u}u + \bar{d}d) + \text{nonscalar terms}\,. \tag{6.4}$$

Note that the convention does not depend on the channel used. The symmetry $SU(3)_L \times SU(3)_R$ guarantees that the constants k that multiply the unit operator in operator products with different flavour or in the product of a vector current with a scalar density are the same (we do not consider the singlet currents).

To the vacuum expectation value only the scalar operators contribute and we therefore have the exact relation

$$\langle 0|A_\mu(x)\,P^+(y)|0\rangle = \frac{z_\mu}{2\pi^2 z^4}\langle 0|\bar{u}u + \bar{d}d|0\rangle\,. \tag{6.5}$$

The intermediate states contributing to the left hand side may be described by a spectral function of the form $p_\mu\,\rho(p^2)$. Since current conservation requires $p^2\,\rho(p^2) = 0$ only the pion contributes:

$$\langle 0|A_\mu|\pi^+\rangle = \mathrm{i} f_\pi p_\mu\,, \qquad \langle 0|P|\pi^+\rangle = g_\pi \tag{6.6}$$

and we get

$$f_\pi g_\pi = -\langle 0|\bar{u}u + \bar{d}d|0\rangle\,. \tag{6.7}$$

Finally, we observe that for nonvanishing quark masses the equation $\partial A = (m_u + m_d)P$ implies the exact relation

$$f_\pi M^2_{\pi^+} = (m_u + m_d) g_\pi\,. \tag{6.8}$$

If both f_π and $\langle 0|\bar{q}q|0\rangle$ tend to finite limits as $m_{\text{quark}} \to 0$ the pion mass must tend to zero in proportion to $(m_u + m_d)^{1/2}$:

$$\lim_{m_{\text{quark}}\to 0} \frac{M^2_{\pi^+}}{m_u + m_d} = -\frac{1}{f^2_\pi}\langle 0|\bar{u}u + \bar{d}d|0\rangle\,. \tag{6.9}$$

The symmetry of the vacuum under $SU(3)_{L+R}$ implies that the expectation values of $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$ coincide in the chiral limit:

$$\langle 0|\bar{u}u|0\rangle = \langle 0|\bar{d}d|0\rangle = \langle 0|\bar{s}s|0\rangle \tag{6.10}$$

and (6.9) requires them to be negative.

Note that the constant g_π and the vacuum expectation values $\langle 0|\bar{q}q|0\rangle$ depend on the renormalization point. The transformation law is contragredient to the transformation law for the quark masses; the quantity $m_u\langle 0|\bar{u}u|0\rangle$ is renormalization group invariant.

If the quark masses do not vanish, the short distance expansion for the product $A_\mu(x)\,P^+(y)$ contains a term proportional to the unit operator $\sim m z_\mu z^{-6}\mathbb{1}$. Since this contribution is more singular than the coefficient of the operator $\bar{u}u + \bar{d}d$ [see (6.4)] the simple argument given above that unambiguously identifies the vacuum expectation value of the quark densities in the chiral limit does not suffice. [The commutator $[A_\mu, P^+]$ contains a c-number Schwinger term.] The spectral function $\rho(p^2)$ associated with $\langle 0|A_\mu P^+|0\rangle$ tends to a constant at high energies (more precisely, it decreases logarithmically like the running quark mass) and it is not a trivial matter to establish an unambiguous connection between the vacuum expectation values $\langle 0|\bar{q}q|0\rangle$ and physical quantities (for a recent paper on the problem see Novikov et al. [77]). We shall not make use of the order parameters $\langle 0|\bar{q}q|0\rangle$ except in the chiral limit and do not discuss the difference between $\langle 0|\bar{u}u|0\rangle$, $\langle 0|\bar{d}d|0\rangle$ and $\langle 0|\bar{s}s|0\rangle$ in the real world.

7. Quark masses as perturbations

In the chiral limit the mass spectrum consists of mass-degenerate multiplets of SU(3); the pseudoscalar octet is massless. If the masses of the light quarks are turned on, these multiplets split. In particular, the members of the pseudoscalar octet drift away from mass zero as is schematically indicated in fig. 1. In this and in the following sections we investigate the expansion of the hadronic energy levels $M_n(\Lambda, m_u, m_d, m_s, m_c, \ldots)$ around the chiral limit $m_u = m_d = m_s = 0$. The quark mass expansion is often referred to as *chiral perturbation theory* [78, 79]. The flavour asymmetries are caused by the quark mass term in the Hamiltonian:

$$H_{QCD} = H_0 + H_1$$

$$H_1 = \int d^3x\, \{m_u \bar{u}u + m_d \bar{d}d + m_s \bar{s}s\}\,. \tag{7.1}$$

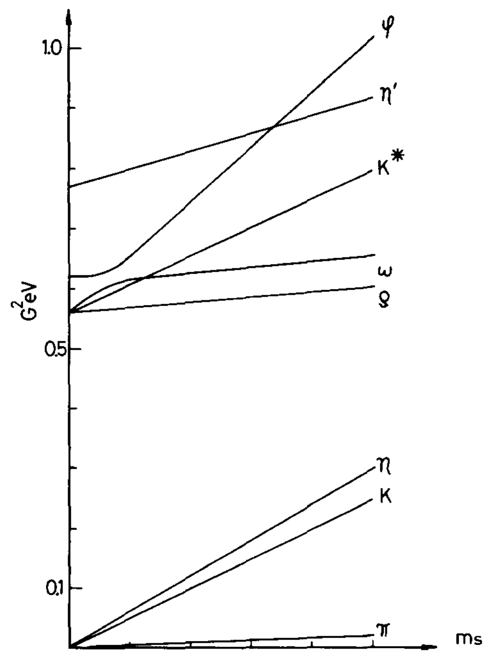

Fig. 1. Square of bound state mass as a function of the quark mass. Schematic plot for $m_u = m_d$ at fixed ratio m_s/m_u.

The expansion of the energy levels in powers of the light quark masses amounts to a perturbative expansion of QCD in powers of H_1. (Note that H_0 treats only the light quarks as massless, the mass terms of the heavy quarks are retained.) In lowest order the perturbation shifts the energy levels by

$$\delta E_n = \langle\psi_n|H_1|\psi_n\rangle/\langle\psi_n|\psi_n\rangle\,. \tag{7.2}$$

Using covariantly normalized plane waves

$$\langle p', n|p, n\rangle = (2\pi)^3\, 2p^0\, \delta^3(\boldsymbol{p}' - \boldsymbol{p}) \tag{7.3}$$

this becomes

$$\delta E_n = \frac{1}{2E_n}\langle p, n|(m_{\mathrm{u}}\bar{u}u + m_{\mathrm{d}}\bar{d}d + m_{\mathrm{s}}\bar{s}s)|p, n\rangle\,. \tag{7.4}$$

Since the perturbation does not affect the momentum of the state we have

$$\delta(M_n^2) = 2E_n\,\delta E_n = \langle p, n|\{m_{\mathrm{u}}\bar{u}u + m_{\mathrm{d}}\bar{d}d + m_{\mathrm{s}}\bar{s}s\}|p, n\rangle\,. \tag{7.5}$$

The expansion of the square of the mass $M_n^2(\Lambda, m_{\mathrm{u}}, m_{\mathrm{d}}, m_{\mathrm{s}}, m_{\mathrm{c}}, \ldots)$ in powers of the quark masses therefore takes the form

$$M_n^2 = A_n + m_{\mathrm{u}}B_n^{\mathrm{u}} + m_{\mathrm{d}}B_n^{\mathrm{d}} + m_{\mathrm{s}}B_n^{\mathrm{s}} + \cdots \tag{7.6}$$

where A_n denotes the (mass)^2 of the level in the chiral limit. The expansion coefficients B_n^{u}, B_n^{d}, B_n^{s} are the matrix elements of the operators $\bar{u}u$, $\bar{d}d$, $\bar{s}s$ in the unperturbed, symmetric state $|p, n\rangle$

$$B_n^{\mathrm{q}} = \langle p, n|\bar{q}q|p, n\rangle\,, \qquad \mathrm{q} = \mathrm{u}, \mathrm{d}, \mathrm{s}\,. \tag{7.7}$$

If there were no heavy quarks, A_n would be given by some pure number times Λ^2, the coefficients B_n^{u}, B_n^{d}, B_n^{s} would be (renormalization point dependent) pure numbers times Λ. In the presence of heavy quarks the pure numbers become functions of the heavy quark mass ratios m_{c}/Λ, $m_{\mathrm{b}}/\Lambda, \ldots$

If A_{n} does not vanish, the expansion (7.6) for M_n^2 may be replaced by the linear mass formula

$$\begin{aligned} &M_n = a_n + m_{\mathrm{u}}b_n^{\mathrm{u}} + m_{\mathrm{d}}b_n^{\mathrm{d}} + m_{\mathrm{s}}b_n^{\mathrm{s}} + \cdots \\ &a_n = A_n^{1/2}\,, \qquad b_n^{\mathrm{q}} = \tfrac{1}{2}A_n^{-1/2}B_n^{\mathrm{q}}\,. \end{aligned} \tag{7.8}$$

As long as A_n is different from zero the linear and quadratic mass formulae only differ by effects of order $(m_{\mathrm{quark}})^2$ and we have no reason to prefer one to the other. For the pseudoscalar mesons the quantity A_n however vanishes. In this case a quark mass expansion is possible only for M_n^2. The quantity $M_\pi(\Lambda, m_{\mathrm{u}}, \ldots)$ e.g. has a square root singularity at $m_{\mathrm{quark}} = 0$. [In the above derivation it is crucial that the states $|p, n\rangle$ are normalized covariantly, such that the expectation value of the scalar $m_{\mathrm{q}}\bar{q}q$ is independent of the momentum of the state. It is not possible to work in the rest frame of the particle because for the pseudoscalar mesons there is no rest frame in the symmetry limit around which we are expanding.] In the following we will consistently be working with mass formulae for M^2.

Note that we are dealing with perturbations of a degenerate spectrum: (i) In the chiral limit the mass levels constitute degenerate multiplets of the flavour group SU(3). (ii) In addition the presence of massless Goldstone bosons in the unperturbed system produces a continuous spectrum starting at the common mass of the multiplet. (The same effect occurs in QED where it is due to photons rather than pions.) We briefly discuss the problems caused by these two types of degeneracies and first consider point (i).

The mesonic bound states transform according to the representations $\underset{\sim}{1}, \underset{\sim}{3}, \underset{\sim}{3}^*$ and $\underset{\sim}{8}$ and the baryon multiplets constitute the representations $\underset{\sim}{1}, \underset{\sim}{3}, \underset{\sim}{3}^*, \underset{\sim}{6}, \underset{\sim}{8}$ and $\underset{\sim}{10}$. To apply perturbation theory we have to choose the states in such a manner that the perturbation does not produce first order transitions between degenerate levels. Since the perturbation has $\Delta T_3 = \Delta Y = 0$ it is convenient to use a basis in which T_3 and Y are diagonal. In this basis mixing within a given multiplet only occurs between states that have the same charge and the same hypercharge. The only multiplet which contains two such states is the octet $(\pi^0/\eta, \Sigma^0/\Lambda)$. We will discuss mixing of these octet states in sections 8 and 9.

Even if two levels are not exactly degenerate in the chiral limit the perturbation will induce mixing if the perturbation is not small in comparison with the energy separation of the unperturbed levels. A mixing phenomenon of this type occurs in all known light meson multiplets with the exception of the Goldstone bosons: the octets are nearly degenerate with the singlets ($\rho - \omega \simeq -6$ MeV, $\delta - S^* \simeq 0$, $A_2 - f \simeq 45$ MeV, $g - \omega_3 \simeq 35$ MeV). The origin of this degeneracy is connected with the OZI rule [80–83]: gluon exchange diagrams treat all flavours alike and hence give all nine flavour combinations the same bound state mass in the chiral limit. In the pseudoscalar channel nonet degeneracy is lifted by the annihilation diagrams responsible for the axial anomaly. In the other channels the degeneracy will be lifted e.g. if there are nearby glueballs with the same J^{PC}. (The OZI rule however also suppresses the coupling of these states to the $\bar{q}q$ multiplets.) The observed nonet degeneracy in the 0^{++} and 1^{--} multiplets suggests that there are no nearby glueballs with these quantum numbers (unless their couplings are very small).

In the case of the qqq bound states the representations $\underset{\sim}{1}, \underset{\sim}{8}, \underset{\sim}{10}$ require the space/spin wave functions to have different symmetry properties. There is no reason to expect the different multiplets to occur at nearly the same mass.

The implications of point (ii) are much more subtle: the presence of massless particles in the chiral limit leads to *infrared divergences* in chiral perturbation theory [84]. They are due to the fact that in the chiral limit the state $n + \pi$ is degenerate with the state n – the corresponding energy denominator in the perturbative expansion of M_n^2 vanishes. A careful treatment of the infrared singularity shows that the terms neglected in (7.6) are in general of order $(m_{\text{quark}})^{3/2}$; in the case of the Goldstone bosons they are of order $m_{\text{quark}}^2 \ln m_{\text{quark}}$ [85, 86]. We will come back to this point in section 10.

8. Masses of the Goldstone bosons

We first consider the quark mass expansion for the pseudoscalar octet. In this case the constant A_n vanishes and we therefore have to work with quadratic mass formulae. For the pion the coefficients $B^{\mathrm{u}}_{\pi^+}$, $B^{\mathrm{d}}_{\pi^+}$ and $B^{\mathrm{s}}_{\pi^+}$ were determined in section 6 – they are related to the vacuum expectation value $\langle 0|\bar{u}u|0\rangle$ in the chiral limit:

$$M^2_{\pi^\pm} = (m_\mathrm{u} + m_\mathrm{d})B + \mathrm{O}(m_\mathrm{q}^2 \ln m_\mathrm{q})$$

$$B = -\frac{2}{f_\pi^2}\langle 0|\bar{u}u|0\rangle \,. \tag{8.1}$$

Applying the same analysis to the kaon we get

$$M^2_{K^\pm} = (m_u + m_s)B + O(m^2_q \ln m_q)$$
$$M^2_{K^0} = M^2_{\bar{K}^0} = (m_d + m_s)B + O(m^2_q \ln m_q) \tag{8.2}$$

with the same constant B (recall that in the chiral limit $f_\pi = f_K$, $\langle 0|\bar{u}u|0\rangle = \langle 0|\bar{d}d|0\rangle = \langle 0|\bar{s}s|0\rangle$).

If $m_u \neq m_d$ the states π^0 and η mix: the perturbation H_1 generates transitions between the unperturbed octet states $|\pi^0\rangle_0$ and $|\eta\rangle_0$. The eigenstates are

$$|\pi^0\rangle = \cos\theta\,|\pi^0\rangle_0 + \sin\theta\,|\eta\rangle_0$$
$$|\eta\rangle = -\sin\theta\,|\pi^0\rangle_0 + \cos\theta\,|\eta\rangle_0$$
$$\mathrm{tg}\,2\theta = \frac{\sqrt{3}}{2}\frac{m_d - m_u}{m_s - \hat{m}}\,; \qquad \hat{m} = \tfrac{1}{2}(m_u + m_d) \tag{8.3}$$

and the physical masses are given by

$$M^2_{\pi^0} = 2\hat{m}B - \tfrac{4}{3}(m_s - \hat{m})B\sin^2\theta/\cos 2\theta$$
$$M^2_\eta = \tfrac{2}{3}(2m_s + \hat{m})B + \tfrac{4}{3}(m_s - \hat{m})B\sin^2\theta/\cos 2\theta\,. \tag{8.4}$$

The mixing angle is proportional to the ratio of the SU(2)-breaking mass difference $m_d - m_u$ to the SU(3)-breaking mass difference $m_s - \hat{m}$. Since this ratio is a small number, the mixing angle is small. Up to terms of order θ^2 the state π^0 is degenerate with $\pi^\pm$ and the mass of the η is given by

$$M^2_\eta = \tfrac{2}{3}(2m_s + \hat{m})B + O\left(\frac{(m_d - m_u)^2}{m_s - \hat{m}}, m^2 \ln m\right). \tag{8.5}$$

The mixing causes the levels to repel. The neutral pion becomes somewhat lighter than $\pi^\pm$:

$$M^2_{\pi^+} - M^2_{\pi^0} \simeq \frac{1}{4}\frac{(m_u - m_d)^2}{m_s - \hat{m}}B\,. \tag{8.6}$$

The ninth member of the pseudoscalar nonet, the η', does not become massless in the chiral limit. In the framework of the perturbation expansion with respect to the light quark masses this state does therefore not play any special role. There is of course no reason for the matrix elements $\langle\eta'|H_1|\eta\rangle$ and $\langle\eta'|H_1|\pi^0\rangle$ to vanish and the perturbation will therefore induce mixing between η and η' as well as between π^0 and η'. The mixing angles are of first order in the quark masses (proportional to $m_s - \frac{1}{2}(m_u + m_d)$ and to $m_d - m_u$ respectively). In the mass spectrum however the presence of the η' makes itself felt only in terms proportional to $(m_{\text{quark}})^2$ which we have neglected in the above discussion. [Numerically, the contribution of the η' is enhanced in comparison with other intermediate states because it is associated with a relatively small energy denominator – the perturbation shifts M^2_η from zero to about 1/3 of $M^2_{\eta'}$. It is however very questionable whether the $\eta\eta'$ mixing angle may be extracted from the mass spectrum by disregarding all other effects of order $(m_{\text{quark}})^2$.]

Spontaneously broken chiral symmetry explains at once why the masses of the pseudoscalar octet are not at all SU(3)-symmetric ($M_\pi^2 : M_K^2 : M_\eta^2 = 1:13:16$) whereas the masses of all other multiplets deviate by less than 20% from their mean values: the squares of the pseudoscalar masses are proportional to the quark masses. There is no reason why the quark masses should show less scattering than the lepton masses do – hence it is natural to find that the Goldstone boson masses also show considerable scattering.

If we ignore higher order corrections for the moment, we may extract the quark mass ratio $m_s : \hat{m}$ either from the ratio

$$M_K^2 : M_\pi^2 = (m_s + \hat{m}) : 2\hat{m} \to m_s : \hat{m} = 25.9 \tag{8.7}$$

or from

$$M_\eta^2 : M_\pi^2 = (2m_s + \hat{m}) : 3\hat{m} \to m_s : \hat{m} = 24.3\,. \tag{8.8}$$

The fact that the two determinations agree remarkably well shows that the Gell-Mann–Okubo formula is well satisfied by the pseudoscalar octet. The asymmetries of the mass values in this multiplet require the s-quark to be about 25 times as heavy as the average of u and d.

The mass formulae for K^- and K^0 also indicate that in order for the K^+ to be lighter than the K^0 the up-quark must be lighter than the down-quark. To extract a reliable value for the ratio $(m_u - m_d) : (m_u + m_d)$ from the observed value of $(M_{K^+}^2 - M_{K^0}^2) : M_\pi^2$ we however need to correct these mass formulae for the energy of the photon cloud which makes a significant contribution to isospin breaking mass differences. This will be done in sections 11 and 12.

9. First order mass formulae for other multiplets

The general form of the quark mass expansion for any hadronic level was given in section 7. To first order the expansion involves four unknown constants A, B^u, B^d, B^s for each one of the states. We now exploit the fact that the unperturbed Hamiltonian is symmetric under the flavour group SU(3). One consequence of this symmetry is that the constants A_n are the same for all members of a given multiplet. Furthermore, the symmetry relates the coefficients B_n^u, B_n^d and B_n^s which stand for the matrix elements of the operators $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$ in the various states of the multiplet. Consider e.g. the baryon octet. In this case the B-coefficients involve three independent matrix elements (F-coupling, D-coupling and singlet). We use this property to express all B-coefficients in terms of the *proton* matrix elements of $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$:

$$\begin{aligned} B^u &= \langle p|\bar{u}u|p\rangle \\ B^d &= \langle p|\bar{d}d|p\rangle \\ B^s &= \langle p|\bar{s}s|p\rangle\,. \end{aligned} \tag{9.1}$$

The first order mass formulae for the baryon octet then read

$$
\begin{aligned}
M_{\mathrm{p}}^2 &= A + m_{\mathrm{u}}B^{\mathrm{u}} + m_{\mathrm{d}}B^{\mathrm{d}} + m_{\mathrm{s}}B^{\mathrm{s}} + \mathrm{O}(m^{3/2})\\
M_{\mathrm{n}}^2 &= A + m_{\mathrm{u}}B^{\mathrm{d}} + m_{\mathrm{d}}B^{\mathrm{u}} + m_{\mathrm{s}}B^{\mathrm{s}} + \mathrm{O}(m^{3/2})\\
M_{\Sigma^+}^2 &= A + m_{\mathrm{u}}B^{\mathrm{u}} + m_{\mathrm{d}}B^{\mathrm{s}} + m_{\mathrm{s}}B^{\mathrm{d}} + \mathrm{O}(m^{3/2})\\
M_{\Sigma^-}^2 &= A + m_{\mathrm{u}}B^{\mathrm{s}} + m_{\mathrm{d}}B^{\mathrm{u}} + m_{\mathrm{s}}B^{\mathrm{d}} + \mathrm{O}(m^{3/2})\\
M_{\Xi^0}^2 &= A + m_{\mathrm{u}}B^{\mathrm{d}} + m_{\mathrm{d}}B^{\mathrm{s}} + m_{\mathrm{s}}B^{\mathrm{u}} + \mathrm{O}(m^{3/2})\\
M_{\Xi^-}^2 &= A + m_{\mathrm{u}}B^{\mathrm{s}} + m_{\mathrm{d}}B^{\mathrm{d}} + m_{\mathrm{s}}B^{\mathrm{u}} + \mathrm{O}(m^{3/2})\,.
\end{aligned}
\tag{9.2}
$$

The states Σ^0 and Λ are mixed according to (8.3) with the same mixing angle. Neglecting terms of order $(m_{\mathrm{u}} - m_{\mathrm{d}})^2/(m_{\mathrm{s}} - \hat{m})$ we obtain

$$
\begin{aligned}
M_{\Sigma^0}^2 &= A + \hat{m}(B^{\mathrm{u}} + B^{\mathrm{s}}) + m_{\mathrm{s}}B^{\mathrm{d}} + \mathrm{O}(m^{3/2})\\
M_{\Lambda}^2 &= A + \tfrac{1}{3}\hat{m}(B^{\mathrm{u}} + 4B^{\mathrm{d}} + B^{\mathrm{s}}) + \tfrac{1}{3}m_{\mathrm{s}}(2B^{\mathrm{u}} - B^{\mathrm{d}} + 2B^{\mathrm{s}}) + \mathrm{O}(m^{3/2})\,.
\end{aligned}
\tag{9.3}
$$

The baryon mass values are left unchanged if the quark masses and the constants A, B^{u}, B^{d}, B^{s} are subject to the following transformations:

$$
\begin{aligned}
&\text{(a)}\quad m_{\mathrm{q}} \to \lambda m_{\mathrm{q}}\,, \qquad B^{\mathrm{q}} \to \lambda^{-1}B^{\mathrm{q}}\,, \qquad A \to A\\
&\text{(b)}\quad m_{\mathrm{q}} \to m_{\mathrm{q}} + \mu\,, \qquad B^{\mathrm{q}} \to B^{\mathrm{q}}\,, \qquad A \to A - \mu(B^{\mathrm{u}} + B^{\mathrm{d}} + B^{\mathrm{s}})\\
&\text{(c)}\quad m_{\mathrm{q}} \to m_{\mathrm{q}}\,, \qquad B^{\mathrm{q}} \to B^{\mathrm{q}} + \sigma\,, \qquad A \to A - \sigma(m_{\mathrm{u}} + m_{\mathrm{d}} + m_{\mathrm{s}})\,.
\end{aligned}
\tag{9.4}
$$

This implies that only 4 combinations of the 7 constants m_{u}, m_{d}, m_{s}, A, B^{u}, B^{d}, B^{s} are relevant for the baryon masses. The mass formulae therefore imply 4 constraints among the 8 mass values. One of these constraints is the Gell-Mann–Okubo formula, the other three are the Coleman–Glashow relations for the splittings within the isospin multiplets. In the absence of information about the value of the constant A it is not possible to extract quark mass ratios like $m_{\mathrm{s}}:\hat{m}$ from the observed masses of the baryons. The same remark applies to all other multiplets whose mass in the chiral limit is not known. The mass of the Goldstone bosons in the chiral limit is known – this is why we do get information about the ratio $m_{\mathrm{s}}:\hat{m}$ from the mass spectrum of the pseudoscalar mesons. What we can extract from the mass spectrum of the other multiplets is the ratio of differences $R = (m_{\mathrm{s}} - \hat{m}):(m_{\mathrm{d}} - m_{\mathrm{u}})$ which measures the strength of isospin breaking in comparison with SU(3)-breaking. We will determine this ratio from different multiplets in section 13 after having analyzed the size of the corrections due to higher order quark mass terms and due to the electromagnetic interaction.

In order for the SU(3) splittings within multiplets not to exceed 20% of the mean mass the quark mass difference $m_{\mathrm{s}} - \hat{m}$ which is responsible for these splittings must be small in comparison with the scale of the strong interaction; since m_{s} is much larger than $\hat{m}$ this implies that all light quark masses m_{u}, m_{d} and m_{s} must be small quantities in comparison with the scale of QCD. The same conclusion results from the observation that the pseudoscalar octet contains the 8 lightest hadrons: the quark mass term does not shift the Goldstone bosons far away from mass zero ($M_{\eta}^2 \simeq \frac{1}{3}M_{\eta'}^2$).

To give the condition "m_{u}, m_{d}, $m_{\mathrm{s}} \ll$ scale of QCD" a quantitative form we should specify what we mean by the scale of QCD. There are various quantities that remain finite in the chiral limit and may serve as mass scales: Λ, f_π, M_ρ, M_{p}, $(\alpha')^{-1/2}$, $\langle p_{\mathrm{T}}\rangle, \ldots$ In fact, these scales are not independent, but are given by some pure number of order one – determined by chiral QCD – times Λ. The scale Λ itself which is closest to the Lagrangian turns out to be too small to serve as a standard with respect to which to measure the size of the SU(3) asymmetry parameter $m_s - \hat{m}$. The quantity f_π is also too small. M_ρ seems to be more adequate although there is no particular reason to prefer this quantity to $(\alpha')^{-1/2}$ or to M_p.

The eightfold way is an approximate symmetry of the strong interaction not because the quarks u, d, s have similar masses, but because their mass differences are small in comparison to M_ρ. Isospin is an even better symmetry, not because $m_u - m_d$ is small in comparison with m_u, but because $m_u - m_d$ is very small in comparison to M_ρ.

The mass of the charmed quark is not a small quantity in comparison with the scale of QCD: the η_c is considerably heavier than the η' ($M^2_{\eta_c} \simeq 10 M^2_{\eta'}$), the charmed baryon $\Lambda_c(2273)$ is more than twice as heavy as the proton. If the mass of the charmed quark is varied from its actual value down to m_s, the Λ_c loses half its weight and moves to $\Lambda(1115)$; if m_c is diminished further and put equal to $\hat{m}$, the Λ_c ends up at the nucleon mass. The difference $m_c - \hat{m}$ must therefore be larger than $m_s - \hat{m}$ by a factor of the order of 8. The difference $m_c - \hat{m}$ is the symmetry breaking parameter of SU(4), the ratio $(m_c - \hat{m}):(m_s - \hat{m})$ measures the strength of SU(4) breaking versus SU(3) breaking. The size of $m_c - \hat{m}$ makes it clear that one cannot get very reliable information from SU(4) symmetry (see section 17 for a brief discussion of SU(4) mass formulae).

For the multiplets $\underline{3}$, $\underline{3}^*$, $\underline{6}$ and $\underline{10}$ the first order mass formulae analogous to (9.2) may be found in appendix A.

The quark mass expansion for the *meson nonets* is complicated by mixing. Consider for definiteness the vector meson nonet. In the chiral limit this multiplet consists of a singlet and an octet with a spacing of order $M_\omega - M_\rho = 6$ MeV. It is not known whether the singlet or the octet is the lower one of the two. When the quark masses are slowly turned on the levels undergo mixing through angles that are sensitive to the details of the small pieces of the interaction that violate the OZI-rule, until the strange quark has become heavy enough for the mass term $m_s\bar{s}s$ to dominate. (Note that the quark mass term produces the splitting $M_\varphi - M_\rho \simeq 240$ MeV which is large in comparison with the original separation of singlet and octet.) The quark mass term dominates as soon as m_s is large in comparison with $|M_1 - M_8| \sim 6$ MeV. For the physical value of m_s the state $|\varphi\rangle$ contains practically only strange quarks. Within the accuracy of first order mass formulae the mixing angle defined by (we choose the phase of the state $|\varphi\rangle$ such that it reduces to $|\bar{s}s\rangle$ for ideal mixing):

$$|\varphi\rangle = -\cos\varphi\, |8, T=0\rangle + \sin\varphi |1\rangle \tag{9.5}$$

has the ideal value $\varphi = 35.3°$ ($\mathrm{tg}\,\varphi = 1/\sqrt{2}$). In (9.5) we have neglected the small component of the state $|\varphi\rangle$ in the direction of the vector $|8, T=1, T_3=0\rangle$ which is caused by the isospin violating piece of the quark mass term and is proportional to $m_u - m_d$. In fact, the transition matrix element $\langle \bar{s}s|\bar{q}mq|\bar{u}u - \bar{d}d\rangle$ which determines the size of the isospin violating component of $|\varphi\rangle$ is forbidden by the OZI-rule. The isospin one component of $|\varphi\rangle$ is therefore very small.

The isospin breaking piece of the perturbation does however have a pronounced effect on the states $|\rho^0\rangle$ and $|\omega\rangle$, because (i) their energy separation is not large in comparison with the size of the perturbation $\frac{1}{2}(m_u - m_d)(\bar{u}u - \bar{d}d)$ and (ii) the OZI-rule does not inhibit transitions between these two states. The interference between ρ^0 and ω can be measured directly in reactions such as $e^+e^- \to \pi^+\pi^-$ by analyzing the behaviour of the cross section in the vicinity of the ω mass. Up to electromagnetic corrections the probability for an ω to decay into $\pi^+\pi^-$ is determined by the ratio $(m_u - m_d):(m_s -$

$\hat{m}$). The reaction $e^+e^- \to \pi^+\pi^-$ therefore provides us with an independent determination of this ratio. The relevant formulae for ρ^0–ω-mixing are collected in appendix B. Furthermore, we give a simple quantum mechanical model describing the interference of two resonances in appendix F.

10. Higher order terms in the quark mass expansion

In the preceding two sections we focused on the lowest order terms in the expansion of the hadron masses $M_n(\Lambda, m_u, m_d, m_s, m_c, \ldots)$ around the chiral limit of the light quarks, $m_u = m_d = m_s = 0$. As mentioned in section 7, the quark mass expansion is not a simple Taylor series, but contains terms that are not analytic in m_u, m_d and m_s [88]. We now discuss these nonanalytic contributions and then make simple estimates of the size of higher order terms in the quark mass expansion in order to determine the accuracy to which quark mass ratios can be extracted from the spectrum of baryons and mesons.

For definiteness we first consider the mass of the nucleon and discuss the expansion around the chiral limit of the u and d quarks, holding m_s fixed at the physical value. In this case the first order mass formula reads

$$M_p^2 = \bar{A} + m_u \bar{B}^u + m_d \bar{B}^d \tag{10.1}$$

where $\bar{A}$, $\bar{B}^u$, $\bar{B}^d$ are unknown functions of Λ, m_s, $m_c, \ldots$ (in the preceding section we also expanded $\bar{A}$, $\bar{B}^u$, $\bar{B}^d$ in powers of m_s). As shown by Langacker and Pagels [89] the next term in the expansion is not of order $(m_{\text{quark}})^2$, but of order $(m_{\text{quark}})^{3/2}$. The structure of the leading nonanalytic terms in the quark mass expansion for the mass of any particle may be obtained from *improved chiral perturbation theory*, a method described in appendix C [90, 91]. One reorders the quark mass expansion by summing leading infrared divergences to all orders in the quark mass. The net result is that the leading nonanalytic terms in $M_n(\Lambda, m_u, m_d, m_s, \ldots)$ are removed by subtracting the lowest order diagram of an effective chiral Lagrangian with physical masses in the propagators. As long as the effective Lagrangian gives the particles the proper masses and satisfies the soft-pion theorems it also exhibits the correct leading nonanalytic terms in the quark mass expansion.

Improved chiral perturbation theory tells us that to remove the leading nonanalytic piece in the quark mass expansion we may proceed in the same manner as in the case of the electromagnetic self-energies: the main contribution to the photon cloud also comes from the Born term diagram (see section 12). We merely have to replace the electric charge by the axial charge and describe pion emission in terms of derivative coupling $\sim \partial_\mu \pi / f_\pi$. The reason behind this simple result is that the effective coupling of pions is weak at small momentum k – the probability amplitude for the quark mass term to generate more than one pion is smaller by powers of k/f_π. What counts in the leading nonanalytic term is only the coefficient of the leading infrared singularity and hence only the lowest order diagram.

The leading nonanalytic term in the proton mass is removed by subtracting the lowest order diagram due to pion emission of an effective chiral Lagrangian with physical masses in the propagators. If $m_u = m_d = \hat{m}$ the quantity $\bar{M}_N$ defined by

$$\bar{M}_N^2 = M_N^2 - \frac{3g_A^2}{32\pi^2 f_\pi^2} J(M_N, M_N, M_\pi) \tag{10.2}$$

in terms of the physical nucleon and pion masses M_N, M_π is free from the leading nonanalytic term in m_u, m_d. The integral $J(M_N, M_N, M_\pi)$ is defined in (C.4) and $f_\pi = 132\,\mathrm{MeV}$, $g_A = 1.25$; expanding this integral in powers of M and rewriting the result in the form (10.1) one obtains

$$M_N^2 = \bar{A} + \hat{m}(\bar{B}^u + \bar{B}^d) - \frac{3g_A^2}{8\pi f_\pi^2}\bar{A}^{1/2}M_\pi^3 + \cdots . \tag{10.3}$$

The last term is nonanalytic in the quark masses, because it grows with the third power of M_π, i.e. is proportional to $(m_u + m_d)^{3/2}$.

It is not difficult to understand the origin of the nonanalytic term. In the chiral limit the nucleon is surrounded by a cloud of virtual pions that extends to infinity. As the quark masses are turned on the pions become heavy and the cloud shrinks, the probability for finding a virtual pion falling off like $\exp(-2M_\pi r)$. The static model provides us with a simple description of the properties of this cloud. Before we look at the static model we should perhaps clarify the significance of a language that uses words like meson cloud, virtual pions and the like in the context of QCD. What we are talking about are nonanalytic terms in the quark mass expansion and what we are claiming is that these nonanalytic terms are the same as those occurring in an effective Lagrangian, the simplest version of which is the static model. This model describes the nucleon in terms of a core and a pion cloud. The model provides us with a reliable effective infrared Lagrangian to the extent that the nucleon is heavy in comparison to the pions and that the inverse radius of the core which plays the role of the ultraviolet cutoff that occurs in a relativistic effective Lagrangian is large in comparison with both the meson mass and the baryon mass difference due to the quark mass term (to $M_n - M_p$ in the case of an expansion in powers of m_u, m_d at fixed m_s and to $M_\Sigma - M_N$ if nonanalytic terms in m_s due to K and η-meson emission are considered). In fact, one may verify that the expressions given for the nonanalytic terms in appendix C on the basis of improved chiral perturbation theory reduce to the meson cloud energies in the static model if one takes the limit $M_N \to \infty$. [The core distribution $\rho(\boldsymbol{x})$ that occurs in the static limit is the Fourier transform of the axial vector form factor $G(-\boldsymbol{q}^2)$.] For this reason we discuss the physics of the nonanalytic terms in the framework of the static model. (In the numerical work we have used the relativistic prescriptions given in appendix C.) The Hamiltonian of the static model is (see e.g. ref. [92]):

$$H = \int \mathrm{d}^3x\, \tfrac{1}{2}[\dot{\boldsymbol{\phi}}^2 + (\nabla \cdot \boldsymbol{\phi})^2 + M_\pi^2 \boldsymbol{\phi}^2] + \frac{g_A}{\sqrt{2}f_\pi}\int \mathrm{d}^3x\, \rho(\boldsymbol{x})\, \boldsymbol{\sigma} \cdot \nabla(\boldsymbol{\phi} \cdot \boldsymbol{\tau}) - \tfrac{1}{2}\Delta M \tau_3 . \tag{10.4}$$

We have used the Goldberger–Treiman relation to express the coupling constant in terms of g_A and f_π; the quantity ΔM stands for the neutron–proton mass difference. It is straightforward to determine the ground state energy to second order in the coupling constant. Neglecting ΔM (i.e. putting $m_u = m_d$) one finds

$$E_\pi = \frac{-3g_A^2}{16\pi f_\pi^2}\int \mathrm{d}^3x\, \mathrm{d}^3y\, \nabla\rho(\boldsymbol{x}) \cdot \nabla\rho(\boldsymbol{y}) \frac{\exp\{-M_\pi|\boldsymbol{x}-\boldsymbol{y}|\}}{|\boldsymbol{x}-\boldsymbol{y}|} . \tag{10.5}$$

The expansion of this expression for the energy of the pion cloud in powers of M_π produces the series $1 - M_\pi r + \frac{1}{2}M_\pi^2 r^2 - \frac{1}{6}M_\pi^3 r^3 + \cdots$. The first term only contributes to the constant $\bar{A}$ that represents the $(\mathrm{mass})^2$ of the nucleon in the chiral limit. The second term disappears upon integration and the third term is proportional to $m_u + m_d$ and is therefore absorbed in the constants $\bar{B}^u$, $\bar{B}^d$. One easily checks that the fourth term indeed gives rise to the nonanalytic contribution given in (10.3).

In the static model the total energy of the pion cloud around the nucleon is $E_\pi = -185$ MeV and the individual terms in the expansion $E_\pi = E_0 + E_1 M_\pi^2 + E_2 M_\pi^3 + \cdots$ amount to $E_0 = -202$ MeV, $E_1 M_\pi^2 = +26$ MeV, $E_2 M_\pi^3 = -13$ MeV. (The corresponding relativistic prescription given in appendix C gives $E_\pi = -121$ MeV; see eq. (C.4) with $M_0 \to M_p$ and $M_p = \bar{M}_p + E_\pi$.)

It is clear that one may give similar expressions for the leading nonanalytic terms in the quark mass expansion with respect to all three light quark masses m_u, m_d, m_s. The leading contribution to the nucleon mass then contains terms proportional to M_K^3 and to M_η^3 in addition to the term proportional to M_π^3 given in (10.3). Taken at face value the leading nonanalytic term proportional to M_K^3 shifts the nucleon mass by -235 MeV, to be compared with $E_2 M_\pi^3 = -13$ MeV. If the *leading* nonanalytic contribution would indeed account for those pieces of the kaon cloud around the nucleon that are not already included in the first order mass formula, then we would have to conclude that the validity of the Gell-Mann–Okubo formula, derived on the basis of first order mass formulae, was an accident.

Fortunately, this is not the case. The static model (or improved chiral perturbation theory) makes it clear that one can trust the expansion of the meson cloud energies in powers of the meson mass only if the meson is light in comparison with the inverse radius of the nucleon core, represented by the Fourier transform $\rho(\boldsymbol{x})$ of the axial vector form factor. An expansion of the energy contributed to the nucleon mass by the virtual K-cloud in powers of M_K would be admissible if M_K were small in comparison with the scale set by the axial vector form factor – this is not the case. (Note that what counts here is not the size of m_s, it is the size of the Goldstone boson masses that contain strange quarks.) The full energy of the kaon cloud around the nucleon only amounts to $E_K = -39$ MeV in the static approximation and to $E_K = -24$ MeV in the relativistic prescription given in appendix C. The formal expansion of the self energy obtained by retaining the terms proportional to M_K^3 and disregarding the higher order terms is large because it fails to reproduce the actual value of the corresponding self-energy diagram. The terms up to M_π^3 represent a good approximation only in the case of the pion cloud whose radius is indeed large in comparison with the scale of QCD.

If the nonanalytic terms in the quark mass expansion are handled according to improved chiral perturbation theory *the* corresponding *corrections to the first order mass formulae are small.* [As discussed in appendix C the singlet and octet pieces of the meson cloud energies may be absorbed in a redefinition of the coefficients A and B^u, B^d, B^s if one desires. What counts is the piece of the cloud energy that does not have octet symmetry.] The corrections are nevertheless significant, because they account for the flavour asymmetries in the baryon masses caused by the fact that different members of the octet get an unequal share of the meson cloud energy. The nucleon cloud e.g. consists mainly of pions whereas the cloud surrounding the Ξ predominantly contains kaons – the contribution to the mass, although small, is rather asymmetric.

In fact, the corrections required by improved chiral perturbation theory explain why earlier determinations of the ratio of SU(2) versus SU(3) symmetry breaking parameters, $(m_d - m_u):(m_s - \hat{m})$, from the baryon masses did not give a consistent picture. The effect that is responsible for these discrepancies is quite amusing and we briefly describe it in the framework of the static model. The energy of the π^+ cloud around the proton is given by

$$E_{\pi^+} = -\frac{g_A^2}{f_\pi^2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3 2\omega(\boldsymbol{k})} \frac{\boldsymbol{k}^2\, G^2(-\boldsymbol{k}^2)}{\omega(\boldsymbol{k}) + \Delta M} \tag{10.6}$$

with $\omega(k)=(M_\pi^2+k^2)^{1/2}$. If we put $\Delta M=0$ this expression coincides with (10.5) (up to a factor 2/3, because we are only looking at π^+ here). As discussed in appendix C we however have to use the physical masses in the energy denominator: $E_n-E_0=M_n+\omega-M_p$ and take the neutron–proton mass difference ΔM into account. In the corresponding expression for the energy of the π^--cloud surrounding the neutron the sign of ΔM is reversed. To lowest order in m_u-m_d the charged pion cloud contributes a negative amount to the neutron–proton mass difference:

$$\Delta E=-\frac{2g_A^2}{f_\pi^2}\,\Delta M\int\frac{d^3k}{(2\pi)^3 2\omega(k)}\,\frac{k^2\,G^2(-k^2)}{\omega(k)^2}\,. \tag{10.7}$$

The cloud leads to an attraction of the two levels – if the masses are purified from nonanalytic contributions the neutron–proton mass difference becomes larger. In the static model the shift amounts to $\Delta E=-0.84$ MeV. We emphasize that the contribution we are talking about is not nonanalytic by itself – it is proportional to m_u-m_d and we could just as well remove it by a suitable redefinition of the constants B^u, B^d. That would indeed guarantee that there is no difference between M_n-M_p and the purified mass difference $\bar{M}_n-\bar{M}_p$ to first order in m_u-m_d; the effect would then however show up e.g. in the mass difference $\bar{M}_\Xi-\bar{M}_\Sigma$, because the contribution of the meson cloud to the mass operator cannot be represented as the sum of a singlet and an octet. To leading order the determination of the ratio $(m_d-m_u):(m_s-\hat{m})$ is of course not affected by shuffling of clouds (the procedure only modifies the quark mass expansion of the baryon masses on the level of $(m_{\text{quark}})^2$).

There are of course also nonanalytic terms in the quark mass expansion of the Goldstone boson masses. Since the constant g_A vanishes for this multiplet the leading infrared singularity is only logarithmic [93]. The corresponding nonanalytic contributions in improved chiral perturbation theory are given in appendix C.

Finally we comment on the *terms of order* $(m_{\text{quark}})^2$ which the above discussion left untouched. To estimate the size of these terms one may simply compare the results of an analysis of the baryon mass formulae for M^2 with an analysis of the corresponding mass formulae for M. A somewhat different method, which also works in the case of the Goldstone bosons, is to add the square of the sum of the quark masses which the particle contains [$(m_1+m_2)^2$ for mesons, $(m_1+m_2+m_3)^2$ for baryons] to the mass formula for M^2 and to check that this modification does not substantially affect the results. We do not claim that this simple recipe accurately describes the terms of order $(m_{\text{quark}})^2$ in the quark mass expansion, but we do believe that it provides us with an estimate of their order of magnitude (note that in the limit of very heavy quarks the bound state mass does tend to the sum of the quark masses).

11. Electromagnetic contributions – renormalized self-energy

The preceding sections dealt with pure QCD, the electromagnetic and weak interactions were turned off. In this approximation the particles π, K, N, Λ, Σ, Ξ are stable. Note that even in the absence of electromagnetic interactions the proton is lighter than the neutron if $m_u<m_d$. (There is no reason why m_u should be equal to m_d if $e=0$.)

If the electromagnetic interaction is turned on, the quarks start emitting and absorbing photons. The π^0 and the Σ^0 become unstable against the decays $\pi^0\to\gamma\gamma$, $\Sigma^0\to\Lambda\gamma$. The cloud of virtual photons surrounding a bound state of quarks contributes to the mass of the state. The order of magnitude of this contribution is given by the fine structure constant times the scale of the strong interaction, i.e. by about 1 MeV. It turns out that the isospin breaking effects produced by the mass difference $m_u \neq m_d$ happen also to be of this order of magnitude – we do not know why this is so. In order to extract information about the size of the mass difference $m_u - m_d$ from the observed mass spectrum it is therefore necessary to first correct the data by subtracting the self energy associated with the photon cloud: to apply the mass formulae established in the preceding sections we need to know where the hadronic levels would be found in a world with $e = 0$.

To lowest order the effective Lagrangian describing the contributions from virtual photons is given by

$$\mathscr{L}_{\mathrm{e.m.}} = -\tfrac{1}{2}e^2 \int \mathrm{d}^4 y\, D(x-y)\, T j^\mu(x)\, j_\mu(y) \tag{11.1}$$

where $D(z) = [\mathrm{i}4\pi^2(z^2 - \mathrm{i}\varepsilon)]^{-1}$ is the photon propagator and $j_\mu(x)$ is the electromagnetic current

$$j_\mu(x) = \bar{q} Q \gamma_\mu q\,.$$

(Q is the charge matrix with eigenvalues 2/3 and −1/3.) The product $j^\mu(x)\, j_\mu(y)$ is too singular at $x = y$ for the above integral to make sense as it stands. The behaviour of the current operator product at short distances may be read off from the expansion

$$T j^\mu(x)\, j_\mu(y) = \sum_n C^n(x-y)\, O^n\,.$$

Only scalar operators O^n survive the integral (11.1). Operators of dimension less than or equal to 4 are accompanied with coefficients $C^n(x-y)$ that produce nonintegrable singularities. In QCD only the operators $\mathbb{1}$, $\bar{u}u$, $\bar{d}d$, $\bar{s}s, \ldots$ and $F^a_{\mu\nu}F^{\mu\nu a}$ belong to this class. More explicitly, retaining only scalar operators and writing $z = x - y$ the expansion reads

$$T j^\mu(x)\, j_\mu(y) = C^{\mathbb{1}}(z)\mathbb{1} + \sum_{\mathrm{q=u,d,\ldots}} C^{\mathrm{q}}(z)\, m_{\mathrm{q}} \bar{q}q + C^{\mathrm{F}}(z)\, F^a_{\mu\nu}F^{\mu\nu a} + \cdots\,. \tag{11.2}$$

The coefficients of the operators $\bar{u}u$, $\bar{d}d, \ldots$ are proportional to the corresponding quark masses – we have explicitly displayed this mass factor in (11.2). At short distances the coefficient $C^{\mathbb{1}}(z)$ behaves like $(z^2)^{-3}$ up to logarithms, while $C^{\mathrm{u}}(z)$, $C^{\mathrm{d}}(z), \ldots$ as well as $C^{\mathrm{F}}(z)$ are proportional to $(z^2)^{-1}$. The terms omitted do not give rise to divergences in the integral. The divergences thus only involve operators that are already present in the QCD Lagrangian – a suitable renormalization of vacuum energy, quark masses and coupling constant removes them. The renormalized effective Lagrangian may be written as

$$\mathscr{L}_{\mathrm{e.m.}} = -\tfrac{1}{2}e^2 \int \mathrm{d}^4 y\, D_\lambda(x-y)\, T j^\mu(x)\, j_\mu(y) + \Delta E \cdot \mathbb{1} + \sum_{\mathrm{q=u,d,\ldots}} \Delta m_{\mathrm{q}}\, \bar{q}q - \frac{\Delta g}{2g^3} F^a_{\mu\nu}F^{\mu\nu a} \tag{11.3}$$

where $D_\lambda(z)$ is a regularized photon propagator. The counter terms ΔE, Δm_{u}, $\Delta m_{\mathrm{d}}, \ldots$ and Δg depend on λ in such a manner as to cancel the divergences of the integral as $\lambda \to \infty$. As usual this requirement does not fix the counter terms uniquely; to obtain an unambiguous expression for the effective electromagnetic Lagrangian a renormalization prescription must be adopted. The splitting of the Hamiltonian into a term that represents pure QCD and a term that describes the electromagnetic contribution depends on this prescription. This implies, in particular, that the values of the quark masses which we are attempting to extract from data will depend on the renormalization prescription.

Fortunately, the dependence of the quark mass values on the renormalization prescription adopted for the electromagnetic contribution is very weak. To lowest order in g the counter term Δm_{q} is given by

$$\Delta m_{\mathrm{q}} = \frac{3e^2}{16\pi^2} Q_{\mathrm{q}}^2 m_{\mathrm{q}} \ln \frac{\lambda^2}{\mu^2} \tag{11.4}$$

where μ is the renormalization point. In the case of the up-quark e.g. a change in μ by a factor 2 changes the value of m_{u} by 1‰ of m_{u}, i.e. by less than 0.01 MeV; for the s quark the same change amounts to 0.25‰ of m_{s}, i.e. to less than 0.08 MeV. It is crucial that the electromagnetic self energy of the quark is proportional to the quark mass – if the renormalization ambiguity in m_{u} was 1‰ of a typical hadronic energy scale rather than 1‰ of m_{u} the dependence on the renormalization prescription would be significant. To lowest order in g the renormalization of the coupling constant is given by

$$\Delta g = -\frac{e^2 g^3}{256\pi^4} \operatorname{tr} Q^2 \ln \frac{\lambda^2}{\mu^2}\,. \tag{11.5}$$

In the chiral limit there is no quark mass counter term; only the operator $F^a_{\mu\nu}F^{\mu\nu a}$ produces a divergence in the electromagnetic self-energy of bound states. Since this operator is proportional to the trace of the energy-momentum tensor [compare (1.1)] its matrix element is known – it is proportional to the (mass)^2 of the state. In the chiral limit the logarithmic divergence in the electromagnetic self-energy of any hadron therefore amounts to a universal shift in the mass scale ($\operatorname{Tr} Q^2 = \frac{4}{9}+\frac{1}{9}+\frac{1}{9}=\frac{2}{3}$):

$$\Delta M = \frac{e^2}{216\pi^2} M \ln \frac{\lambda^2}{\mu^2}\,. \tag{11.6}$$

The logarithm is accompanied by a very small coefficient; for the proton e.g. a change in the renormalization scale μ by a factor 2 shifts the self-energy by only 0.06 MeV. In the chiral limit the electromagnetic self energy of the Goldstone bosons is finite to lowest order in e^2 ($M = 0$ implies $\Delta M = 0$). Since the neutral axial currents remain conserved in the presence of the electromagnetic interaction, π^0, K^0, $\bar{\mathrm{K}}^0$ and η stay massless. The charged axial currents are not conserved in QCD; $\pi^\pm$ and $\mathrm{K}^\pm$ obtain a finite mass of order eM_ρ (c.f. eq. (12.9)).

12. Energy of the photon cloud

The energy of the photon cloud is the expectation value of $-\int \mathrm{d}^3x\, \mathscr{L}_{\mathrm{e.m.}}$. To evaluate this expectation value we need to know the matrix element ($\nu = p \cdot q$)

$$T(q^2, \nu) = \tfrac{1}{2}\mathrm{i} \int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}qx} \langle \mathrm{p}| T j^\mu(x) j_\mu(0) |\mathrm{p}\rangle \tag{12.1}$$

for the particle in question. As shown by Cottingham [94], the imaginary part of this matrix element at spacelike values of q is related to the cross section for *electron scattering*. The information contained in the structure functions for electron scattering in fact suffices to determine the entire matrix element T [95] provided $\mathrm{Im}\, T$ does not contain fixed poles [ν-independent contributions, or, more generally, contributions of the type $\nu^n \beta_n(q^2)$ with integer n]. We do not know of a proof that fixed poles of this sort are absent in QCD, but we assume that this is the case, such that the energy of the photon cloud surrounding the proton can in principle be worked out from data on elastic and inelastic electron–proton scattering. The divergences of the electromagnetic effective Lagrangian of course also show up in the energy of the photon cloud: the *Cottingham formula* must be renormalized [96]. The algorithm which we set up [97] to express the renormalized electromagnetic self-energy in terms of the cross section for electron scattering is only valid in the limit $g = 0$ (canonical scaling). To our knowledge an analogous renormalized Cottingham formula that only involves the structure functions for electron scattering and is consistent with the high energy behaviour of these structure functions in QCD is not available in the literature. [Note that if subtracted fixed q^2 dispersion relations are used, an algorithm allowing one to calculate the subtraction $T(q^2, 0)$ in terms of the structure functions is needed.]

The construction of a properly renormalized Cottingham formula is of theoretical interest rather than of practical importance for the following reason: the difference between the electromagnetic self-energies of particles belonging to the same isospin multiplet, say between proton and neutron is finite in the chiral limit $m_u = m_d = 0$. The real world is very close to this limit so that the divergence and the associated renormalization problems are associated with very small coefficients. The reason for the finiteness of the self-energy differences in the chiral limit was given in the last section – it is a consequence of the fact that the quark mass counter terms Δm_u, Δm_d are proportional to the quark masses m_u, m_d respectively and hence vanish in the limit $m_u = m_d = 0$. In fact, consider the difference between the proton and the neutron matrix elements of $Tj^\mu(x) j_\mu(y)$. The corresponding short distance expansion involves the matrix elements of the operators $\mathbb{1}$, $\bar{u}u$, $\bar{d}d$, $\bar{s}s$, $F^a_{\mu\nu}F^{\mu\nu a}, \ldots$ The unit operator – the disconnected piece – may be disregarded because it amounts to a universal energy shift for all states, including the vacuum. The difference of the matrix elements belonging to the isoscalar operators $\bar{s}s$, $F^a_{\mu\nu}F^{\mu\nu a}, \ldots$ between proton and neutron is proportional to $m_u - m_d$ (in the limit $m_u = m_d$ isospin is an exact symmetry of QCD, hence the matrix elements of isoscalar operators are the same for all members of an isospin multiplet) and therefore vanishes for $m_u - m_d = 0$. Finally, in the case of the operators $\bar{u}u$ and $\bar{d}d$ the expansion coefficient is proportional to m_u and m_d respectively; the contribution from these operators to the short distance expansion hence also vanishes in the limit $m_u = m_d = 0$. This shows that all singular terms in the short distance expansion of the difference between the proton and neutron matrix elements of $Tj^\mu(x) j_\mu(y)$ vanish in the limit $m_u = m_d = 0$; in that limit the difference of the electromagnetic energies of proton and neutron is therefore finite and may be worked out by using the unrenormalized Cottingham formula.

The electromagnetic self-energies of baryons and mesons were investigated by many authors (see Zee [98], Cini and Stichel [99] for a review up to 1972). We have reanalyzed the various contributions to the Cottingham formula in 1975, using the data on inelastic scattering available at that time (see also Gunion [100]). We found that the only sizeable contribution to the proton–neutron mass difference comes from the Born term; resonance region, Regge region and deep inelastic scattering contribute very little. The fact that the deep inelastic region does not contribute significantly is closely related to the observation made in the last section: the logarithmic divergence which is associated with these contributions has a tiny coefficient, even if one looks at the proton self-energy itself rather than at the proton–neutron difference, for which the deep inelastic contributions are further suppressed.

The *Born term* is determined by the electromagnetic form factors of the nucleon which we approximate by the dipole formula with $F_1^n = 0$:

$$G_E^p(t) = G(t)\,; \qquad G_M^p(t) = \mu^p G(t)$$

$$G_E^n(t) = \frac{t}{4M^2}\mu^n G(t)\,; \qquad G_M^n(t) = \mu^n G(t)$$

$$G(t) = (1 - t/m^2)^{-2}\,. \tag{12.2}$$

Using the values $m^2 = 0.71\,\mathrm{GeV}^2$, $\mu^p = 2.79$, $\mu^n = -1.91$ one finds

$$M_p^{\mathrm{Born}} = 0.63\,\mathrm{MeV}\,, \qquad M_n^{\mathrm{Born}} = -0.13\,\mathrm{MeV}\,. \tag{12.3}$$

Note that for a heavy particle the magnetic contribution is small (of order M^{-2}) and the Born term reduces to the electrostatic energy of a charge distribution of which $G_E(-\boldsymbol{q}^2)$ is the Fourier transform:

$$M^\gamma = \frac{\alpha Q^2}{4\pi^2}\int \frac{d^3q}{\boldsymbol{q}^2}[G_E(-\boldsymbol{q}^2)]^2\,. \tag{12.4}$$

For a baryon of charge $Q = 1$ this amounts to 0.96 MeV.

We have performed a new determination of the Born terms for the other members of the baryon octet on the basis of the available experimental information about the magnetic moments. The magnetic moments of Λ, Σ^+, Σ^-, Ξ^0 and Ξ^- are measured directly: $\mu^\Lambda = -0.730 \pm 0.006$, $\mu^{\Sigma^+} = 2.95 \pm 0.16$, $\mu^{\Sigma^-} = -1.80 \pm 0.32$, $\mu^{\Xi^0} = -1.68 \pm 0.08$, $\mu^{\Xi^-} = -2.61 \pm 1.06$ (in units of $e/2M$ where M is the mass of the particle). The magnetic moment of the Σ^0 then follows from isospin symmetry: $\mu^{\Sigma^0} = \frac{1}{2}(\mu^{\Sigma^+} + \mu^{\Sigma^-}) = 0.58 \pm 0.18$. Approximating the form factors again by the dipole formula and using the central experimental values for the magnetic moments we obtain the results given in column 1 of table 2. Note that the Σ^0 self-energy contains two Born terms: one is due to the transition $\Sigma^0 \to \Sigma^0 + \gamma \to \Sigma^0$, the other to the transition $\Sigma^0 \to \Lambda + \gamma \to \Sigma^0$. The second contribution develops an imaginary part which manifests itself in the decay $\Sigma^0 \to \Lambda + \gamma$. The known rate of this decay implies the value $\mu^{\Sigma\Lambda} = 2.31 \pm 0.26$ for the magnetic transition moment (in units of $e/2M_\Sigma$). The real part of the second contribution amounts to $M_{\Sigma\Lambda}^{\mathrm{Born}} = -0.20\,\mathrm{MeV}$.

The uncertainties in the magnetic moments do not affect the self-energies substantially: for comparison column 2 of table 2 contains the results that one obtains if one uses the old SU(3) predictions [101] for the magnetic moments ($\mu^\Lambda = \frac{1}{2}\mu^n$, $\mu^{\Sigma^+} = \mu^p$, $\mu^{\Sigma^0} = -\frac{1}{2}\mu^n$, $\mu^{\Sigma^-} = -(\mu^p + \mu^n) = \mu^{\Xi^-}$, $\mu^{\Xi^0} = \mu^n$; $\mu^{\Sigma\Lambda} = -\frac{1}{2}\sqrt{3}\mu^n$) rather than the experimental values. In column 3 we quote the values obtained by Coleman and Schnitzer [102] who used two different form factor models to estimate the uncertainties in their analysis. Within the errors their results are consistent with the information about the electromagnetic form factors available today.

As mentioned above the data on inelastic electron scattering show that the Born term dominates the electromagnetic self-energy of the nucleon: the phase space factors in the Cottingham formula suppress the inelastic contributions. We estimated [97] the size of the contributions from inelastic intermediate states to the proton–neutron mass difference to be less than 0.3 MeV. We think that this estimate also applies to the other members of the baryon octet: we assume that the Born terms given in column 1 of table 2 represent reliable values for the total electromagnetic self-energy to within ± 0.3 MeV.

Table 2
Estimates for the photon cloud energies of the baryons. Column 1 contains the Born terms calculated with the measured magnetic moments and column 2 gives the corresponding values calculated with the SU(3) predictions for the magnetic moments. Column 3 quotes the values given by Coleman and Schnitzer and in column 4 we give estimates based on the simple quark model ansatz described in the text.

	Born term exp 1	Born term SU(3) 2	Coleman and Schnitzer 3	Quark model 4
p	0.63	0.63		
n	−0.13	−0.13		
Σ^+	0.70	0.73		
Σ^0	−0.21	−0.13		
Σ^-	0.87	0.94		
Ξ^0	−0.07	−0.09		
Ξ^-	0.79	0.95		
p − n	0.76	0.76	1.1/1.4	1.0 ± 0.4
$\Sigma^+ - \Sigma^-$	−0.17	−0.21	−0.7/−1.8	-0.3 ± 0.7
$\Xi^0 - \Xi^-$	−0.86	−1.04	−1.25/−1.6	-1.3 ± 0.6
$\Sigma^+ + \Sigma^- - 2\Sigma^0$	1.98	1.92	2.1/2.0	2.4 ± 0.5

A different estimate is based on the *quark model* picture of the photon cloud [103–105]. One separates the self-energy into a contribution coming from photon exchange between two of the three quarks and a contribution that accounts for diagrams of the quark self-energy type:

$$M^\gamma = \alpha\langle 1/r\rangle(Q_2Q_3 + Q_3Q_1 + Q_1Q_2) + C(Q_1^2 + Q_2^2 + Q_3^2) \tag{12.5}$$

where Q_1, Q_2, Q_3 are the quark charges. As discussed in the preceding section the self-energy of the bound state must be renormalized. The counter term does not have the structure of the above formula (in the chiral limit the divergent part of the mass shift is flavour independent). We have however also seen that the deep inelastic contributions which produce this divergence are tiny. Within the accuracy of quark model estimates it is not a crime to neglect terms that are not of the structure (12.5).

The first term in (12.5) may be estimated on the basis of simple models for the wave function, which give, typically

$$\begin{aligned} &\langle 1/r\rangle = 330/390\ \text{MeV} &&\text{(Itoh et al. [124])}\\ &\langle 1/r\rangle = 255\ \text{MeV} &&\text{(Isgur [125])} \end{aligned} \tag{12.6}$$

for the average inverse distance of two quarks in a baryon. In perturbation theory the second term is proportional to αm_{quark} and hence vanishes in the chiral limit. Perturbation theory does however not properly account for the chirality changing piece of the quark propagator; the constant C contains nonperturbative contributions of order α times the scale of QCD. We do not attempt to determine a reliable value for the constant C, but will be content with a rough estimate. We note that the entire self-energy of the proton and of the Σ^+ are given by C. The Born term estimates for the proton and Σ^+ self-energies are $C \simeq 0.63$ MeV and $C \simeq 0.70$ MeV respectively. In an electrostatic picture the self-energy is $C \simeq \frac{1}{2}\alpha|1/r|$, where $|1/r|$ is the mean inverse distance weighted with the charge distribution. For a charge distribution corresponding to the dipole formula we have $C \simeq 0.96$ MeV and hence $|1/r| \simeq 260$ MeV. We allow the constant C to have any value in the range $0 < C < 1.5$ MeV. We consider this estimate very generous as the upper limit would require the Born term to represent less than half of the actual value of the renormalized self-energy of the proton (the direct comparison with the Cottingham formula indicates that contributions from intermediate states other than the proton are substantially smaller than the Born term).

The estimates for $\langle 1/r \rangle$ and C given above lead to the values given in column 4 of table 2. Within the errors that are of the order of 0.5 MeV the simple quark model estimates described above agree with the Born terms.

The literature contains several quark model calculations of the photon cloud energy that are more sophisticated and do not rely on the ansatz (12.5) but instead use detailed quark wave functions (refs. [108–119]; for more recent work see refs. [120–125] and the references quoted in these papers). In particular, the magnetic interaction among the quarks has been estimated. Note that some of these estimates lead to large electromagnetic contributions to $\Sigma^+ - \Sigma^-$ and to $\Xi^0 - \Xi^-$. [The rule according to which the quark magnetic moments are given by $e/2m_{\text{constituent}}$ with $m_{\text{const}}(\text{u}) \simeq 350$ MeV, $m_{\text{const}}(\text{s}) \simeq 550$ MeV imply large SU(3) asymmetries in the magnetic self-energy.] In the following we shall not make use of the quark model estimates, but use the experimental information from electron scattering on protons and neutrons, extrapolated to the other members of the baryon octet in what we consider a conservative manner.

Our final estimates for the photon cloud energies of p–n, $\Sigma^+ - \Sigma^-$ and $\Xi^0 - \Xi^-$ are given in column 2 of table 3: we use the Born terms with the measured magnetic moments and assess the uncertainties due to inelastic contributions at ± 0.3 MeV. Subtracting this from the observed mass differences we obtain the corresponding values in pure QCD (column 3 of table 3). In the case of the combination $\Delta M_\Sigma = M_{\Sigma^+} + M_{\Sigma^-} - 2M_{\Sigma^0}$ the analogous estimate of the electromagnetic self-energy gives 1.98 ± 0.30 MeV, whereas the simple quark model described above leads to 2.4 ± 0.5 MeV. The fact that these values agree with the observed mass difference $(\Delta M_\Sigma)^{\text{Exp}} = 1.78 \pm 0.14$ MeV is a welcome

Table 3
Isospin breaking mass differences. For p − n, $\Sigma^+ - \Sigma^-$ and $\Xi^0 - \Xi^-$ the electromagnetic self-energies are estimated with the Cottingham formula: the central value is the Born term, the error bar is an estimate of the size of inelastic contributions. For $\text{K}^+ - \text{K}^0$ the electromagnetic self-energy is calculated from Dashen's theorem, for $\text{D}^+ - \text{D}^0$ we have used a simple quark model ansatz. Column 3 is the difference between columns 1 and 2, except for $\Sigma^+ + \Sigma^- - 2\Sigma^0$ and $\pi^+ - \pi^0$ where column 3 is the input: in pure QCD these mass differences are very small, of order $(m_u - m_d)^2$. (For direct estimates of the corresponding electromagnetic self-energies see text.)

	Experiment 1	Electromagnetic 2	QCD 3
p − n	−1.29	0.76 ± 0.30	−2.05 ± 0.30
$\Sigma^+ - \Sigma^-$	−7.98 ± 0.08	−0.17 ± 0.30	−7.81 ± 0.31
$\Xi^0 - \Xi^-$	−6.4 ± 0.6	−0.86 ± 0.30	−5.5 ± 0.7
$\Sigma^+ + \Sigma^- - 2\Sigma^0$	1.78 ± 0.14	1.78 ± 0.14	±0.02
$\pi^+ - \pi^0$	4.60	4.6 ± 0.1	±0.1
$\text{K}^+ - \text{K}^0$	−4.01 ± 0.13	1.27 ± 0.30	−5.28 ± 0.33
$\text{D}^+ - \text{D}^0$	5.0 ± 0.8	1.7 ± 0.5	3.3 ± 0.9

consistency check: in pure QCD this combination has a very small value of order $(m_u - m_d)^2$. The order of magnitude of ΔM_Σ in QCD is given by $\Sigma^0 - \Lambda$ mixing which produces the mass splitting $\Delta M_\Sigma \simeq -2\theta^2(M_\Sigma - M_\Lambda) = -0.015\,\mathrm{MeV}$. The contributions of order $(m_{\mathrm{quark}})^2$ and the nonanalytic terms are also of this size. (The situation is very similar to the case of $\pi^+ - \pi^0$, see below.)

For the *Goldstone bosons* a nonrelativistic quark model analysis in terms of $\langle 1/r\rangle$ does not make sense; the mass difference between the π^+ and π^0 would require $\langle 1/r\rangle \sim 1.2\,\mathrm{GeV}$ which is outrageous. We see no reason why a model that fails for the pion should be valid for the kaon.

In the chiral limit current algebra implies the *low energy theorem* [126]:

$$(M^2_{\pi^+})^\gamma = \frac{3\alpha}{4\pi f^2_\pi}\int_0^\infty \mathrm{d}s\, s \ln\frac{\mu^2}{s}\,[\rho^{\mathrm{V}}(s) - \rho^{\mathrm{A}}(s)]\,; \qquad (M^2_{\pi^0})^\gamma = 0 \tag{12.7}$$

where ρ^{V} and ρ^{A} are the vector and axial vector spectral functions normalized as

$$\rho^{\mathrm{V}}(s) = f^2_\rho\, \delta(M^2_\rho - s) + \cdots \tag{12.8}$$

with $f_\rho = (204 \pm 11)$ MeV and $f_\pi = 132\,\mathrm{MeV}$. On account of the second Weinberg sum rule (which is valid in massless QCD) the expression (12.7) is independent of the renormalization point μ. Saturating the integral (12.7) with ρ and A_1 and eliminating the A_1 parameters f_{A_1}, M_{A_1} by using the analogous saturation of the Weinberg sum rules one obtains

$$(M^2_{\pi^+})^\gamma = \frac{3\alpha}{4\pi} M^2_\rho \left(\frac{f_\rho}{f_\pi}\right)^2 \ln\frac{f^2_\rho}{f^2_\rho - f^2_\pi} \tag{12.9}$$

which gives $M^\gamma_{\pi^+} = (4.9 \pm 0.2)$ MeV for the value of f_ρ quoted above. The result is very close to the observed mass difference $M_{\pi^+} - M_{\pi^0} = 4.60\,\mathrm{MeV}$. [The error bar does not account for the error committed in approximating the spectral functions ρ^{V} and ρ^{A} with single narrow resonances. A more accurate analysis of the ρ-contribution that includes finite width effects leads to the value $M^\gamma_{\pi^+} = 6.1 \pm 0.8\,\mathrm{MeV}$ [127].]

If the quark masses are allowed to be different from zero, perturbation theory shows that the sum rule (12.7) diverges with a coefficient proportional to αm^2. In fact the operator product expansion for the difference $V_\mu V_\nu - A_\mu A_\nu$ contains a term proportional to $m\bar{q}q$ – the integral occurring in (12.7) therefore involves a divergence proportional to $\alpha m\langle 0|\bar{q}q|0\rangle$. The formula (12.7) itself is only valid up to terms of order αm. The corrections appearing in a renormalized version of this formula are however expected to be tiny, because the masses m_u, m_d are very small in comparison to the scale of QCD. [A renormalized formula would be of limited practical use, because there is no direct experimental information on $\rho^{\mathrm{A}}(s)$ and it is not easy to improve the rough approximation by a single narrow resonance used in (12.9).]

A direct evaluation of the electromagnetic self-energy of the pion on the basis of the *Cottingham formula*, using theoretical models both for the elastic and inelastic contributions was performed by Socolow [128]. Approximating the Born term with a ρ-dominated form factor $F(t) = (1 - t/M^2_\rho)^{-1}$ one finds

$$(M_{\pi^+} - M_{\pi^0})^{\mathrm{Born}} = 4.3\,\mathrm{MeV}\,. \tag{12.10}$$

According to Socolow the inelastic contributions increase this to 4.9 MeV to within an error estimated at ±1 MeV. [Note that the ρ-dominance model overestimates the data [129]; a realistic evaluation of the Born term with the measured form factor would lead to a smaller value.]

In *pure QCD* the mass difference $M_{\pi^+} - M_{\pi^0}$ is very small. As mentioned in section 8 the first order mass formula contains a contribution of order $(m_u - m_d)^2/(m_s - \hat{m})$:

$$M_{\pi^+} - M_{\pi^0} = M^{\gamma}_{\pi^+} - M^{\gamma}_{\pi^0} + \frac{1}{16}\left(\frac{m_d - m_u}{m_s - \hat{m}}\right)^2 \left(\frac{m_s}{\hat{m}} - 1\right) M_{\pi} . \tag{12.11}$$

The contribution from $m_u \neq m_d$ amounts to a tiny shift of 0.11 MeV. The higher order terms in the quark mass expansion produce corrections that are of the same size: (i) the leading nonanalytic correction given in (C.7) is of order −0.05 to −0.1 MeV (for $\mu = 1$ GeV the correction cancels the first order term) and (ii) the corrections of order $(m_{\text{quark}})^2$ are also expected to be of this size (the additivity rule leads to a correction of order $-(m_u - m_d)^2/2M_{\pi} \sim -0.05$ MeV). We conclude that the first order mass formula cannot be trusted – it only indicates the order of magnitude of $M_{\pi^+} - M_{\pi^0}$ in pure QCD. Up to this uncertainty of the order of 0.1 MeV the observed value for $M_{\pi^+} - M_{\pi^0}$ must therefore come from electromagnetism:

$$M^{\gamma}_{\pi^+} - M^{\gamma}_{\pi^0} = 4.6 \pm 0.1\ \text{MeV} . \tag{12.12}$$

This value is consistent both with the estimate obtained from the current algebra sum rule (where the main uncertainty stems from the fact that the behaviour of the axial vector spectral function is not known) and with the estimate based on the Cottingham formula.

For the electromagnetic self-energy of the kaon current algebra implies the low energy theorem [130]:

$$(M^2_{K^+})^{\gamma} = (M^2_{\pi^+})^{\gamma} , \qquad (M^2_{K^0})^{\gamma} = 0 . \tag{12.13}$$

Using the above value for $M^{\gamma}_{\pi^+} - M^{\gamma}_{\pi^0}$ this gives

$$M^{\gamma}_{K^+} - M^{\gamma}_{K^0} = 1.27 \pm 0.30\ \text{MeV} . \tag{12.14}$$

Dashen's theorem is strictly valid only in the chiral limit: in the real world the relations (12.13) are subject to corrections of order αm_s (and $\alpha m_{\text{quark}} \ln m_{\text{quark}}$) which are difficult to evaluate. The error bar given in (12.14) represents the typical order of magnitude of SU(3) × SU(3) breaking effects. [The leading infrared singularities responsible for the *nonanalytic terms* of order $\alpha m_{\text{quark}} \ln m_{\text{quark}}$ were worked out by Langacker and Pagels [131]. To obtain the proper expression for the corresponding nonanalytic terms one should use improved chiral perturbation theory. It does not make sense to replace all infrared logarithms by a universal number involving some mean meson mass. One first has to verify that the leading nonanalytic term indeed represents a good approximation to the quantity in question; if this is the case ICPT implies that one has to weigh the different contributions with the logarithm of the physical mass of the corresponding virtual meson. We did not perform such a calculation for the electromagnetic self-energies. On the basis of the analysis given in appendix C (in the context of the corresponding nonanalytic contributions to the baryon and meson mass formulae) we expect that these corrections will again turn out to be small, although the contributions obtained by using universal logarithms are large. Note that the estimates of the electromagnetic kaon self-energy based on the Cottingham formula $M^{\gamma}_{K^+} - M^{\gamma}_{K^0} = 2.9 \pm 1.0$ MeV [128] are also considerably larger than what is implied by Dashen's theorem. This analysis should be repeated with the experimental information available today.]

We see no reason why symmetry breaking should be unusually large in the electromagnetic self-energies. In the following we use the value (12.14) based on Dashen's theorem.

For the *D-mesons* the quark model described above gives

$$M^{\gamma}_{D^+} - M^{\gamma}_{D^0} = \tfrac{2}{3}\alpha\langle 1/r\rangle - \tfrac{1}{3}C . \tag{12.15}$$

Using the estimates given by Lane and Weinberg [105] on the basis of nonrelativistic potential models [$\langle 1/r\rangle = 350$ to 450 MeV] and the crude estimate for the constant C given above one obtains (see also [106a])

$$M^{\gamma}_{D^+} - M^{\gamma}_{D^0} = 1.7 \pm 0.5 \text{ MeV} \tag{12.16}$$

whereas the Born term with $F(t) = (1 - t/M^2_\rho)^{-1}$ is

$$M^{\text{Born}}_{D^+} - M^{\text{Born}}_{D^0} = 1.5 \text{ MeV} \tag{12.17}$$

close to the electrostatic limit $\alpha M_\rho/4 \simeq 1.4$ MeV for a heavy particle; the corresponding mean inverse distance weighted with the charge distribution is $|1/r| = \tfrac{1}{2}M_\rho \sim 390$ MeV. For an analysis of the inelastic contributions to the electromagnetic self-energy of the D mesons see Boal and Wright [132]. We do not discuss these estimates in detail because at the present level of our understanding the uncertainties involved do not allow us to extract an accurate value for the ratio $R = (m_s - \hat{m}):(m_d - m_u)$ of quark masses. For the same reason we do not discuss *isospin breaking mass differences of resonances*. In fact, we do not understand the values $K^*_0 - K^*_+ = 6.7 \pm 1.2$ MeV and $D^*_+ - D^*_0 = 2.6 \pm 1.8$ MeV given in the particle tables: although the photon cloud contributions are expected to be smaller than for the corresponding pseudoscalar states [107, 106b] the corrected values are not in our opinion in good agreement with the mass formula (B.8) and with the relation analogous to (14.13). To make use of the experimental information on small differences of this sort it is crucial that one understands the electromagnetic corrections applied in the determination of the pole positions (see Zidell, Arndt and Roper [133] for a discussion of these problems in the case of the Δ resonance).

13. Quark mass ratios from meson and baryon masses

We are now ready to determine the two ratios $m_u : m_d : m_s$ from the mass formulae for the pseudoscalar mesons and for the $\tfrac{1}{2}^+$ baryon octet. We first determine the value of the *ratio* $m_s : \hat{m}$ from the masses of π, K and η. A comparison of the first order mass formulae for π, K and for π, η (corrected for electromagnetic effects according to Dashen's theorem) gives $m_s : \hat{m} = 25.9$ and 24.3 respectively. There are two kinds of corrections to be discussed in order to assess the reliability of these values: nonanalytic corrections (terms of order $(m_{\text{quark}})^2 \ln m_{\text{quark}}$) and contributions of order $(m_{\text{quark}})^2$.

The leading *nonanalytic corrections* account for the self-energies of the meson clouds [we define the energy of the meson cloud around the pion, ΔM_π, as the difference between the physical mass M_π and the purified mass $\bar{M}_\pi$ given in (C.6): $M_\pi = \bar{M}_\pi + \Delta M_\pi$]. If the cutoff μ is given a small value ($\mu = 0.5\,\mathrm{GeV}$) the energies of the clouds around π, K and η are small ($\Delta M_\pi = -1.6\,\mathrm{MeV}$, $\Delta M_\mathrm{K} = 3.4\,\mathrm{MeV}$, $\Delta M_\eta = -8\,\mathrm{MeV}$). If μ is taken large ($\mu = 1\,\mathrm{GeV}$), we obtain $\Delta M_\pi = +1.2\,\mathrm{MeV}$, $\Delta M_\mathrm{K} = -21\,\mathrm{MeV}$, $\Delta M_\eta = -22\,\mathrm{MeV}$. For $\mu = 1\,\mathrm{GeV}$ the correction in the ratio $m_\mathrm{s}:\hat{m}$ amounts to about +3 units ($m_\mathrm{s}:\hat{m} = 28.8$ and 26.8 from π, K and π, η respectively); for smaller values of the cutoff the shift is correspondingly smaller.

Consider now the *corrections of order* $(m_\mathrm{quark})^2$. In the case of pion and kaon the additivity rule according to which we should add the term $(m_1 + m_2)^2$ to the first order mass formula for M^2 may give a reasonable estimate for the order of magnitude of these corrections. For a strange quark mass of order 130 to 170 MeV the correction tends to lower the value of the ratio $m_\mathrm{s}:\hat{m}$ by 2 to 3 units. The terms of order $(m_\mathrm{quark})^2$ thus act in the direction opposite to the leading nonanalytic contribution – the two corrections roughly cancel. In the case of the eta it is not clear how to use the additivity rule. One may use the picture that 2/3 of the time the eta consists of two strange quarks, 1/3 of the time of two u or d quarks and weigh the correction $(m_1 + m_2)^2$ accordingly. This prescription lowers the value of $m_\mathrm{s}:\hat{m}$ by about 3 to 6 units and thus overcompensates the leading nonanalytic correction. The corrections to the first order formula preserve the octet character of the mass operator to a remarkable degree of accuracy, the simple modification of the additivity rule given above does not. We do not try to improve on this prescription; instead we observe that the leading nonanalytic contribution and the terms of order $(m_\mathrm{quark})^2$ act in opposite directions with an amplitude of the same order. We conclude that it makes sense to determine the ratio $m_\mathrm{s}:\hat{m}$ from the uncorrected physical masses ($M_{\pi^+}^\mathrm{QCD} \simeq M_{\pi^0}^\mathrm{QCD} \simeq 135.0\,\mathrm{MeV}$, $M_{\mathrm{K}^+}^\mathrm{QCD} \simeq 492.4\,\mathrm{MeV}$, $M_{\mathrm{K}^0}^\mathrm{QCD} \simeq 497.7\,\mathrm{MeV}$, $M_\eta^\mathrm{QCD} \simeq 548.8\,\mathrm{MeV}$) and to use the size of the leading nonanalytic contribution as an estimate of the error bar to be attached to this ratio. In this manner we obtain

$$m_\mathrm{s}/\hat{m} = 25.0 \pm 2.5 \tag{13.1}$$

for the average of the two determinations.

We now consider the *ratio of the SU*(2) *breaking piece to the SU*(3) *breaking piece* of the quark mass term, which we numerically analyze in terms of the ratio

$$R = \frac{m_\mathrm{s} - \hat{m}}{m_\mathrm{d} - m_\mathrm{u}}. \tag{13.2}$$

In the language of the symmetry breaking parameters ε_3 and ε_8 that characterize the Gell-Mann–Oakes–Renner model the ratio R stands for

$$R = \frac{\sqrt{3}}{2}\frac{\varepsilon_8}{\varepsilon_3}. \tag{13.3}$$

In terms of this ratio the first order mass formula for $\mathrm{K}^+ - \mathrm{K}^0$ may be written in the form

$$R = \frac{M_\mathrm{K}^2 - M_\pi^2}{M_{\mathrm{K}^0}^2 - M_{\mathrm{K}^+}^2}. \tag{13.4}$$

Using the value for $(M_{K^0} - M_{K^+})^{QCD}$ derived from Dashen's theorem this gives

$$R = 43.4 \pm 2.7\,. \tag{13.5}$$

We again analyze the corresponding leading nonanalytic correction and the contributions of order $(m_{\text{quark}})^2$. The nonanalytic correction shifts R downwards by 4 units if $\mu = 1$ GeV (by less than that if μ is smaller). The phenomenon observed in the determination of the ratio $m_s : \hat{m}$ occurs also here: the correction of order $(m_{\text{quark}})^2$, estimated on the basis of the additivity rule, counterbalances the leading nonanalytic correction such that we may again use the uncorrected value (13.5) and use the size of the leading nonanalytic correction to estimate the error bar to be attached to R. The result is quoted in table 4.

Next we consider the baryon mass differences for which the first order formulae may be written as

$$\begin{aligned} R &= \frac{\frac{1}{2}(M_\Xi - M_N) - \frac{3}{4}(M_\Sigma - M_\Lambda)}{M_n - M_p} \\ R &= \frac{M_\Xi - M_N}{M_{\Sigma^-} - M_{\Sigma^+}} \\ R &= \frac{\frac{1}{2}(M_\Xi - M_N) + \frac{3}{4}(M_\Sigma - M_\Lambda)}{M_{\Xi^-} - M_{\Xi^0}} \end{aligned} \tag{13.6}$$

providing us with three independent measurements of the ratio R. The three relevant *isosopin breaking baryon mass differences*, corrected for electromagnetic self-energies, are given in table 3. With these values the first order mass formulae give

Table 4
Measurements of the ratio $R = (m_s - \hat{m})/(m_d - m_u)$. Column 1 contains the uncertainties in the experimental data and in the calculation of the electromagnetic corrections. Column 2 is our estimate of the uncertainties due to higher order terms in the quark mass expansion. (In the case of $\rho - \omega$ mixing these terms have not been analyzed. We have increased the corresponding error bar from ± 3 to ± 5 in order not to bias the average which is calculated by treating the five values given in column 3 as independent measurements).

	$\Delta R^{\text{exp},\gamma}$ 1	$\Delta R^{\text{h.o.}}$ 2	Result 3
$K^+ - K^0$	± 2.7	± 3	43 ± 4
$p - n$	± 7.5	± 6	51 ± 10
$\Sigma^+ - \Sigma^-$	± 1.7	± 3	43 ± 4
$\Xi^0 - \Xi^-$	± 5.4	± 3	42 ± 6
$\rho - \omega$ mixing	± 3		44 ± 5
Average			43.5 ± 2.2

$$
\begin{array}{ll}
\text{linear} & \text{quadratic} \\
R_{\rm N} = 64 \pm 9 & R_{\rm N} = 76 \pm 11 \\
R_{\Sigma} = 49 \pm 2 & R_{\Sigma} = 46 \pm 2 \\
R_{\Xi} = 44 \pm 6 & R_{\Xi} = 38 \pm 5
\end{array}
\tag{13.7}
$$

where we have given the results both for the linear mass formula (13.6) and for the corresponding quadratic version; within the accuracy of first order mass formulae there is no reason to prefer one to the other. The values differ by up to a factor 2. It is evident that the first order mass formulae do not provide us with a reliable determination of R and that we have to investigate the corrections to see whether the observed mass differences are at all consistent with the framework advocated here.

The higher order terms in the quark mass expansion were discussed in section 10 on the basis of improved chiral perturbation theory. According to this discussion the leading asymmetries due to the *nonanalytic terms* in the quark mass expansion are eliminated if one replaces the physical masses M_n by the corresponding purified masses $\bar{M}_n$ defined in (C.5). In the quark mass expansion of $\bar{M}_n$ there are of course still higher order terms that are neglected in the first order mass formulae, but their effect should be smaller, because the terms proportional to $(m_{\rm quark})^{3/2}$ that are present in the physical masses have been eliminated. Inserting the purified masses into the linear mass formula (13.6) we obtain

$$
\begin{array}{ll}
R_{\rm N} = 54 \pm 8 & R_{\rm N} = 48 \pm 7 \\
R_{\Sigma} = 45 \pm 2 & R_{\Sigma} = 41 \pm 2 \\
R_{\Xi} = 44 \pm 6 & R_{\Xi} = 41 \pm 5
\end{array}
\tag{13.8}
$$

where the first value is based on SU(3) symmetric axial vector couplings, the second on SU(3) symmetric meson-baryon coupling constants (see appendix C, table 6).

The scattering of these values is indeed considerably smaller than for the uncorrected mass values. This shows that a substantial fraction of the discrepancy in the values of R obtained from the raw physical masses is due to the effect described in section 10 in the framework of the static effective Lagrangian: the energies of the pion, kaon and eta clouds are rather different, not only because they have completely different radii, but also because the mass differences of the corresponding virtual baryons produce sizeable differences in the energies of the clouds.

The difference between the quadratic and the linear mass formulae is a rough estimate for the size of the uncertainties due to the *remaining higher order terms.* The first order formulae for the square of the masses generally lead to larger deviations than the first order formulae linear in the masses. We first look at the deviations from the Gell-Mann–Okubo formula

$$
\Delta = \frac{2{\rm N} + 2\Xi - 3\Lambda - \Sigma}{\Xi - {\rm N}}\,. \tag{13.9}
$$

Using the raw physical masses in pure QCD the linear formula gives $\Delta = -0.070$, the quadratic one implies $\Delta = +0.093$. The corresponding numbers for the purified masses [SU(3)-symmetric axial vector couplings] are $\Delta = -0.084$ and $\Delta = +0.061$ for the linear and quadratic version of (13.9) respectively. The deviations are perfectly consistent with the size of the corrections to be expected from higher order terms in the quark mass expansion. Estimating the terms of order $(m_{\rm quark})^2$ with the additivity rule [i.e. writing $\bar{M}_n^2 = \hat{M}_n^2 + (m_1 + m_2 + m_3)^2$ and using the values of $\hat{M}_n$ rather than $\bar{M}_n$] the quadratic formula gives $\Delta = -0.005$ for $m_{\rm s} = 130\,\text{MeV}$ and $\Delta = -0.057$ for $m_{\rm s} = 170\,\text{MeV}$: the terms of order $(m_{\rm quark})^2$

obtained from the additivity rule essentially transform the quadratic mass formulae into the corresponding linear ones.

These observations are confirmed by an analysis of the ratio R. If one uses the quadratic mass formulae with the purified masses (symmetric axial vector couplings) one finds $(R_{\mathrm{N}}, R_{\Sigma}, R_{\Xi}) = (63, 43, 38)$ instead of the values $(54, 45, 44)$ given above on the basis of linear mass formulae. Correcting the quadratic mass formulae for terms of order $(m_{\mathrm{quark}})^2$ with the additivity rule one finds $(58, 43, 40)$ if $m_s = 130\,\mathrm{MeV}$ and $(54, 43, 41)$ if $m_s = 170\,\mathrm{MeV}$. The terms of order $(m_{\mathrm{quark}})^2$ obtained from the additivity rule again essentially convert the quadratic mass formulae into the linear ones.

On the basis of the above discussion we conclude that the values of the ratio R extracted from Σ and Ξ are affected very little by higher order terms in the quark mass expansion, whereas the value obtained from the neutron–proton mass difference is strongly affected, both by the nonanalytic contributions and by the terms of order $(m_{\mathrm{quark}})^2$: the value $R_{\mathrm{N}} = 76 \pm 11$ that follows from the quadratic mass formula if one uses the physical masses (corrected for the electromagnetic self-energy, of course) is shifted to $R_{\mathrm{N}} = 63 \pm 9$ if one subtracts the nonanalytic contributions [SU(3) symmetric axial vector coupling] and is further reduced to $R_{\mathrm{N}} = 54 \pm 8$ if the terms of order $(m_{\mathrm{quark}})^2$ are taken from the additivity rule with $m_s = 170\,\mathrm{MeV}$. In table 4 we quote the values obtained from the linear mass formulae, using the purified masses [mean values in (13.8)]. Columns 1 and 2 contain the error bars due to the uncertainties in the values of the masses in pure QCD and in the higher order terms of the quark mass expansion respectively. The three values of R obtained from the spectrum of the baryon octet are not only consistent with one another, but are also in perfect agreement with the independent determination of R from the meson spectrum. (For other recent estimates see refs. [135–144].)

14. Quark mass ratios from other sources

An independent direct measurement of the ratio $R = (m_s - \hat{m}) : (m_d - m_u)$ is based on *$\rho - \omega$ mixing* [145–152]. The effect of an off-diagonal term in the $\rho\omega$ mass matrix can be seen in a very clean manner in the reaction $e^+e^- \to \pi^+\pi^-$ which exhibits a shoulder near the mass of the ω [147, 153, 154]. From these and other experiments the particle data group gives

$$\Gamma_{\omega\to\pi^+\pi^-}/\Gamma_{\omega}^{\mathrm{TOT}} = 0.014 \pm 0.002\,.$$

Using equation (B.12) of appendix B this becomes

$$M_{\rho\omega}^{\mathrm{Exp}} = -2.22 \pm 0.16\,\mathrm{MeV} \tag{14.1}$$

for the off-diagonal element of the mass matrix (the phase of $M_{\rho\omega}$ is observed to be close to 180°). The electromagnetic interaction of course also produces ρ–ω-transitions. The process $\rho \to \gamma \to \omega$ contributes

$$M_{\rho\omega}^{\gamma} = \frac{e^2}{2M_\rho} F_\rho F_\omega = 0.43 \pm 0.04\,\mathrm{MeV} \tag{14.2}$$

where F_ρ and F_ω are the coupling constants of these mesons to the electromagnetic current

$$\langle 0|j_\mu|\rho^0\rangle = \varepsilon_\mu F_\rho M_\rho\,; \qquad F_\rho = \frac{1}{\sqrt{2}} f_\rho = 144 \pm 8\,\mathrm{MeV}$$

$$\langle 0|j_\mu|\omega\rangle = \varepsilon_\mu F_\omega M_\omega\,; \qquad F_\omega = 51 \pm 3\,\mathrm{MeV}\,. \tag{14.3}$$

Other electromagnetic contributions are estimated to be negligibly small [147]. The bulk of the experimental value must therefore come from the quark mass difference $m_u - m_d$. Inserting the above values for $M_{\rho\omega}^{\mathrm{Exp}} - M_{\rho\omega}^{\gamma}$ into the mass formula (B.14) this gives

$$R = 44 \pm 3 \tag{14.4}$$

where the error bar only includes the errors given above in the values of $M_{\rho\omega}^{\mathrm{Exp}}$, $M_{\rho\omega}^{\gamma}$. We do not attempt to estimate the higher order terms in the quark mass expansion to assess the accuracy of the first order mass formula (B.14). [Compare also Langacker [152] who obtains $R = 48 \pm 10$ from a somewhat different weighting of the same data.]

Several authors (Segré and Weyers [155]; Genz [156]; Langacker [157]; Ioffe [158]; Voloshin [159]) have shown that *the decays* $\psi' \to \psi + \pi^0$ *and* $\psi' \to \psi + \eta$ provide information about symmetry breaking: both decays are forbidden if the quark masses vanish. The amplitude of the first reaction is proportional to $m_u - m_d$ (apart from an electromagnetic contribution of order e^2), the amplitude for the second reaction is proportional to $m_s - \hat{m}$. Ioffe and Shifman [160] have shown that the ratio of rates can unambiguously be calculated on the basis of PCAC with the result

$$r = \frac{\Gamma_{\psi' \to \psi + \pi^0}}{\Gamma_{\psi' \to \psi + \eta}} = \frac{27}{16} \frac{1}{R^2} \left(\frac{p_\pi}{p_\eta}\right)^3 . \tag{14.5}$$

In fact, it is not even necessary to invoke PCAC. To show this we consider the limit $m_{\mathrm{quark}} \to 0$ at fixed ratios $m_u : m_d : m_s$. First order perturbation theory in the quark masses implies that the amplitude for the decay $\psi' \to \psi + \pi^0$ is given by

$$T_{\psi' \to \psi + \pi^0} = -\langle \psi \pi^0 \ \mathrm{out}|\bar{q} m q|\psi'\rangle \tag{14.6}$$

where the matrix element is to be evaluated with the unperturbed states. Since in the unperturbed system the octet of Goldstone bosons is degenerate, there is in general no physical difference between π^0 and η in that limit. In the present case perturbation theory shows that the first order mass formula (8.4) unambiguously selects the limiting states. The matrix elements of the operator $\bar{q} m q$ may then be worked out by using SU(3) for the octet basis ${}_0\langle \pi^0|$, ${}_0\langle \eta|$. The result is

$$\frac{T_{\psi' \to \psi + \pi^0}}{T_{\psi' \to \psi + \eta}} = \frac{\sqrt{3} \cos\theta + 2 \sin\theta R}{2 \cos\theta R - \sqrt{3} \sin\theta}\,; \qquad \mathrm{tg}\, 2\theta = \frac{\sqrt{3}}{2R}\,. \tag{14.7}$$

This determines the ratio of rates for any value of R. Since R is large, the angle θ is small and we obtain the result of Ioffe and Shifman given above.

The average of the recent data [161, 162] for the branching ratios is

$$\begin{aligned} &B(\psi' \to \psi + \pi^0) = (0.10 \pm 0.03)\% \\ &B(\psi' \to \psi + \eta) = (2.3 \pm 0.4)\% \end{aligned} \tag{14.8}$$

which gives

$$r_{\text{exp}} = 0.043 \pm 0.015\,. \tag{14.9}$$

The relation (14.5) then implies

$$R = 28^{+7}_{-4}$$

less than the value $R = 43.5$ given in table 4 by about two standard deviations. The theoretical prediction for r with $R = 43.5$ is

$$r_{\text{th}} = 0.018\,. \tag{14.10}$$

[This prediction is based on SU(3) and hence is subject to corrections of order m_s; also the contribution of the electromagnetic interaction to the decay $\psi' \to \psi + \pi^0$ is neglected. We see no reason for these corrections to be large, but to our knowledge their size has not been estimated in the literature.]

A further source of information on symmetry breaking is *η-decay*. In particular, the decays $\eta \to \pi^+\pi^-\pi^0$ and $\eta \to \pi^0\pi^0\pi^0$ have been identified as symmetry breaking probes long ago (refs. [163, 164]; for a review of the early literature see [165]). Using PCAC for pions only (m_u, $m_d \to 0$, m_s fixed) one finds good agreement with the data on the slope parameter of the Dalitz plot. Also, the direct measurements of the ratio $B = \Gamma_{\eta\to\pi^0\pi^0\pi^0}/\Gamma_{\eta\to\pi^+\pi^-\pi^0}$ [167, 168] which give $B = 1.46 \pm 0.14$ are in good agreement with the prediction $B = \frac{3}{2}$ (up to small corrections for the difference in Dalitz plot slopes and phase space) that follows from the isospin properties of the symmetry breaking quark mass term. (Note however that the overall fit of all data on η-decays provided by the particle data group implies $B = 1.268 \pm 0.060$. This disagrees with the isospin prediction by more than 3 standard deviations.)

The predictions involving PCAC for the eta are not in good agreement with the data. In our opinion, none of these predictions is more reliable than the prediction for the decay rate ($f_\eta \simeq f_\pi$):

$$\Gamma_{\eta\to\gamma+\gamma} = \frac{\alpha^2}{96\pi^3}\frac{m_\eta^3}{f_\eta^2} = 170\ \text{eV} \tag{14.11}$$

which disagrees with the data [166]:

$$\Gamma^{\text{Exp}}_{\eta\to\gamma+\gamma} = (324 \pm 46)\ \text{eV} \tag{14.12}$$

by about a factor two. The absolute rate for this process (which determines the other rates via their branching ratio) is difficult to measure. The value decreased by more than a factor two a couple of years ago [166]. On the theoretical side the theorem (14.11) is only valid in the limit $m_{\text{quark}} \to 0$; one has to allow for corrections of order $m_{\text{quark}} \ln m_{\text{quark}}$ as well as for corrections of order m_s (which may well be dominated by $\eta\eta'$-mixing, see e.g. refs. [170, 171]). The higher order corrections appear to be substantial; to our knowledge the uncertainties involved in these corrections have however not been investigated. In view of both the theoretical and the experimental uncertainties it appears to be difficult to determine symmetry breaking parameters from η-decays in a reliable manner and we do not discuss the subject further here. The reader is referred to the literature which may be traced from the papers of Neveu and Scherk [169], Dittner, Bondi and Eliezer [165], Langacker and Pagels [172], Weinberg [173], Crewther [174], Raby [175], Dominguez and Zepeda [176], Roiesnel and Truong [177].

The measured *mass difference* $M_{D^+} - M_{D^0} = 5.0 \pm 0.8$ MeV also provides information about isospin breaking. In contrast to the K^+–K^0 mass difference the electromagnetic self-energy however has the same sign as the contribution from $m_d - m_u$. The uncertainty in the value of the mass difference in QCD (see table 3) is therefore rather large: $(M_{D^+} - M_{D^0})^{QCD} = 3.3 \pm 0.9$ MeV. Furthermore, the corresponding SU(3)-breaking mass difference $M_F - M_D = 164 \pm 60$ MeV based on $M_F = 2030 \pm 60$ MeV [178] needs experimental confirmation. Using the mean values the first order mass formula

$$M_{D^+} - M_{D^0} = \frac{m_d - m_u}{m_s - \hat{m}} (M_F - M_D) \tag{14.13}$$

gives $R = 50$. In view of the large uncertainties both in $(M_{D^+} - M_{D^0})^{QCD}$ and in $M_F - M_D$ the error is large, of order ± 20. One may replace the uncertain observed value of the mass difference $M_F - M_D$ by a theoretical estimate, based on SU(4) or on other dynamical assumptions to reduce the error bars in the result. It is however clear that the $D^+ - D^0$ mass difference does not provide reliable information about the ratio R at this stage.

15. Light quark masses from QCD sum rules

The first determination of the light quark masses within the framework of QCD was given by Vainshtein, Voloshin, Zakharov, Novikov, Okun and Shifman [179]. They considered the two-point-function for two axial divergences:

$$P(q^2) = \mathrm{i} \int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}qx} \langle 0 | T\, \partial A^+(x)\, \partial A(0) | 0 \rangle \tag{15.1}$$

$$A_\mu = \bar{u} \gamma_\mu \gamma_5 d\,.$$

Asymptotic freedom guarantees that the behaviour of $P(q^2)$ at large values of q^2 is given by perturbation theory. To zeroth order in the strong coupling constant the imaginary part of $P(q^2)$ is given by $(s \gg \hat{m}^2)$:

$$\mathrm{Im}\, P^{\mathrm{pert}}(s) = \frac{3}{2\pi} s \hat{m}^2\,; \qquad \hat{m} = \tfrac{1}{2}(m_u + m_d)\,. \tag{15.2}$$

Using renormalization group arguments Vainshtein et al. then show that the leading asymptotic behaviour of Im P in QCD is given by the same expression, provided $\hat{m}$ is identified with the running quark mass $\hat{m}(s)$ defined in (3.3). At sufficiently large values of s this expression should provide a good approximation to the actual imaginary part of P (see below for a more accurate high energy representation that includes first order corrections). At low energies the imaginary part does not follow a smooth curve such as (15.2) nor does it produce the threshold at $s = 4\hat{m}^2$ characteristic of perturbation theory. Nonperturbative effects instead generate a pole at $s = M_\pi^2$, a continuum starting at $s = 9M_\pi^2$, a bump at $s = M_{\pi'}^2$ etc. (for experimental evidence on π' see refs. [182, 183]):

$$\mathrm{Im}\, P(s) = \pi f_\pi^2 M_\pi^4\, \delta(M_\pi^2 - s) + \cdots . \tag{15.3}$$

To confront the information contained in the high energy representation (15.2) with the known behaviour of the amplitude in the low energy region Vainshtein et al. assume that the asymptotic behaviour (15.2) is valid for $s > s_0$ where s_0 is low enough for the interval $0 < s < s_0$ to be dominated by the pion contribution alone. The sum rule

$$\int_0^{s_0} \mathrm{d}s\, \{\mathrm{Im}\, P(s) - \mathrm{Im}\, P^{\mathrm{pert}}(s)\} \simeq 0 \tag{15.4}$$

then implies

$$\hat{m}(s_0) \simeq \frac{2\pi}{\sqrt{3}} \frac{f_\pi M_\pi^2}{s_0} . \tag{15.5}$$

With the estimate $s_0 \simeq 1.5\,\mathrm{GeV}^2$ this gives

$$\hat{m}(s_0) \simeq 6.5\,\mathrm{MeV} .$$

A value for s_0 somewhere in the interval between $M_\pi^2 = 0.02\,\mathrm{GeV}^2$ and $M_{\pi'}^2 \simeq 2.3\,\mathrm{GeV}^2$ appears to be reasonable in the sense of duality and finite energy sum rules. It is however clear that the numerical value is very sensitive to the choice of s_0 and that somewhat different values of s_0, say $s_0 = 1.2\,\mathrm{GeV}^2$ or $s_0 = 1.8\,\mathrm{GeV}^2$ ($\hat{m} \simeq 8\,\mathrm{MeV}$, $\hat{m} \simeq 5\,\mathrm{MeV}$) are also acceptable.

The most extensive analysis of the QCD sum rules in this channel was performed by Becchi, Narison, de Rafael and Yndurain [184] and by Narison and de Rafael [185]. These authors circumvent the lack of information about the behaviour of the imaginary part in the intermediate energy region by resorting to *inequalities.* Since Im P is positive, to saturate with the pion contribution alone means to get a lower bound on the value of a convergent dispersion integral. Furthermore, they improve the accuracy of the high energy representation by including the first order (two loop) correction [186–191]. For the absorptive part their result amounts to

$$\mathrm{Im}\, P^{\mathrm{pert}}(s) = \frac{3}{2\pi}\, s\, \hat{m}(s)^2 \left[1 + \frac{17}{3}\frac{\alpha_s(s)}{\pi} - \frac{2}{s}\{m_u^2(s) + m_d^2(s) - m_u(s)\, m_d(s)\} + \cdots\right] \tag{15.6}$$

where $m(s)$ is the running mass to two loop order and $\alpha_s(s)$ is the running coupling constant.

In the most recent version of their work Narison and de Rafael use the Laplace transform technique [181]. Let

$$L(M) = \frac{1}{\pi}\int_0^\infty \mathrm{d}s\, \mathrm{e}^{-s/M^2}\, \mathrm{Im}\, P(s) . \tag{15.7}$$

The pion contribution requires

$$L(M) \geq L^{\pi}(M) = f_{\pi}^{2} M_{\pi}^{4} \exp\{-M_{\pi}^{2}/M^{2}\} \tag{15.8}$$

for all values of the variable M. The perturbative asymptotic representation equivalent to (15.6) reads

$$L^{\text{pert}}(M) = \frac{3}{2\pi^{2}} M^{4} \hat{m}(M)^{2} \left[1 + \left(\frac{11}{3} + 2\gamma_{\mathrm{E}}\right)\frac{\alpha_{s}(M^{2})}{\pi} - \frac{2}{M^{2}}\{m_{\mathrm{u}}^{2}(M) + m_{\mathrm{d}}^{2}(M) - m_{\mathrm{u}}(M)\, m_{\mathrm{d}}(M)\} + \cdots\right] \tag{15.9}$$

where $\gamma_{\mathrm{E}} = 0.577\ldots$ is Eulers constant.

In addition to using an improved perturbative high energy representation they also include the *leading nonperturbative correction* which may be extracted from the operator product expansion:

$$L^{\text{nonpert}}(M) = \frac{3}{2\pi^{2}} \hat{m}(M)^{2} \left\{\frac{1}{3}\langle \boldsymbol{B}^{2}\rangle + \frac{2\pi^{2}}{3} f_{\pi}^{2} M_{\pi}^{2}\right\} \tag{15.10}$$

where $\langle \boldsymbol{B}^{2}\rangle$ is the vacuum expectation value of the square of the gluon field strength, introduced by Vainshtein, Zakharov and Shifman [180]:

$$\langle \boldsymbol{B}^{2}\rangle = \tfrac{1}{4}\langle 0|F_{\mu\nu}^{a} F^{\mu\nu a}|0\rangle = \pi^{2}\langle 0|\frac{\alpha_{s}}{\pi} G_{\mu\nu}^{a} G^{\mu\nu a}|0\rangle \sim 0.12\ \mathrm{GeV}^{4}\,.$$

Narison and de Rafael then study the inequality $L^{\pi} \leq L^{\text{pert}} + L^{\text{nonpert}}$, which gives a lower bound on the quark mass for any value of M. For large values of M the bound is very weak as it decreases with M^{-2}. Letting M become smaller the bound becomes more stringent until one reaches a value M_0 below which the bound again weakens. Picking the "optimal" value M_0 they arrive at the bounds

$$\hat{m}(1\ \mathrm{GeV}) \geq \begin{cases} 9.4 \pm 2.2\ \mathrm{MeV}\,; & \Lambda = 70\ \mathrm{MeV}\,; \quad M_0 = 600\ \mathrm{MeV} \\ 7.2 \pm 1.8\ \mathrm{MeV}\,; & \Lambda = 140\ \mathrm{MeV}\,; \quad M_0 = 650\ \mathrm{MeV} \\ 5.3 \pm 1.2\ \mathrm{MeV}\,; & \Lambda = 210\ \mathrm{MeV}\,; \quad M_0 = 750\ \mathrm{MeV}\,. \end{cases} \tag{15.11}$$

(The error bars are given by Narison and de Rafael on the basis of the size of the corrections to the leading asymptotic behaviour; we have converted the bounds given for the renormalization group invariant mass into the corresponding bounds for the running mass at 1 GeV according to table 1.)

As pointed out by Hubschmid and Mallik [192] the crucial question is whether the asymptotic representation may indeed be used down to values of M of the order of 600 or 700 MeV. The question was studied by Eidelman, Kurdadze and Vainshtein [193] in the case of the two point function associated with the vector current where the behaviour of the imaginary part is known from e^+e^--annihilation data. In that channel the analogous asymptotic representation is indeed consistent with the data down to M-values of this order of magnitude. Saturating the Laplace transform at $M = M_\rho$ with the ρ-meson alone one obtains an estimate for the ρ coupling constant

$$f_{\rho} = \frac{\sqrt{e}}{2\pi} M_{\rho}\left\{1 + \frac{1}{6}\frac{\langle \boldsymbol{B}^{2}\rangle}{M_{\rho}^{4}} + \mathrm{O}(\alpha_{s})\right\} \tag{15.12}$$

which is in fair agreement with the data. Assuming that the asymptotic representation for the Laplace transform of $\langle 0|\partial A^+ \, \partial A|0\rangle$ also holds at $M = M_\rho$ and saturating it with the pion contribution Narison and de Rafael obtain the estimate

$$\hat{m}(M_\rho) = \sqrt{\frac{e}{6}}\frac{f_\pi M_\pi^2}{f_\rho M_\rho}\{1 + O(\alpha_s)\} = 10.7\,\text{MeV} \tag{15.13}$$

which for $\Lambda = 140\,\text{MeV}$ is equivalent to $\hat{m}$ (1 GeV) $\simeq 10\,\text{MeV}$. The formula is similar to the old SU(6) result (see section 1), the numerical value is twice as large.

It is difficult to estimate the accuracy of the asymptotic representation of the Laplace transform at values of M of the order of M_ρ or lower on the basis of experience with the vector channel: the qualitative behaviour of the imaginary part is different in the two cases. For the vector current the imaginary part tends to a constant at high energies whereas for the axial divergence it grows essentially linearly with s. At any rate the contribution to the Laplace transform from intermediate states other than the pion are strongly suppressed at these values of M such that one should expect the inequality (15.11) to be almost saturated. We consider the values given in (15.11) as rough estimates of the quark mass rather than as reliable lower bounds. To get a reliable bound one should not extrapolate the high energy representation of the Laplace transform to values of M below 1 GeV. If one assumes the high energy representation only to be valid down to 1 GeV then the *bounds of Narison and de Rafael*, expressed in terms of the running mass at 1 GeV become

$$\hat{m}(1\,\text{GeV}) \geq \begin{cases} 5.1\,\text{MeV}\,, & \Lambda = 70\,\text{MeV} \\ 4.7\,\text{MeV}\,, & \Lambda = 140\,\text{MeV} \\ 4.3\,\text{MeV}\,, & \Lambda = 210\,\text{MeV}\,. \end{cases} \tag{15.14}$$

Hubschmid and Mallik avoid using the high energy representation down to small values of M. Instead they improve the low energy representation of the amplitude by including contributions from higher intermediate states. They parametrize these contributions in terms of a narrow radial excitation of the pion at $M_{\pi'} \simeq 1.5\,\text{GeV}$ and allow the coupling strength of this state to vary within a rather broad range of values. Working with finite energy and dispersion sum rules rather than with the Laplace tranform they find that the model satisfies the constraints imposed by QCD provided the quark mass is in the range

$$\hat{m}\ (1\,\text{GeV}) \simeq 5.7 \pm 2.2\,\text{MeV}\,. \tag{15.15}$$

[Since they work at relatively large values of q^2 the result is not sensitive to the value of Λ or of $\langle \boldsymbol{B}^2\rangle$.]

We have carried out an independent *evaluation of the Laplace transform* with the aim of getting a conservative estimate of the range of $\hat{m}$ values that are consistent with the QCD sum rule for the axial divergence. Consider the Laplace transform of $\operatorname{Im} P(s)$ at $M = 1\,\text{GeV}$. This quantity gets sizeable contributions only from the range $0 < \sqrt{s} < 2.5\,\text{GeV}$. Around $\sqrt{s} \simeq 2\,\text{GeV}$ the asymptotic formula (15.6) should start to become a reliable representation of the actual imaginary part. [Note that the quantity $2\pi \operatorname{Im} P^{\text{pert}}(s)/3s$ is almost constant in the range $1.5\,\text{GeV} < \sqrt{s} < 2.5\,\text{GeV}$. If Λ is of order 140 MeV then this quantity varies from $1.25\,\hat{m}^2\,(1\,\text{GeV})$ to $\hat{m}^2\,(1\,\text{GeV})$ as $\sqrt{s}$ grows from 1.5 GeV to 2.5 GeV.] *The problem is to obtain a reliable estimate of the imaginary part in the region below* $\sqrt{s} \leq 2$ *GeV*. The continuum starts at $\sqrt{s} = 3M_\pi$ – the contribution of the *three pion intermediate state* can be worked out from current algebra. In the chiral limit it is given by

$$\operatorname{Im} P^{3\pi}(s) = \frac{1}{384\pi^3 f_\pi^2} M_\pi^4 s \,. \tag{15.16}$$

The contribution grows linearly with s like the perturbative imaginary part, however with a much smaller coefficient. For $\hat{m}$ (1 GeV) = 5 MeV the ratio $\operatorname{Im} P^{3\pi}/\operatorname{Im} P^{\text{pert}}$ is 0.15, for $\hat{m}$ (1 GeV) = 10 MeV it is four times smaller. In the real world the three pion contribution is further suppressed by phase space [at $\sqrt{s} = 1$ GeV the suppression still amounts to a factor of two]. We conclude from this that the imaginary part does not receive significant contributions before the threshold for resonance formation ($M_\rho + M_\pi = 0.9$ GeV, $M_{\pi'} \sim 1.3$ GeV) is reached: apart from the one-pion intermediate state the imaginary part practically vanishes below $\sqrt{s} = 1$ GeV. The critical region is the range 1 GeV $< \sqrt{s} <$ 2 GeV. We parametrize the imaginary part in terms of π, π' and a continuum contribution:

$$\operatorname{Im} P(s) = \pi f_\pi^2 M_\pi^4 \{\delta(s - M_\pi^2) + r\,\delta(s - M_{\pi'}^2)\} + \theta(s - s_0) \operatorname{Im} P^{\text{pert}}(s) \,. \tag{15.17}$$

The *radial excitation of the pion* is weighted with the ratio r of the wave functions at the origin:

$$r = \left| \frac{\langle 0|\bar{u}\gamma_5 d|\pi'\rangle}{\langle 0|\bar{u}\gamma_5 d|\pi\rangle} \right|^2 . \tag{15.18}$$

In fig. 2 we show the value of the running quark mass at 1 GeV that one obtains by calculating the Laplace transform of (15.17) at M and equating the result with the asymptotic representation $L^{\text{pert}}(M) + L^{\text{nonpert}}(M)$ given in (15.9) and (15.10). The renormalization group invariant scale is taken at Λ = 140 MeV with $N_f = 3$ [as can be seen from (15.14) the precise value of Λ is not essential; the power

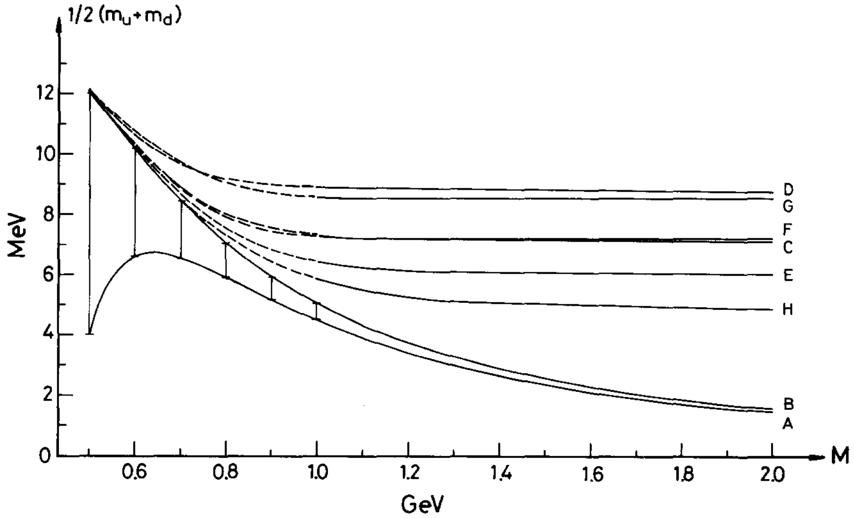

Fig. 2. Quark mass from QCD sum rules for Λ = 140 MeV; m_u and m_d are the running masses in the $\overline{\text{MS}}$ scheme at μ = 1 GeV, M is the variable in the Laplace transform. Curves A and B are lower bounds obtained by different algebraic treatment of the first order corrections: the vertical bars are an estimate of the errors due to higher order contributions. Curves C, D, E, F, G and H are obtained by using a series of models for the behaviour of the imaginary part in the region 1 GeV $< \sqrt{s} <$ 2 GeV.

correction proportional to $\langle \boldsymbol{B}^2 \rangle$ amounts to less than 10% for $M > 1\,\mathrm{GeV}$ and might just as well be dropped]. If the model for the imaginary part is a good approximation we should get the same value of $\hat{m}$ (1 GeV) for all values of M for which the asymptotic representation holds. Since we do not trust this representation below $M = 1\,\mathrm{GeV}$ we have indicated the corresponding quark mass values by dashed lines.

Consider first the *bounds* of Narison and de Rafael. In our model these bounds correspond to $s_0 = \infty$, $r = 0$ (curve B). With the accuracy of the perturbative calculation there is no distinction between an expression of the form $(1 + \alpha/\pi)^{-1/2}$ and $(1 - \frac{1}{2}\alpha/\pi)$. De Rafael and Narison use a somewhat different algebraic form for the perturbative expression to obtain a curve like A. We take the difference between A and B as an estimate for the errors due to higher order perturbative contributions. Now consider the contributions from intermediate states other than the pion. To get a *rough estimate of the range of parameters* s_0, $M_{\pi'}$ and r of interest we observe that a linear π-trajectory with the same slope as the ρ-trajectory and with π' as the first spin zero daughter predicts $M_{\pi'}^2 = M_\pi^2 + 2(\alpha')^{-1}$, i.e. $M_{\pi'} = 1.5\,\mathrm{GeV}$. If the continuum threshold is taken half way between $M_{\pi'}^2$ and the next recurrence we get $\sqrt{s_0} = 1.8\,\mathrm{GeV}$. According to *duality and finite energy sum rules* (FESR) the π' peak should balance the area between $\frac{1}{2}(M_\pi^2 + M_{\pi'}^2)$ and s_0; this leads to $r \simeq 8$. The resulting value of the quark mass is given in curve C. Between $M = 1\,\mathrm{GeV}$ and $M = 2\,\mathrm{GeV}$ the quark mass indeed stays practically constant at a value $\hat{m}$ (1 GeV) = 7.2 MeV.

If one keeps the continuum threshold fixed, the main uncertainty in the model is the size of the π' peak, measured by r. The value $r \simeq 8$ suggested by FESR may at first sight appear to be unreasonably large. In fact, the analogous ratio of say the vector coupling constants for ψ and its radial excitation ψ' is known to be small (the measured decay rates into e^+e^- give $r_\psi = (f_{\psi'}^2 : f_\psi^2) \simeq 0.5$). If one estimates r in the framework of nonrelativistic potential models one also gets numbers of order one. Why do FESR suggest a much larger value in the pion channel? The reason for this is most easily seen if one replaces the imaginary part by a sequence of equally spaced narrow resonances: since at high energies the imaginary part grows linearly with s the residues r_n of these resonances have to grow linearly with n. The value of r_n must increase by $\Delta r = 6\hat{m}^2(\pi f_\pi M_\pi^2\, \alpha')^{-2}$ from resonance to resonance; for $\hat{m} = 5\,\mathrm{MeV}$ this gives $\Delta r \simeq 3$, for $\hat{m} = 10\,\mathrm{MeV}$ the increase is four times larger: $\Delta r \simeq 12$ ($\hat{m} = 7\,\mathrm{MeV}$ amounts to $\Delta r \simeq 6$). For very heavy quarks the excitation energy of the first excited state is small compared to the rest energy of the ground state. There is no dramatic increase in r_n for the first couple of excited states – the linear increase in the imaginary part only develops once the excitation energy is of the order of the rest energy. In the pion channel nonrelativistic models fail, because the excitation energy $M_{\pi'} - M_\pi$ is large in comparison with the rest energy of the ground state and it is not clear how to handle kinematic factors like $M_{\pi'} : M_\pi \simeq 10$.

It is certainly legitimate to question whether FESR provide us with a good estimate for the size of the imaginary part below 2 GeV. Any model that saturates the sum rule may be regarded as acceptable. To show the *dependence of the quark mass value on the parameters of the model* we give several different sets of parameters which all lead to a consistent picture with $\hat{m}$ being roughly independent of the point at which the Laplace transform is evaluated. The parameters defining the curves D, E, F, G, H are given in table 5. If the imaginary part is taken too small in the intermediate energy region the sum rule does not saturate. If one wants to have nothing but a narrow resonance below 2 GeV then saturation only occurs for sizeable values of r – the lower the mass of the resonance, the larger r has to be (models E, F, H). Curves D and G are almost indistinguishable: it does not matter whether the imaginary part is carried only by a narrow resonance with $r = 8$ at $M_{\pi'} = 1.3\,\mathrm{GeV}$ or whether $\frac{3}{4}$ of the resonance contribution is spread over the interval from 1.1 to 1.6 GeV. We consider models D and G as upper

Table 5
Values of the parameters s_0, $M_{\pi'}$ and r used in fig. 2. The running quark mass at 1 GeV is determined by comparing the Laplace transform in the region 1 GeV < M < 2 GeV with the asymptotic representation.

	$\sqrt{s_0}$ (GeV)	$M_{\pi'}$ (GeV)	r	$\frac{1}{2}(m_u + m_d)$ (MeV)
A, B	∞		0	
C	1.8	1.5	8	7.2
D	1.6	1.3	8	8.8
E	2	1.65	8	6.2
F	2	1.6	12	7.3
G	1.1	1.3	2	8.6
H	2	1.75	5	5.2

limits for the imaginary part in the region 1 GeV $< \sqrt{s} <$ 2 GeV (note that in model G the continuum starts with full strength at $\sqrt{s_0} = 1.1$ GeV, a small π' peak sitting on top of it). Model H describes the opposite extreme, for which there is nothing but a π' peak of modest size in this region. On the basis of these models we conclude that our present knowledge about the behaviour of the two-point function $\langle 0|\partial A^+\ \partial A|0\rangle$ in *QCD is consistent with a quark mass value around 7 MeV*:

$$\hat{m}\,(1\ \text{GeV}) = (7 \pm 2)\ \text{MeV}\,. \tag{15.19}$$

In the absence of reliable information about the behaviour of the imaginary part in the intermediate energy region it is not possible to reduce the error bar substantially. *Quark masses of the order of* 10 *MeV* or higher would imply that the imaginary part of the amplitude $\langle 0|T\,\partial A^+\ \partial A|0\rangle$ contains very large contributions from the intermediate energy region below 2 GeV. The presence of such contributions would be a most interesting phenomenon by itself as these contributions would presumably produce *large corrections to PCAC* (for a discussion of PCAC violations due to excited states of the pion see [194–196] and the references quoted in these papers).

It is straightforward to apply the same analysis to the *K-meson channel.* The ITEP group obtains the estimate $m_s\,(1.6\ \text{GeV}) \simeq 110$ MeV which for Λ in the range 100 to 140 MeV is equivalent to

$$m_s\,(1\ \text{GeV}) \simeq 120\ \text{MeV}\,. \tag{15.20}$$

The bounds of Narison and de Rafael for $M \geq 1$ GeV are

$$m_s\,(1\ \text{GeV}) + \hat{m}\,(1\ \text{GeV}) \geq \begin{cases} 130\ \text{MeV}\,, & \Lambda = 70\ \text{MeV} \\ 120\ \text{MeV}\,, & \Lambda = 140\ \text{MeV} \\ 110\ \text{MeV}\,, & \Lambda = 210\ \text{MeV} \end{cases} \tag{15.21}$$

(the corresponding "optimal" bounds for the running mass at 1 GeV are 210, 160 and 130 MeV respectively. As discussed above these optimal bounds are obtained by using the high energy representation at low values of M; it is difficult to estimate the error bars in that procedure). Using the estimates for the imaginary part given above we obtain

$$m_s\,(1\ \text{GeV}) = 180 \pm 50\ \text{MeV}\,. \tag{15.22}$$

(The positions of the thresholds and of the radial excitation K′ are expected to be at slightly higher values of s than in the pion channel. Based on the observation that $M_{\mathrm{K}}^2 - M_\pi^2$ is roughly equal to $M_{\mathrm{K}^*}^2 - M_\rho^2$ we have shifted the values of s according to $s_{\mathrm{K}} = s_\pi + M_{\mathrm{K}}^2 - M_\pi^2$ which leads to $M_{\mathrm{K}'} - M_{\pi'} \simeq 80\,\mathrm{MeV}$.)

We conclude that the QCD sum rules are consistent with a strange quark mass in the interval $130\,\mathrm{MeV} < m_{\mathrm{s}} < 230\,\mathrm{MeV}$. A strange quark mass below 130 MeV is in conflict with the bounds (15.21) which we argued to be on the safe side, whereas a strange quark mass larger than 230 MeV calls for very large contributions from the intermediate energy region. The sum rules are perfectly consistent with the ratio $m_{\mathrm{s}}:\hat{m} \simeq 25$ obtained from the spectrum of the pseudoscalar mesons in section 13.

One may obtain information about the quark mass values from *other two-point functions* than those discussed above. The problem with these other channels is that there is even less information about the behaviour of the imaginary part than in the case of the axial divergence where at least the coupling strength of the pion and of the kaon are known. The analysis is therefore necessarily rather crude and does not permit one to extract very reliable information about the quark masses at this stage.

In a recent paper Mallik [197] has analyzed the QCD sum rules for all two-point functions associated with the scalar, vector, tensor, axial and pseudoscalar quark densities (see also Craigie and Stern [198]). This analysis not only shows that quark mass estimates very similar to those proposed by Okubo [19] may be obtained within QCD, it also demonstrates that some of the SU(6)-relations among the vector, tensor and axial coupling constants that lead to the value $\hat{m} = 5.4\,\mathrm{MeV}$ (see section 1) indeed follow if one saturates two-point function sum rules in the manner described above. In view of the fact that one has to eliminate unknowns by combining saturated sum rules the numerical results of course involve larger uncertainties than in the case of the sum rule for the axial divergence. Within the errors associated with the saturation scheme the quark mass values given by Mallik are perfectly consistent with the estimates given above.

Finally, for the purpose of illustration rather than with the intention of reviewing the subject in any detail, we briefly indicate how an analysis of the two-point function associated with the *baryonic "currents"* $q(x)\,q(x)\,q(x)$ leads to information about the order parameter $\langle 0|\bar{q}q|0\rangle$ and hence about the size of the quark masses [199]. For definiteness we consider some parity even spin $\frac{1}{2}$ field $\eta(x)$ with the quantum numbers of uud, e.g.:

$$\eta(x) = \gamma_\mu\,\gamma_5\,d_{\mathrm{a}}(x)\,(u_{\mathrm{b}}^{\mathrm{T}}(x)\,C\,\gamma^\mu\,u_{\mathrm{c}}(x))\,\varepsilon_{\mathrm{abc}} \tag{15.23}$$

where a, b, c are colour indices and C is the charge conjugation matrix with $\gamma_\mu^{\mathrm{T}} = -C\gamma_\mu C^{-1}$. Lorentz invariance implies a spectral representation of the form

$$\langle 0|\eta(x)\,\bar{\eta}(y)|0\rangle = \frac{1}{\mathrm{i}}\int_0^\infty \mathrm{d}s\,\{\rho_1(s)\,\mathrm{i}\partial\!\!\!/ + \rho_2(s)\}\Delta^{(+)}(x-y;s) \tag{15.24}$$

where $\Delta^{(+)}(z;s)$ is the positive frequency part of the free Green's function for a scalar field of mass $\sqrt{s}$. In perturbation theory (massless quarks) the spectral function $\rho_2(s)$ vanishes to all orders because the propagators of the field d which enters the diagrams at x and leaves at y always contain an odd number of γ-matrices. In the real world $\rho_2(s)$ is obviously different from zero. The proton e.g. produces the contribution

$$\begin{aligned}\rho_1(s) &= \kappa\,\delta(s - M_{\mathrm{p}}^2) + \cdots\\ \rho_2(s) &= \kappa M_{\mathrm{p}}\,\delta(s - M_{\mathrm{p}}^2) + \cdots.\end{aligned} \tag{15.25}$$

Only a parity degenerate spectrum could be consistent with $\rho_2(s)=0$. In the short distance behaviour of the amplitude $\langle 0|\eta(x)\,\bar\eta(y)|0\rangle$ the contribution of the spectral function $\rho_2(s)$ is governed by the vacuum expectation values of the operators with lowest dimension that carry chirality; the leading short distance contribution is again proportional to $\langle 0|\bar qq|0\rangle$. It is therefore possible to extract information about the size of this quantity if one assumes the asymptotic representation for the Laplace transforms of $\rho_1(s)$ and $\rho_2(s)$ to be valid at $M\simeq M_{\mathrm p}$ and to be saturated by the proton intermediate state. Similar arguments may of course be applied to the two-point functions associated with spin 3/2 fields. We quote one of the ensuing relations between baryon masses and the order parameter $\langle 0|\bar qq|0\rangle$, due to Ioffe:

$$(M_\Delta)^3=\frac{40\pi^2}{3}\left(1+\sqrt{\tfrac{2}{5}}\right)[-\langle 0|\bar uu|0\rangle]\,. \tag{15.26}$$

As pointed out by Chung, Dosch, Kremer and Schall [200], the result is not independent of the interpolating field one works with. We nevertheless find it most remarkable that the values of the parameter $\langle 0|\bar qq|0\rangle$ which one extracts in this manner from the baryon spectrum are consistent with the estimates described in the previous sections. With $M_\Delta=1232\,\mathrm{MeV}$ the quark mass value that one obtains from the relation (15.26) is

$$\hat m\approx 9.5\,\mathrm{MeV}\,. \tag{15.27}$$

16. Heavy quarks

As was shown by Novikov et al. [201] the masses of heavy quarks may be extracted from e^+e^- data by investigating the *QCD sum rules for the vacuum polarization amplitude.* Since there are several excellent reviews on this subject [201–204, 206–208] we will not describe the analysis in detail. There is only one minor point which we find useful to add: we wish to express the values extracted from e^+e^- data in terms of the corresponding *running quark mass in the $\overline{\mathrm{MS}}$ scheme.* The sum rules are usually evaluated in terms of the mass function $M(p^2)$ introduced by Georgi and Politzer [209]. One writes the quark propagator in the form

$$S(p)=Z(p^2)\,[M(p^2)-\not p]^{-1} \tag{16.1}$$

and determines the value of the quantity $M(p^2)$ at the spacelike point $p^2=-m^2_{\mathrm{quark}}$. The functions $M(p^2)$ and $Z(p^2)$ may be expressed in terms of the running mass $m(\mu)$ and running coupling constant $g(\mu)$ order by order; they depend on the gauge. To lowest order the explicit expression reads (see e.g. ref. [210])

$$\begin{aligned}
M(p^2)&=m(\mu)\left\{1+\frac{\alpha_s}{\pi}(a+\lambda b)+\mathrm{O}(\alpha_s^2)\right\}\\
Z(p^2)&=1-\frac{\alpha_s}{3\pi}\lambda(a-3b+\tfrac{2}{3})+\mathrm{O}(\alpha_s^2)\\
a&=\frac{4}{3}-\ln\frac{m^2}{\mu^2}+\frac{1}{p^2}(m^2-p^2)\ln\frac{m^2-p^2}{m^2}\\
b&=\frac{(m^2-p^2)}{3p^2}\left\{-1-\frac{m^2}{p^2}\ln\frac{m^2-p^2}{m^2}\right\}
\end{aligned} \tag{16.2}$$

where λ is the gauge parameter (Landau gauge: $\lambda = 0$, Feynman gauge: $\lambda = 1$). To give a value for the quantity $M(p^2)$ at some point p^2 in a suitable gauge is equivalent to giving a value for the running mass $m(\mu)$ at some scale μ. In contrast to $M(p^2)$ the running mass is a gauge invariant notion. We compare the value of the quantity $M(p^2)$ in the Landau gauge at the spacelike point $p^2 = -m^2$ with the running mass $m(\mu)$ at the point $\mu = m$. The equation

$$m(m) = m \tag{16.3}$$

[where $m(\mu)$ is the running mass in the $\overline{\mathrm{MS}}$ scheme with N_f flavours] unambiguously defines a quark mass parameter m to all orders in perturbation theory – we denote this quantity by $m(m)$. The quarks c and b are sufficiently heavy for perturbation theory to work at $\mu = m_c$ or $\mu = m_b$; for the light quarks u, d, s the quantity $m(m)$ is an academic notion, because a perturbative calculation of the function $m(\mu)$ at small scales of order $\mu = 5$ MeV or $\mu = 150$ MeV does not appear to be feasible. [To be precise, we should specify the number of flavours to be used in the $\overline{\mathrm{MS}}$ scheme that defines the function $m(\mu)$. We expect the dependence on N_f to be very weak (see below); for definiteness we define the quantity $m_c(m_c)$ in the $\overline{\mathrm{MS}}$ scheme with 3 flavours, whereas $m_b(m_b)$ refers to the $\overline{\mathrm{MS}}$ scheme with 4 flavours.]

To lowest order the *relation between the Georgi–Politzer function* at $p^2 = -m^2$ in the Landau gauge *and the running mass* at scale m may be read off from (16.2):

$$M(-m^2) = m(m)\left\{1 + \frac{\alpha_s}{\pi}\left(\frac{4}{3} - 2\ln 2\right) + \mathrm{O}(\alpha_s^2)\right\}. \tag{16.4}$$

The numerical coefficient in front of α_s is very small; for $\alpha_s < 0.3$ the difference between $M(-m^2)$ and $m(m)$ is less than 5‰ – small in comparison with the errors of the method used to determine the value of $M(-m^2)$.

In the case of the b quark mass *the main uncertainty stems from the value of α_s used in the analysis of the sum rules.* Since the perturbative asymptotic representation of the sum rule is only known to first order in α_s the method does not allow one to express the value of this parameter in terms of the known running coupling constant $\alpha_s(\mu)$. One has to calculate the second order contributions to find out which value of α_s to use in the first order expression. The value depends on the quantity one is looking at. In the context of Laplace transformed sum rules (exponential moments) the value of α_s to be used depends on the scale M at which the Laplace transform is evaluated – there is no universal prescription (like the prescription that $\alpha_s(2m_b) \simeq 0.17$ should be used). At a low value of M the sum rule is almost saturated by the ground state; in this case the effective coupling strength is presumably determined by the size of the system, or, equivalently, by the level splitting [$\alpha_s = 0.25$ to $\alpha_s = 0.30$] rather than by its mass (see e.g. refs. [211, 202]). The result for $m_b(m_b)$ increases with the value of α_s used in the analysis of the sum rule ($\alpha_s = 0.17$ leads to $m_b(m_b) = 4.19$ GeV whereas $\alpha_s = 0.25$ gives $m_b(m_b) = 4.26$ GeV). With the estimates for $M(-m^2)$ given by Novikov et al. [201], Voloshin [211], Reinders et al. [204, 205], Guberina et al. [206], Bertlmann [218], Iwao [212] and Shifman [208] we obtain the following values for the running masses at scale m:

$$m_c(m_c) = 1.27 \pm 0.05\ \mathrm{GeV}$$
$$m_b(m_b) = 4.25 \pm 0.10\ \mathrm{GeV}\,. \tag{16.5}$$

The quark mass values found within nonrelativistic models for the bound state motion (for an up-to-date review see ref. [213]) are considerably higher than these current quark masses. Unfortunately the connection of the potential model parameters to the Lagrangian of the theory is not easy to establish in the case of the quarks that are presently available for experimentation. However, if nature does provide us with a very heavy t-quark then the bound states of this particle will be a sensitive probe for measuring the parameters of the QCD Lagrangian. For very heavy quarks the perturbative nonrelativistic bound state picture is self-consistent. The levels are given by the *Balmer formula*

$$M_n = 2M_q - \frac{1}{4n^2} M_q(\tfrac{4}{3}\alpha_s)^2 . \tag{16.6}$$

For very heavy quarks this formula should be very accurate: both perturbative corrections (of order $M_q\alpha_s^3$) and nonperturbative corrections (of order $\langle B^2\rangle M_q^{-3}\alpha_s^{-4}$, see below) are small if the mass of the quark is sufficiently large. The quark mass M_q that appears in this formula may be calculated in terms of the running mass as a power series in the running coupling constant, by solving the equation

$$M(M_q^2) = M_q \tag{16.7}$$

which determines the position of the pole in the propagator and the start of the cut: M_q is the *dressed mass of the quark* (in contrast to the function $M(p^2)$ itself, the dressed quark mass is a gauge invariant notion). The concept of mass for an isolated, dressed quark of course conflicts with the confinement of colour. The above definition of the quantity M_q may work order by order in perturbation theory – it fails in the real world where the propagator is not supposed to have a pole. The propagator $S(p)$ of a heavy quark only looks like the free fermion Green's function as long as p^2 stays away from the region $p^2 = M_q^2$, where nonperturbative methods have to be used to get a reliable representation for the propagator. Within perturbation theory the physical mass of the quark however appears to be a well-defined concept (in particular Tarrach [214] has shown that M_q is infrared finite to two loops).

To lowest order the *connection between the running mass and the mass to be used in the Balmer formula* may be read off from (16.2):

$$M_q = m(m)\{1 + \tfrac{4}{3}\alpha_s/\pi + O(\alpha_s^2)\} . \tag{16.8}$$

It is clear that one needs the next order term in this expansion both to determine what value for α_s to use and to have sufficient accuracy to consider binding effects – which is what the Balmer formula does. To our knowledge the higher order corrections have not yet been calculated.

Using the numerical value (16.5) the *mass of the dressed b quark* becomes

$$M_b = \begin{cases} 4.6 \pm 0.1\ \text{GeV}, & \alpha_s = 0.2 \\ 4.8 \pm 0.1\ \text{GeV}, & \alpha_s = 0.3 . \end{cases} \tag{16.9}$$

The value of the dressed b quark mass obtained in this manner is consistent with the value obtained within a nonrelativistic treatment of the bound state motion, provided the leading nonperturbative contributions are accounted for. The leading nonperturbative corrections were first calculated by Voloshin [211] who finds

$$M_b = 4.795 \pm 0.025\ \text{GeV} \tag{16.10}$$

for $\alpha_s = 0.30 \pm 0.03$. As shown by one of us (Leutwyler [215]) the *leading nonperturbative correction to the Balmer formula* may be written in the form

$$M_{nl} = 2M_q - \frac{1}{4n^2} M_q(\tfrac{4}{3}\alpha_s)^2 + \frac{\langle \boldsymbol{B}^2\rangle n^6}{M_q^3[\tfrac{4}{3}\alpha_s]^4}\varepsilon_{nl} \tag{16.11}$$

where ε_{nl} is a rational number of order one ($\varepsilon_{10} = 1.468$, $\varepsilon_{20} = 1.585$, $\varepsilon_{21} = 0.998$). The formula (16.11) only applies if the quark mass is sufficiently heavy such that the nonperturbative effects proportional to $\langle \boldsymbol{B}^2\rangle$ do not distort the wave function too strongly. To estimate the size of the nonperturbative correction we first look at the $b\bar{b}$ ground state in perturbation theory. To lowest order the levels are given by the uncorrected Balmer formula. The perturbative corrections are of order $M_b\alpha_s^3$. In QED the analogous correction can be absorbed by a proper normalization of e simultaneously for all levels of positronium (one simply has to identify e with the charge of an isolated electron). This is not the case in QCD because the diagrams in which the exchanged gluon virtually dissociates into a gluon pair is infrared divergent if the gluon momentum vanishes. (There is a similar effect in muonium where the vacuum polarization due to electron pairs produces a correction proportional to $M_\mu\alpha^5(M_\mu/M_e)^2$ which explodes as $M_e \to 0$. For massless electrons the correction is of order α^3 rather than α^5.) If one wants to absorb the terms of order $M_b\alpha_s^3$ in the uncorrected Balmer formula by a renormalization of α_s then one has to choose a different value of α_s for every level of the system. The value of α_s to be used is the running coupling constant at a scale of the order of the inverse Bohr radius of the orbit in question [216]

$$\alpha_s \simeq \alpha_s(\tfrac{2}{3}M_b\alpha_s/n)\,. \tag{16.12}$$

For the ground state ($n = 1$) this gives $\alpha_s \simeq 0.3$, $M_b = 4.83\,\text{GeV}$. The corresponding Bohr radius is $a_{\text{Bohr}} = (M_b\tfrac{2}{3}\alpha_s)^{-1} \simeq (1\,\text{GeV})^{-1}$, the kinetic energy amounts to $E_{\text{kin}} = 193\,\text{MeV}$, the expectation value of the Coulomb potential is twice as large: $E_{\text{Cb}} = -386\,\text{MeV}$. With the same value of α_s the nonperturbative correction [evaluated with $\langle \boldsymbol{B}^2\rangle = 0.12\,\text{GeV}^4$ as given by Vainshtein et al. [217]] shifts the quark mass to $M_b = 4.79\,\text{GeV}$ [compare (16.10); the kinetic energy and the expectation value of the Coulomb potential are almost unaffected; the nonperturbative correction amounts to $E_{\text{np}} = +62\,\text{MeV}$]. If we instead pick $\alpha_s = 0.25$ in the uncorrected Balmer formula then we get $M_b = 4.80\,\text{GeV}$, $E_{\text{Cb}} = -266\,\text{MeV}$ and $a_{\text{Bohr}} \simeq (800\,\text{MeV})^{-1}$. For $\alpha_s = 0.25$ and $\langle \boldsymbol{B}^2\rangle = 0.12\,\text{GeV}^4$ the leading nonperturbative correction shifts the quark mass to $M_b = 4.73\,\text{GeV}$ [the leading nonperturbative term amounts to $E_{\text{np}} = +135\,\text{MeV}$, i.e. to 50% of the Coulomb potential – the net binding energy vanishes]. It is clear that a framework which treats the nonperturbative effects as small corrections is not reliable for the $b\bar{b}$ ground state if the effective coupling strength for the bound state motion is below $\alpha_s \lesssim 0.25$: the mass of the b quark is then too light and we have to expect a sizeable distortion of the wave function. The pressure exerted by the vacuum fields tends to squeeze the system and hence to increase the inverse Bohr radius. Since the ground state realizes a minimum of the energy we nevertheless expect the formula (16.11) to give a respectable estimate for the energy of the state, despite the fact that the wave function that underlies it is too broad. [This is confirmed in a recent paper by Bertlmann [218] who finds $\alpha_s = 0.25$ for the effective coupling strength; using the value $\langle \boldsymbol{B}^2\rangle = 0.16\,\text{GeV}^4$ (obtained from an analysis of the $c\bar{c}$ system) and working with exponential moments he obtains $M_b = 4.71\,\text{GeV}$. If one uses the values $\alpha_s = 0.25$, $\langle \boldsymbol{B}^2\rangle = 0.16\,\text{GeV}^4$ in (16.11) one indeed obtains $M_b = 4.70\,\text{GeV}$.]

What the formula (16.11) clearly shows is that the *excited states* of the $b\bar{b}$ system cannot be described in terms of corrections to a positronium-like bound state: the nonperturbative term explodes with the sixth power of the principal quantum number. It ruins QCD perturbation theory unless the quark is very much heavier than the b quark. In the case of the $c\bar{c}$ system the nonperturbative effects win already in the ground state and it is not clear whether the perturbative notion of a dressed c quark mass ($M_c \simeq 1.4$ GeV) is of any use.

To compare the masses of the heavy quarks with the running masses of the light quarks at the scale $\mu = 1$ GeV we have to convert the above values into the corresponding *running masses* m_c, m_b at $\mu = 1$ GeV. In the case of m_c this is a straightforward matter. The renormalization factor $m_c(1.27\text{ GeV}) : m_c(1\text{ GeV})$ may be extracted from table 1. With a value of Λ in the range $100\text{ MeV} < \Lambda < 200\text{ MeV}$ (3 flavours) one finds

$$m_c(1\text{ GeV}) = 1.35 \pm 0.05\text{ GeV}\,. \tag{16.13}$$

In the case of m_b there is a complication: the change in scale involves crossing the threshold for $c\bar{c}$ production. To our knowledge the connection between the running mass in the $\overline{\text{MS}}$ scheme for N_f flavours (appropriate for scales large in comparison to all quark masses) and the corresponding running mass for $N_f - 1$ flavours (appropriate below the threshold for the heaviest quark) is not available in the literature [see Ovrut and Schnitzer [219] and Bernreuther and Wetzel [220] for the corresponding connection between the values of Λ]. If one evaluates the renormalization factor $r = m(1) : m(4.25)$ for $N_f = 3$ according to (3.4) one finds $r = 1.26$, 1.32, 1.38 for $\Lambda_3 = 100$, 150 and 200 MeV respectively. To calculate the renormalization factor for $N_f = 4$ one has to adjust the value of Λ; according to Bernreuther and Wetzel [220] the corresponding values of Λ are $\Lambda_4 = 76$, 120 and 165 MeV. Using these values in (3.4) one finds $r = 1.25$, 1.31, 1.38 respectively. This shows that the mass renormalization factors are almost unaffected by the presence of an additional quark flavour, but do of course depend on the value of Λ, which is still not known very accurately.

The running b quark mass at $\mu = m_b$ is determined on the basis of sum rules that only include first order corrections. Since the first order mass renormalization is independent of the number of quark flavours the method is not accurate enough to require a distinction between different definitions of the running mass. Using the value $m_b(m_b) = 4.25 \pm 0.10$ GeV the running b quark mass at $\mu = 1$ GeV becomes

$$m_b(1\text{ GeV}) = \begin{cases} 5.3 \pm 0.1\text{ GeV}\,, & \Lambda_3 = 100\text{ MeV} \\ 5.9 \pm 0.1\text{ GeV}\,, & \Lambda_3 = 200\text{ MeV}\,. \end{cases} \tag{16.14}$$

17. Quark masses from SU(4) symmetry

The mass of the charmed quark is of the same order as the scale of QCD. It is therefore doubtful whether one may treat it as a perturbation and compare bound states containing charmed quarks with the analogous light quark bound states, say ψ with φ or Λ_c with the nucleon. *To test* the assumption that $SU(4)$ is an approximate symmetry one may *compare the* ψ *and* φ *wave functions at the origin*, which are measured in the decays into e^+e^-. If SU(4) is an approximate symmetry at infinite momentum then $f_\psi \simeq f_\varphi$; experimentally $f_\psi = (1.69 \pm 0.14) f_\varphi$. The symmetry is obviously rather strongly broken in comparison with SU(3) for which the prediction $f_\varphi \simeq f_\rho$ is well obeyed: $f_\varphi = (1.11 \pm 0.07) f_\rho$. The perturbation caused by the mass difference $m_c - \hat{m}$ which plays the role of the SU(4) symmetry-breaking parameter is not very small and one should therefore not expect SU(4) predictions to be very accurate.

The pattern of SU(4) breaking observed in the meson and baryon masses can nevertheless be understood on the basis of first order SU(4) mass formulae [221–230]. The pattern is determined by the ratio of SU(4) breaking to SU(3) breaking quark mass differences

$$C = (m_c - \hat{m})/(m_s - \hat{m}) . \tag{17.1}$$

The value of this ratio is particulary interesting, because it relates the light quark masses to the charmed quark mass which is known rather accurately. If the quantity C can be extracted from the observed meson and baryon masses the measured value $m_c\,(1\,\mathrm{GeV}) = 1.35 \pm 0.05\,\mathrm{GeV}$ then determines the running light quark masses at the scale $\mu = 1\,\mathrm{GeV}$. [As discussed in section 13 the analogous ratio $R = (m_s - \hat{m}):(m_d - m_u)$ determines the size of the SU(2) splittings in terms of the SU(3) splittings; we disregard isospin breaking for the moment and put $m_u = m_d$.]

The *first order SU*(4) *mass formulae* (see appendix A) imply the following relations among the observed masses:

$$C = \frac{\Lambda_c - N}{\Lambda - N} = \frac{\Sigma_c - N}{\Sigma - N} = \frac{D - \pi}{K - \pi} = \frac{F - K}{K - \pi} = \frac{D^* - \rho}{K^* - \rho} = \frac{F^* - K^*}{K^* - \rho} . \tag{17.2}$$

[Except for the relations involving the Goldstone bosons π, K, D, F – for which one has to use M^2 – the first order quark mass expansion does not distinguish between mass formulae linear in M and mass formulae for M^2.] The isoscalar particles undergo mixing. The 16-plet of vector mesons is nearly ideally mixed [see appendix B for a discussion of the OZI rule in the case of SU(3) nonets] and the corresponding first order mass formulae for φ and ψ imply the further relation

$$C = (\psi - \rho)/(\varphi - \rho) . \tag{17.3}$$

The 16-plet of pseudoscalars is not ideally mixed – we do not discuss the SU(4) mass relations for η, η' and η_c here.

Consider first the *linear mass relations*. The observed masses $\Lambda_c(2273)$, $\Sigma_c(2430)$, $D^*(2007)$, $F^*(2140)$ and $\psi(3097)$ imply $C_{\Lambda_c} = 7.6$, $C_{\Sigma_c} = 5.9$, $C_{D^*} = 10.6$, $C_{F^*} = 10.8$ and $C_\psi \doteq 9.5$ respectively. The scattering of these values is consistent with what one should expect: the ratio of the largest to the smallest value is $C_{F^*} : C_{\Sigma_c} = 1.8$, comparable to the asymmetry in the vector coupling constants mentioned above. Using the average

$$C = 8.9^{+1.9}_{-3.0} \tag{17.4}$$

of the above five values and using the ratio $m_s : \hat{m} = 25$ the value $m_c\,(1\,\mathrm{GeV}) = 1.35 \pm 0.05\,\mathrm{GeV}$ leads to the following estimates for the light quark masses:

$$\hat{m}\,(1\,\mathrm{GeV}) = 6.3^{+3.4}_{-1.3}\,\mathrm{MeV} , \qquad m_s\,(1\,\mathrm{GeV}) = 160^{+80}_{-30}\,\mathrm{MeV} . \tag{17.5}$$

As mentioned above it does not make sense to use linear mass relations for the light pseudoscalars. Using *quadratic mass relations* rather than linear ones throughout we get $C_{\Lambda_c} = 11.8$, $C_{\Sigma_c} = 9.3$, $C_D = 15.3$, $C_F = 17.2$, $C_{D^*} = 17.7$, $C_{F^*} = 19.6$, $C_\psi = 20.5$. As was to be expected the scattering is substantially larger here (relative splittings in M^2 are twice as large as relative splittings in M). What is worse is that the value of C_ψ obtained from the quadratic mass relation differs from the one obtained from the linear mass formula by almost a factor of two. It is however not difficult to understand the origin of this discrepancy: the bulk of M_ψ comes from the rest energy of the c-quarks – if a mass formula neglects this contribution it is bound to fail. The first order SU(4) mass formula for the square of the mass of the 16-plet of vector mesons may be written as (OZI rule, ideal mixing):

$$M_{\mathrm{v}}^2 = \hat{A} + \hat{B}(m_1 + m_2) \tag{17.6}$$

where m_1 and m_2 are the masses of the two quarks contained in the state in question. The formula obviously neglects the term $(m_1 + m_2)^2$ that corresponds to the quark rest energy. In the case of light quark bound states the terms of order $(m_{\text{quark}})^2$ in the quark mass expansion of M^2 are small corrections, for heavy quarks they contain the bulk of the bound state mass. [For very heavy quarks M^2 is given by $(m_1 + m_2)^2$ up to effects of order $v^2/c^2 \sim \alpha_s^2$.]

The linear SU(4) mass formula

$$M_{\mathrm{v}} = a + b(m_1 + m_2) \tag{17.7}$$

does not have this obvious shortcoming: if b is close to 1 then the linear mass formula may be approximately valid even if one or both of the quarks are heavy. It is therefore not a priori unreasonable to use first order formulae linear in the mass for particles containing heavy quarks, but it is unreasonable to use quadratic first order formulae. In the preceding sections we have shown that in the case of light quark bound states crude estimates (additivity rule) for the terms of order $(m_{\text{quark}})^2$ tend to transform the quadratic mass formulae into the corresponding linear ones. One may of course apply the same analysis also to the SU(4) relations. Since one is however not looking at small corrections here the results are rather sensitive to the manner in which one analyzes the terms of order $(m_{\text{quark}})^2$ and it is not clear that one learns a great deal from this analysis. We find it more instructive to briefly discuss an extremely *simple ansatz* that allows one to understand the main features of the observed pattern of symmetry breaking even if the quark masses are not small [232]. For the Goldstone bosons we take a quadratic SU(4) mass formula of the type

$$M_{\mathrm{P}}^2 = \hat{A} + \hat{B}(m_1 + m_2) + \hat{C}(m_1 + m_2)^2 . \tag{17.8}$$

[We do not consider the isoscalar states η, η', η_c which are complicated by mixing.] Since the mass of the multiplet vanishes in the chiral limit, the constant $\hat{A}$ is zero. In the opposite extreme of very heavy quarks the mass of the bound state must tend to the sum of the quark masses. For the mass formula (17.8) to have this property the constant $\hat{C}$ must be equal to 1:

$$M_{\mathrm{P}}^2 = \hat{B}(m_1 + m_2) + (m_1 + m_2)^2 . \tag{17.9}$$

For the vector mesons we take the first order SU(4) formula (17.7) linear in M and again require that this formula has the right behaviour for very heavy quarks:

$$M_{\mathrm{v}} = M_0 + m_1 + m_2 . \tag{17.10}$$

[We could just as well also use a formula of the type (17.8) for the vector mesons; this would introduce an additional free parameter that could be tuned to get even better agreement with the observed spectrum. The linear formula (17.10) corresponds to the choice $\hat{A} = M_0^2$, $\hat{B} = 2M_0$, $\hat{C} = 1$ in (17.8).] We then note that the differences $M_\rho^2 - M_\pi^2 = 0.58\,\mathrm{GeV}^2$, $M_{K^*}^2 - M_K^2 = 0.55\,\mathrm{GeV}^2$, $M_{D^*}^2 - M_D^2 = 0.55\,\mathrm{GeV}^2$ are essentially flavour independent. The mass formulae (17.9) and (17.10) have this property, provided $\hat{B} = 2M_0$:

$$M_P^2 = 2M_0(m_1 + m_2) + (m_1 + m_2)^2\,. \qquad (17.11)$$

Apart from the quark masses m_u, m_d, m_s, m_c the mass formulae for the pseudoscalar and vector multiplets thus involve a single unknown, the mass M_0 of the vector mesons in the chiral limit, given by $M_0^2 = M_\rho^2 - M_\pi^2$. We may therefore determine the quark masses $\hat{m}$, m_s, m_c (recall that we disregard the mass difference $m_u - m_d$) from the mass formulae for π, K and D with the result

$$\hat{m} = 6.2\,\mathrm{MeV}\,, \qquad m_s = 140\,\mathrm{MeV}\,, \qquad m_c = 1250\,\mathrm{MeV}\,, \qquad m_s/\hat{m} = 22.7\,, \qquad C = 9.3\,. \qquad (17.12)$$

The linear mass formula for the vector mesons then of course reproduces the mass values of K* and D* rather accurately – this is built in. What is not trivial is that the simple linear formula (17.10) also gives acceptable values for the masses of $\varphi(1020)$, $\psi(3097)$ and F*(2140): $M_\varphi = 1044\,\mathrm{MeV}$, $M_\psi = 3256\,\mathrm{MeV}$, $M_{F^*} = 2150\,\mathrm{MeV}$.

The model is too crude to allow us to specify the precise normalization of the quark mass parameters in terms of the quark masses that appear in the Lagrangian. In view of this the fact that the absolute values of the masses one obtains from the model are consistent with the estimates for the running masses at a scale of order 1 GeV may not appear to be very significant. Note that in the linear mass formula (17.10) one might just as well think of some sort of constituent masses that remain finite in the chiral limit – what counts there is only the property that the quark mass differences have the right value. In the mass formula (17.11) for the Goldstone bosons it is however evident that the ansatz does not involve constituent masses: the mass formula can only be right if the quark masses that appear there measure the breaking of chiral symmetry. Since the algebraic structure in the chiral limit is correct we believe that the model does allow one to estimate quark mass *ratios* in a meaningful manner. The estimate for $m_s : \hat{m}$ is indeed essentially the one given in section 8. The value $C = 9.3$ based on the masses of ρ, π, K, D confirms the estimate $C = 8.9^{+1.9}_{-3.0}$ derived above from the masses of Λ_c, D* and ψ. If the value $C = 9.3$ is correct, then the absolute values of the running light quark masses at $\mu = 1\,\mathrm{GeV}$ follow from the measured quantity $m_c\,(1\,\mathrm{GeV}) = 1.35\,\mathrm{GeV}$ and the value $m_s : \hat{m} = 25$ given in section 13:

$$\hat{m}\,(1\,\mathrm{GeV}) = 6.0\,\mathrm{MeV}\,; \qquad m_s\,(1\,\mathrm{GeV}) = 150\,\mathrm{MeV}\,. \qquad (17.13)$$

Finally, the ratio $R = 43.5$ determines the masses m_u and m_d individually:

$$m_u\,(1\,\mathrm{GeV}) = 4.4\,\mathrm{MeV}\,, \qquad m_d\,(1\,\mathrm{GeV}) = 7.7\,\mathrm{MeV}\,. \qquad (17.14)$$

18. Quark masses from grand unified theories

In the preceding sections we have exclusively considered low energy phenomena and have exploited the fact that at low energies QCD + QED represents an accurate effective Lagrangian. In that Lagrangian the fermion masses are external parameters. To understand the size of these parameters one has to go beyond the effective low energy theory and *study the degrees of freedom that are frozen at low energies.*

One set of degrees of freedom that certainly plays a central role in the mass problem is known: the *electroweak gauge fields*. The electroweak interaction can be formulated in a consistent manner only if the Lagrangian of the theory does not contain fermion mass terms. In the standard framework based on SU(3) × SU(2) × U(1) the presence of mass terms in the low energy effective Lagrangian is associated with a spontaneous breakdown of the SU(2) gauge symmetry which gives mass not only to the fermions but at the same time also to the W and Z gauge fields. In fact, the electroweak SU(2) group cannot be a symmetry of the ground state if the dynamics of the strong interaction produces a fermion condensate: the vacuum expectation value $\langle 0|\bar{u}u|0\rangle = 2\langle 0|\bar{u}_{\mathrm{R}}u_{\mathrm{L}}|0\rangle$ necessarily breaks SU(2), because the left- and right-handed fields transform in a different manner under this group. It does however not seem to be possible to understand the observed mass scales on the basis of the SU(2) asymmetries due to the quark condensate alone. Other degrees of freedom than those contained in the observed fermions and the SU(3) × SU(2) × U(1) gauge fields seem to be involved in the mass problem.

The *Higgs model* demonstrates that there exist renormalizable Lagrangians which do reduce to the proper effective theory at low energies. It suffices to introduce a doublet of scalar fields with a suitable self interaction to get spontaneous breakdown of SU(2). The observed fermion mass matrix can be arranged by giving the Yukawa couplings of the scalar fields with the fermions suitable strength. In this form the Higgs model however only demonstrates that there exists a theoretically consistent framework which does reproduce the low energy phenomena – it does not shed any light on the fact that m_e is so different from m_μ or on the origin of the flavour asymmetries caused by $m_u < m_d < m_s < m_c < m_b < m_t$. These flavour asymmetries are introduced into the Lagrangian by hand in terms of widely different Yukawa coupling constants.

There are *other theoretical indications for the need of a larger framework*:

(i) The standard model does not explain why the proton and the electron have exactly opposite electric charge (quarks and leptons transform in a different manner under the U(1) group; the standard model does not require the eigenvalues of the U(1) generators to be related in such a manner that say the d quark carries one third of the electric charge of the electron).

(ii) In the standard model it is a mystery why there remained an appreciable number of protons, neutrons and electrons after the big bang was over; since the model separately conserves quark number and lepton number it calls for a tiny excess of quarks over antiquarks in the very beginning (alternatively, the universe must have managed to keep islands of quarks and antiquarks separated well enough for them to escape annihilation – this possibility looks even more mysterious).

(iii) The values of the three coupling constants in SU(3) × SU(2) × U(1) are arbitrary – theoretically, the model would also make sense if these constants were different from what is observed.

Grand unified theories [234, 235] propose a remarkably simple and beautiful solution to these puzzles. In the minimal model [235] the group SU(3) × SU(2) × U(1) is a low energy relic of the group SU(5) that unifies all known interactions with the exception of gravity in a gauge field theory containing a single coupling constant. Since the generator of the U(1) subgroup is among the generators of the simple group SU(5) its eigenvalues are integer multiples of the eigenvalues that occur in the fundamental representation. This explains why the electric charge is quantized. The irreducible representations $\underline{5}^*$ and $\underline{10}$ unify quarks and leptons; the interactions mediated by the gauge fields of SU(5) transform quarks into leptons and therefore allow antiprotons to convert into electrons. The conversion may be sufficiently inhibited at low energies to have escaped detection, but yet be sufficiently fast in the early hot phase to explain why a small *CP* violating component in the mass matrix will prevent the baryons from disappearing altogether after the universe had become too cold for baryon pair formation. For the model to be realistic it is essential that the SU(5) symmetry breaks down at a mass scale for which the value of the effective coupling constant $g^2/4\pi$ is close to $\frac{1}{42}$. This value appears to be too small for a spontaneous breakdown to occur within the theory itself. One may help it a little by introducing a scalar field ϕ that transforms according to the 24 dimensional adjoint representation of SU(5) and that has the proper self interaction to make it energetically favourable for the ground state to be asymmetric under SU(5), $\langle 0|\phi|0\rangle \neq 0$. This expectation value gives mass to the gauge fields X, Y that transform quarks into leptons (the fermions remain massless because the left-handed fields transform differently from the right-handed fields under the center of SU(5) – the Higgs fields are invariant under the center and hence cannot produce any left–right transitions). If at the scale M_X the effective coupling constant $g^2/4\pi$ is close to $\frac{1}{42}$ then one can understand why α_s, α and θ_W have the values observed at low energy: for the strong interaction to bind light quarks, to form a fermion condensate and all that the effective coupling constant α_s must be of order one. Since this value has to grow out of $\frac{1}{42}$ the size of the bound states produced by the strong interaction is larger than the Compton wave length of X and Y by about 14 orders of magnitude (refs. [236, 237]; see Antoniadis, Kounnas and Roiesnel [238] for more precise statements and references to the recent literature):

$$M_x \simeq 1.5 \times 10^{15} \Lambda_{\overline{MS}} . \tag{18.1}$$

The fine structure constant decreases during this huge interval from $\frac{3}{8} \cdot \frac{1}{42} = \frac{1}{112}$ to $\frac{1}{137}$ and $\sin^2 \theta_W$ decreases from the SU(5) Clebsch–Gordon value $\frac{3}{8}$ to 0.22.

What remains a mystery in this scenario is why the symmetry breakdown of SU(2) necessary if SU(3) is to produce a fermion condensate occurs at a mass scale M_W, $M_Z \sim 100$ GeV, 12 orders of magnitude smaller than M_X, but two orders of magnitude larger than the scale of the strong interaction bound states (Gildener [239]; see Ellis [240] for an excellent review of the problem).

Georgi and Glashow [235] proposed that the second breakdown is associated with a scalar Higgs field transforming according to the representation $\underline{5}$ that happens to get a relatively small vacuum expectation value which breaks SU(2) and leaves us with the effective low energy theory SU(3) × U(1). The W and Z gauge fields get the masses needed to reproduce the Fermi constant if the vacuum expectation value has the proper size and the fermions obtain masses in proportion to the Yukawa couplings. The symmetry group asserts that the Yukawa coupling constants are determined by two free parameters for each one of the three generations (more precisely, two for each pair of generations). The masses of the u, c and t quarks and the masses m_e, m_μ and m_τ may be used to tune these constants. The symmetry then predicts

$$m_d = m_e , \qquad m_s = m_\mu , \qquad m_b = m_\tau . \tag{18.2}$$

[Quite apart from the question of how to get the proper scale for M_W and M_Z any Higgs model which treats the three fermion generations as independent suffers from the defect mentioned above: some of the flavour asymmetries are still introduced by hand in the form of different values for the Yukawa coupling constants associated with the three generations.] The masses are renormalization dependent; since the leptons are not surrounded by gluons their masses change much less as the scale is changed. If relations like (18.2) hold at very short distances at which all degrees of freedom of SU(5) are in action ($\mu \gtrsim M_X$) then the b quark must be considerably heavier than the τ-lepton at low energies. In order of magnitude the observed ratio

$$m_b(m_b)/m_\tau = 2.38 \pm 0.06 \tag{18.3}$$

indeed agrees with the calculated renormalization effect in leading log approximation [241]. A more precise calculation of the renormalization effect appears to give a value for the ratio $m_b : m_\tau$ that is somewhat larger than what is observed [242, 243].

To analyze the relation for m_s we consider the ratio $m_b : m_s$ which in the minimal subtraction scheme is renormalization independent. With the value $m_b\,(1\,\text{GeV}) = 5.6 \pm 0.4\,\text{GeV}$ given in section 16 the SU(5) relation

$$m_s = \frac{m_\mu}{m_\tau}\, m_b \tag{18.4}$$

implies

$$m_s\,(1\,\text{GeV}) = 330 \pm 25\,\text{MeV} \tag{18.5}$$

a value that is substantially higher than the estimates given in the preceding sections. Note that with the ratio $m_s : \hat{m} = 25.0 \pm 2.5$ the above value corresponds to $\hat{m}\,(1\,\text{GeV}) = 13.2 \pm 1.8\,\text{MeV}$.

Finally, the relation

$$m_s/m_d = m_\mu/m_e = 206.8 \tag{18.6}$$

may be compared directly with the quark mass ratios obtained in sections 13 and 14 which imply

$$m_s/m_d = 19.6 \pm 1.6 \tag{18.7}$$

in obvious disagreement with (18.6). Since the masses m_d and m_e are very small it is conceivable that the relation (18.6) is affected by some small residual interaction e.g. due to a heavy fermion with a mass that may be as large as the Planck mass [244]. Georgi and Jarlskog [245] instead propose to replace the relations (18.2) by

$$m_d = 3m_e\,, \qquad m_s = \tfrac{1}{3} m_\mu\,, \qquad m_b = m_\tau \tag{18.8}$$

and show that these relations may be arranged together with the familiar formula for the Cabibbo angle [246, 247]

$$\mathrm{tg}^2\,\theta_c = m_d/m_s \tag{18.9}$$

if one considers a richer spectrum of Higgs particles and chooses the Yukawa couplings appropriately. [Note, however, that it is difficult to tune the Cabibbo angle "naturally"; see Gatto, Morchio and Strocchi [248]; Sartori [249] and the references quoted in these papers.] We do not discuss the theoretical status of the above relations [250–253], but simply compare them with the quark mass values obtained in the preceding sections. The prediction (18.5) is lowered by a factor of 3 and now gives $m_s\,(1\,\text{GeV}) = 110 \pm 8\,\text{MeV}$, which is somewhat low. The ratio $m_s : m_d$ is modified by a factor 9 to $m_s : m_d = 23$, which is somewhat high, but is not inconsistent with the measured value (18.7).

It is clear that the present understanding of the mass problem in the framework of grand unified theories does not give reliable predictions about the fermion masses. Many basic problems have yet to be solved: (i) It is not clear why the curvature of space-time is determined by the physical energy-momentum tensor, which measures the difference of the actual local energy density and the energy density of the asymmetric, physical ground state – in the absence of a principle that guarantees this the theory is likely to predict a cosmological constant of order GM_X^4. Unless gravity is modified well before the Planck length is reached it would generate curvature of order $(10^{-20}\,\mathrm{cm})^{-2}$ or more and ruin even the most beautiful scenario. (ii) A theory that embeds the effective low energy Lagrangian $\mathrm{SU}(3)\times\mathrm{U}(1)$ in a simple gauge group suffering spontaneous breakdown leads to heavy monopoles [254, 255]. The physical significance of these exotic objects is not understood. (iii) It is not clear that all the fields appearing in the effective low energy Lagrangian are in fact still elementary at distances of order M_X^{-1} – they may turn out to be composite already at a much larger scale. In particular, the fields responsible for the spontaneous breakdown of SU(2) may have an interesting life of their own with a characteristic scale perhaps of order 1 TeV (see e.g. refs. [256–261]). (iv) The occurrence of two scales M_W, M_X is difficult to understand in terms of a hand made Higgs potential. Whether a consistent realistic theory emerges if one associates some of the forces responsible for symmetry breakdown with radiative corrections [262, 264] remains to be seen. (v) Radiative corrections may also be at the origin of some of the fermion masses [263]. The smallness of ratios like $m_e : m_\mu$ or $m_u : m_c$ suggests that only some of the fermions get their mass directly from the same source as W and Z. At any rate a solution of the generation problem is not in sight.

The idea that the asymmetries in the effective low energy Lagrangian are not relics of an asymmetry in the underlying theory, but are there as a consequence of the fact that the state of lowest energy happens not to be symmetric is too beautiful to be wrong – the available partial realizations of this idea provide us with instructive models showing how this could happen; if these models are however equipped with all the parameters necessary to reproduce the proper low energy theory they become too ugly to be right.

19. Summary and conclusions

1. At low energies ($\mu \ll 100$ GeV) all degrees of freedom except the quarks, the gluons, the leptons, the photon and gravity are frozen. Low energy phenomenology is consistent with the standard picture according to which QCD + QED is a very accurate effective low energy theory (unless one considers very large systems for which gravity cannot be neglected).

2. The effective low energy theory involves two coupling constants g, e and the mass matrices of the quarks and leptons as external parameters. The theory is renormalizable and may be characterized by the on-shell values of e, m_{lept} and by the running coupling constant g and the running quark mass matrix at scale μ.

3. We have reviewed the information about the eigenvalues m_u, m_d, m_s, m_c, m_b of the quark mass matrix. We use the $\overline{MS}$ *scheme* and normalize at the running scale $\mu = 1$ GeV. [Since the renormalization group invariant scale Λ is small, a change in μ produces only a small change in the value of $m(\mu)$: for $\Lambda < 200$ MeV the value of $m(\mu)$ differs from m (1 GeV) by less than 10% if $0.80\,\mathrm{GeV} < \mu < 1.3\,\mathrm{GeV}$, compare table 1.] In contrast to the value of the running mass at a given scale the renormalization group invariant mass $\bar{m}$ is sensitive to the value of Λ which is not known to sufficient accuracy. [If we wanted to quote renormalization group invariant masses we would always have to give several numbers for a representative set of values of Λ.]

4. To first order in e^2 the *effective Lagrangian* describing the properties of hadronic bound states is given by

$$\mathscr{L}_{\mathrm{eff}} = \mathscr{L}_{\mathrm{QCD}} - \frac{e^2}{2}\int \mathrm{d}^4x\, D_\lambda(x-y)\, Tj^\mu(x)\, j_\mu(y) + \Delta E\mathbb{1} + \sum_q \Delta m_q\, \bar{q}q - \frac{\Delta g}{2g^3} F^a_{\mu\nu}F^{\mu\nu a}\,. \tag{19.1}$$

The interaction due to photon exchange is ultraviolet divergent and must be *renormalized.* As it is familiar from QED the counter terms ΔE, Δm_q, Δg are not unique – the total Hamiltonian of QCD + QED is uniquely defined by the constants e, m_{lept}, $g(\mu)$, $m_{\text{quark}}(\mu)$, the splitting into pure QCD and an effective electromagnetic interaction is not. Numerically, the inherent ambiguity in the notion of a quark mass in pure QCD is however not significant: A change of the renormalization point that defines the renormalized electromagnetic interaction by a factor two affects the quark mass only by about 1‰ for charge $\frac{2}{3}$ and by $\frac{1}{4}$‰ for charge $\frac{1}{3}$ – this is far beyond the accuracy to which quark masses can be determined at present.

5. In the limiting case of a theory involving only *massless quarks* the counter term Δm_q is absent. Apart from the renormalization ΔE of the vacuum energy the divergent piece of the electromagnetic interaction is absorbed in a suitable shift of the strong coupling constant, i.e. in a change of the renormalization group invariant scale Λ. A change in the scale defining the renormalized electromagnetic self-energy amounts to a *universal multiplicative renormalization of all physical masses* in pure QCD (see section 11). A change in the scale by a factor two changes the mass of the proton in pure QCD by 0.06 MeV. If m_u and m_d vanish all *isospin breaking mass differences are finite* and may be worked out on the basis of the unrenormalized Cottingham formula.

6. In the real world with $m_u \neq m_d$ the Cottingham formula for isospin breaking mass differences such as $M_n - M_p$ must be renormalized. The fact that the divergence only has a tiny coefficient proportional to αm_u, αm_d is consistent with the analysis of the Cottingham formula: the contributions from the deep inelastic region that are responsible for the divergence are indeed invisibly small.

7. We have reanalyzed the *electromagnetic self-energies* of the qqq and $q\bar{q}$ ground states. The resulting values for the isospin breaking mass differences in pure QCD are given in table 3. There is an important check: in pure QCD the mass differences $\Sigma^+ + \Sigma^- - 2\Sigma^0$ and $\pi^+ - \pi^0$ vanish up to terms of order $(m_u - m_d)^2$ [$\Sigma^0 - \Lambda$ and $\pi^0 - \eta$ mixing and higher order terms in the quark mass expansion produce a contribution estimated at $(\Sigma^+ + \Sigma^- - 2\Sigma^0)_{\text{QCD}} = \pm 0.02$ MeV, $(\pi^+ - \pi^0)_{\text{QCD}} = \pm 0.1$ MeV]. The calculated electromagnetic contributions to these mass differences indeed agree with the observed values within the errors of the calculation.

8. In pure QCD the masses of the bound states are determined by Λ and by the quark masses: $M_n(\Lambda, m_u, m_d, m_s, m_c, \ldots)$. Since the masses of the light quarks u, d, s turn out to be small in comparison to their typical kinetic energy it makes sense to expand the mass of the bound states in powers of m_u, m_d, m_s, keeping $m_c, m_b, \ldots$ fixed at their physical values. For the mass of the proton, e.g., the *quark mass expansion* is of the form

$$M_p = a + b m_{\text{quark}} + \cdots \tag{19.2}$$

(the first order expansion for M_p^2 is of the same form – first order mass formulae do not distinguish between M and M^2). In the case of the pseudoscalar octet the mass vanishes in the chiral limit $m_u = m_d = m_s = 0$. The expansion for M_π takes the form

$$M_\pi^2 = B m_{\text{quark}} + \cdots \tag{19.3}$$

(a quark mass expansion holds only for M_π^2; the quantity M_π has a square root singularity at $m_{\text{quark}} = 0$).

First order mass formulae such as (19.2) and (19.3) indeed give a satisfactory account of the SU(3) asymmetries in the bound state masses due to $m_u < m_d < m_s$. In particular, the first order mass formulae imply the Gell-Mann–Okubo relations which are very well satisfied. Furthermore, the ratio

$$R = (m_s - \hat{m})/(m_d - m_u)\,; \qquad \hat{m} = \tfrac{1}{2}(m_u + m_d) \tag{19.4}$$

determines the size of the SU(2) splittings produced by $m_u \neq m_d$ in comparison with the SU(3) splittings generated by $m_s \neq \hat{m}$: if R is known then the mass differences $p - n$, $\Sigma^+ - \Sigma^-$, $\Xi^0 - \Xi^-$ and $K^+ - K^0$ can be calculated from the observed SU(3) splittings $\Xi - N$, $\Sigma - \Lambda$ and $K^2 - \pi^2$. The value of R that fits $K^+ - K^0$ roughly agrees with $\Sigma^+ - \Sigma^-$ and $\Xi^0 - \Xi^-$. This value however leads to a $p - n$ mass difference that is considerably larger than what is observed. Quite apart from the need to resolve this discrepancy one has to study the *higher order terms in the quark mass expansion* to see to what extent first order mass formulae can be trusted.

9. As pointed out by Li and Pagels the quark mass expansion is not a simple Taylor series – the terms neglected in (19.2) and (19.3) are not of order $(m_{\text{quark}})^2$. In the chiral limit around which one is expanding the theory contains massless physical particles (Goldstone bosons) which generate *infrared singularities.* The coefficients in the formal Taylor expansion are infrared divergent. The proper quark mass expansion for the proton mass e.g. has the structure

$$M_p = a + bm_{\text{quark}} + c(m_{\text{quark}})^{3/2} + \cdots . \tag{19.5}$$

The size of the *nonanalytic term* is determined by current algebra in terms of f_π, g_A, M_π, M_K and M_η. Numerically, the nonanalytic term is a disaster: it is as big as the first order term and there is every reason to expect that the higher order terms in the quark mass expansion are as big as the term retained in (19.5). If one uses the quark mass expansion in this raw form (chiral perturbation theory) one does not get meaningful results. To obtain a reliable approximation scheme *chiral perturbation theory must be improved* by reordering the expansion and summing up leading infrared singularities that occur to all orders in the quark mass (see appendix C). A simple coherent method that achieves this is based on the notion of an effective chiral Lagrangian. As long as the effective Lagrangian gives the particles the proper masses and satisfies the soft-pion theorems it also produces the correct leading nonanalytic terms in the quark mass expansion. To purify the mass of the proton of its leading nonanalytic terms it suffices to subtract the self-energy diagram corresponding to a suitable effective Lagrangian. If the *nonanalytic terms* are handled in this manner one finds that the corrections to the first order mass formulae *are indeed small.* They are nevertheless significant as they account for the flavour asymmetries in the mass spectrum caused by the fact that different members of a multiplet get an unequal share of meson cloud energy. In fact, the corrections required by improved chiral perturbation theory not only lead to a perfectly coherent approximation scheme, but explain why earlier determinations of the ratio R did not give a consistent picture.

10. Apart from the nonanalytic contributions the quark mass expansion also contains regular higher order terms, present even if the masses of the bound states were analytic in m_u, m_d, m_s. We give simple estimates of these contributions which show up e.g. in the difference between the mass formulae for M and for M^2.

11. Armed with these estimates of the higher order terms in the quark mass expansion we then analyze the two *quark mass ratios* $m_u : m_d : m_s$ on the basis of the observed meson and baryon masses. We determine the ratio R on the basis of five independent manifestations of isospin breaking ($K^+ - K^0$, $p - n$, $\Sigma^+ - \Sigma^-$, $\Xi^0 - \Xi^-$ and $\rho - \omega$ mixing, see table 4). The results cluster around $R = 43.5$ which is within the estimated uncertainty of each one of the five values. Treating these values as independent measurements we obtain

$$R = 43.5 \pm 2.2 \tag{19.6}$$

(the individual error bars range from ±4 to ±10). Note that the nonanalytic terms in the quark mass expansion produce a substantial correction to the R-value obtained from $p-n$; the value extracted from the bare first order mass formula is inconsistent with (19.6). The information on the ratio R derived from ψ' decay and from the $D^+ - D^0$ mass difference is not inconsistent with the value 43.5, but is too uncertain to give comparable accuracy. We do not attempt to extract a value of R from the decay $\eta \to 3\pi$, because the theoretical prediction for the rate of the decay $\eta \to 2\gamma$ (which normalizes the other experimental rates) disagrees with the available data.

The value of the ratio $m_s : \hat{m}$ may be extracted from the observed masses of π, K and η. On the basis of an analysis of the corresponding higher order terms in the quark mass expansion we obtain the value

$$m_s/\hat{m} = 25.0 \pm 2.5 . \tag{19.7}$$

The above results for R and $m_s : \hat{m}$ imply the following values for related quark mass ratios:

$$\begin{aligned} &m_d/m_u = 1.76 \pm 0.13 , && m_u/\hat{m} = 0.72 \pm 0.03 \\ &m_s/m_d = 19.6 \pm 1.6 , && m_d/\hat{m} = 1.28 \pm 0.03 \\ &m_s/m_u = 34.5 \pm 5.1 , && (m_u - m_d)/(m_u + m_d) = -0.28 \pm 0.03 . \end{aligned} \tag{19.8}$$

[Note that if the u quark mass would vanish the two ratios $m_s : \hat{m}$ and R would be related by $m_s : \hat{m} = 1 + 2R$, in total disagreement with the above values which give 25.0 ± 2.5 and 88.0 ± 4.4 for the left- and right-hand sides of this relation.]

12. The baryon masses also provide us with an estimate of the *σ-term in pion–nucleon scattering*. It turns out that the higher order terms in the quark mass expansion increase the value of the σ-term by about 10 MeV. The result is

$$\sigma = \frac{1}{1-y}\,\sigma_0 , \qquad \sigma_0 = 35 \pm 5\,\text{MeV} \tag{19.9}$$

where y measures the proton expectation value of the operator $\bar{s}s$:

$$y = \frac{2\langle p|\bar{s}s|p\rangle}{\langle p|\bar{u}u + \bar{d}d|p\rangle} . \tag{19.10}$$

The value of y is not known; it is correlated with the value M_0 of the nucleon mass in the chiral limit [$y = 0$ corresponds to $M_0 = 870$ MeV, for $y = 0.2$ one finds $M_0 = 670$ MeV]. We consider the estimate $y < 0.2$ which leads to $\sigma < 50$ MeV as very conservative. The observed value $\sigma(2M_\pi^2) = 65$ MeV is not consistent with our analysis (for details see appendix D, where we also discuss isospin breaking in the σ-terms).

13. The *absolute value of the light quark masses* is not known very accurately. We have performed a new evaluation of the *QCD sum rules* for the axial divergence and show that the sum rules are consistent with the following running masses at scale $\mu = 1$ GeV:

$$\hat{m}\,(1\,\text{GeV}) = 7 \pm 2\,\text{MeV} , \qquad m_s\,(1\,\text{GeV}) = 180 \pm 50\,\text{MeV} \tag{19.11}$$

[the quantity $\hat{m}(1\,\text{GeV})$ stands for $\frac{1}{2}\{m_u(1\,\text{GeV}) + m_d(1\,\text{GeV})\}$]. Smaller values of $\hat{m}$, m_s are inconsistent with the bounds given by Narison and de Rafael, larger values call for an exceedingly large spectral function in the intermediate energy region between 1 GeV and 2 GeV which would represent a remarkable phenomenon in itself.

The evaluation of other QCD sum rules that provide information about light quark masses is subject to larger uncertainties, because there is less information about the behaviour of the spectral function than in the case of the axial divergence. To within the rather large error bars the resulting quark mass estimates are consistent with the above values.

14. With the values for the two ratios $m_u : m_d : m_s$ given above the estimate $\hat{m}$ (1 GeV) = (7 ± 2) MeV amounts to the following *results for* m_u, m_d *and* m_s:

$$\begin{aligned} m_u\,(1\,\mathrm{GeV}) &= 5.1 \pm 1.5\,\mathrm{MeV} \\ m_d\,(1\,\mathrm{GeV}) &= 8.9 \pm 2.6\,\mathrm{MeV} \\ m_s\,(1\,\mathrm{GeV}) &= 175 \pm 55\,\mathrm{MeV}\,. \end{aligned} \tag{19.12}$$

The corresponding *renormalization group invariant masses* may be read off from table 1 (the value of Λ refers to the $\overline{\mathrm{MS}}$ scheme with 3 flavours):

$$\begin{array}{ll} \Lambda = 100\,\mathrm{MeV} & \Lambda = 200\,\mathrm{MeV} \\ \bar{m}_u = 7.6 \pm 2.2\,\mathrm{MeV} & \bar{m}_u = 6.3 \pm 1.9\,\mathrm{MeV} \\ \bar{m}_d = 13.3 \pm 3.9\,\mathrm{MeV} & \bar{m}_d = 11.0 \pm 3.2\,\mathrm{MeV} \\ \bar{m}_s = 260 \pm 80\,\mathrm{MeV} & \bar{m}_s = 215 \pm 65\,\mathrm{MeV}\,. \end{array} \tag{19.13}$$

15. According to Gell-Mann, Oakes and Renner the pion mass is determined by the product of the quark mass $\hat{m}$ and the *order parameter* $\langle 0|\bar{u}u|0\rangle = \langle 0|\bar{d}d|0\rangle$:

$$M_\pi^2 f_\pi^2 = -4\hat{m}\langle 0|\bar{u}u|0\rangle + \cdots\,. \tag{19.14}$$

The measured values of M_π and f_π therefore fix the order parameter once the quark mass $\hat{m}$ is known. With the value $\hat{m}$ (1 GeV) = 7 ± 2 MeV we obtain

$$\begin{aligned} \langle 0|\bar{u}u|0\rangle &= -(1.1 \pm 0.3) \times 10^{-2}\,\mathrm{GeV}^3 \\ &= -(225 \pm 25\,\mathrm{MeV})^3\,. \end{aligned} \tag{19.15}$$

16. The masses of the *c and b quarks* are known much more precisely than the masses of u, d and s. The values obtained from the QCD sum rules involving the electromagnetic current may be written in the form

$$\begin{aligned} m_c(m_c) &= 1.27 \pm 0.05\,\mathrm{GeV} \\ m_b(m_b) &= 4.25 \pm 0.10\,\mathrm{GeV}\,. \end{aligned} \tag{19.16}$$

[The function $m_c(\mu)$ is the running c quark mass, the quantity $m_c(m_c)$ is the solution to the equation $m_c(\mu) = \mu$.] The corresponding values at scale μ = 1 GeV are

$$\begin{aligned} m_c\,(1\,\mathrm{GeV}) &= 1.35 \pm 0.05\,\mathrm{GeV} \\ m_b\,(1\,\mathrm{GeV}) &= \begin{cases} 5.3 \pm 0.1\,\mathrm{GeV}\,; & \Lambda = 100\,\mathrm{MeV} \\ 5.9 \pm 0.1\,\mathrm{GeV}\,; & \Lambda = 200\,\mathrm{MeV}\,. \end{cases} \end{aligned} \tag{19.17}$$

17. For very heavy quarks the perturbative notion of a *dressed quark mass* M_q (the mass that appears in the Balmer formula) is a meaningful concept. The information about the value of M_b is reviewed in section 16 – the nonperturbative effects in bound states containing c quarks are presumably too large for the perturbative notion of a dressed c quark mass to be useful.

18. In contrast to m_u, m_d, m_s the mass of the c-quark is not small in comparison with the characteristic scale of bound states. One therefore has no right to expect the *SU*(4) *mass relations* (which are obtained by treating $m_c - \hat{m}$ as a perturbation) to be reliable. An analysis of these mass relations nevertheless leads to a remarkably coherent picture which allows one to extract an estimate of the ratio

$$C = (m_c - \hat{m})/(m_s - \hat{m}) \tag{19.18}$$

that measures the size of SU(4) splittings in comparison to SU(3) splittings:

$$C = 9 \pm 2 \,. \tag{19.19}$$

[It is difficult to assess the systematic error of the analysis – we think that the error bars given in (19.19) represent a realistic estimate for the range of C values that are consistent with broken SU(4).] With the measured value of the running c-quark mass at 1 GeV and with the ratios $m_u : m_d : m_s$ given above the estimate (19.19) implies the following values for the running light quark masses at 1 GeV:

$$\begin{aligned} &\hat{m}\,(1\,\text{GeV}) = 6.2 \pm 1.9\,\text{MeV}\,, \qquad && m_d\,(1\,\text{GeV}) = 7.9 \pm 2.4\,\text{MeV}\,, \\ &m_u\,(1\,\text{GeV}) = 4.5 \pm 1.4\,\text{MeV}\,, && m_s\,(1\,\text{GeV}) = 155 \pm 50\,\text{MeV}\,. \end{aligned} \tag{19.20}$$

These values are perfectly consistent with those obtained from QCD sum rules.

19. The recent developments in the numerical evaluation of QCD correlation functions on a *lattice* (for an excellent review see Hasenfratz [265, 266]) have made it possible to calculate observable quantities in terms of the parameters of the Lagrangian. The numerical results [267–271] are impressive. The evidence given for spontaneous breakdown of chiral symmetry is rather convincing. The numerical values for the order parameter $\langle 0|\bar{q}q|0\rangle$ and for the quark masses however have to be taken with some caution, because the lattice regularization violates chiral symmetry ab initio; a more detailed analysis is required to clearly separate spontaneous chiral symmetry breaking from the residual symmetry breaking effects due to the regularization. It is not a trivial matter to connect the lattice quark mass parameters with the running quark masses of the continuum theory. (The quark mass values given by different groups differ by a factor two.) Mass ratios are scale independent and may be less sensitive to the uncertainties of the method. If the ratios of the lattice quark mass parameters given by Hasenfratz et al. ($\hat{m} : m_s : m_c \simeq 5.6 : 144 : 1213$) can be taken as a measurement of the ratios of the masses that occur in the Lagrangian of QCD then the known value (19.17) of the running c-quark mass at 1 GeV implies $\hat{m}$ (1 GeV) = 6.2 MeV, m_s (1 GeV) = 160 MeV, in remarkable agreement with the values given above (The two ratios of lattice quark masses amount to $m_s : \hat{m} \simeq 26$, $C \simeq 8.7$.) At any rate the numbers given for the light quark masses confirm the standard picture underlying this review and it may well turn out that in the future lattice calculations will provide us with the most accurate method to measure the parameters of the QCD Lagrangian.

20. Many interesting attempts to find a nonperturbative approximation scheme for QCD that does not rely on computer-based numerical methods have been made. Some of these *dynamical schemes* are indeed consistent with a spontaneously broken chiral symmetry and a non-vanishing vacuum expectation value of $\bar{q}q$ whose size can be estimated (see e.g. refs. [272–277]). The resulting quark mass estimates agree with the values given above to within a factor of two. More work is needed to establish that massless QCD indeed implies a nonvanishing value of $\langle 0|\bar{q}q|0\rangle$. In fact, the literature also contains a considerable number of papers that are based on the (explicit or implicit) assumption that the vacuum expectation value of $\bar{q}q$ vanishes in the chiral limit (see appendix E). Even if this alternative is theoretically not very attractive we emphasize that it is not ruled out on phenomenological grounds. A clean calculation of the order parameter $\langle 0|\bar{q}q|0\rangle$ in massless QCD would decide whether or not QCD indeed realizes a spontaneously broken chiral symmetry in the form proposed by Gell-Mann, Oakes, Renner, Glashow and Weinberg, and would at the same time provide us with an approximate value for the mean mass of the u and d quarks if the answer is affirmative.

21. Like the lepton masses the quark masses are scattered over more than three orders of magnitude. Leaving the neutrinos aside the first generation (e, u, d) populates the interval from 0.5 MeV to 10 MeV, the second (μ, c, s) covers the range from 100 MeV to 1.4 GeV, the third (τ, t, b) extends from 1.7 GeV to the mass of the t quark whose value is not known yet. The leptons are lighter than the quarks of the same generation. Otherwise the distribution of the masses within the generations does however not indicate any conspicuous systematic features [normalizing the running quark masses at 1 GeV the ratios are $(m_e : m_u : m_d) \simeq (1:9:16)$; $(m_\mu : m_c : m_s) \simeq (1:13:1.5)$; $(m_\tau : m_t : m_b) \simeq (1:?:3.1)$]. The order of magnitude of the ratio $m_b : m_\tau$ may be understood on the basis of SU(5) *grand unification*. The other ratios can be accommodated, but there are no convincing arguments to show why they could not just as well have any other random values – the origin of the observed pattern of flavour symmetry breaking remains mysterious.

Acknowledgements

It is a pleasure to acknowledge informative discussions with M. Böhm, I. Files, H. Fritzsch, F. Jegerlehner, S. Mallik, E. de Rafael, M. Roos, J. Stern, T.N. Truong, D. Wyler and with our colleagues at the institute, in particular with P. Minkowski. We have made a serious effort to give references to the literature, but we are aware of the fact that our list is unlikely to be complete, particularly since it is long. We apologize for the omissions. (The material contained in sections 15, 17 and 18 only concerns those aspects of QCD sum rules, SU(4) symmetry and grand unified theories that are of immediate relevance to this review; the references were selected accordingly.)

Appendix A. Other multiplets: 3, 3*, 6 and 10

A.1. SU(3) symmetry

Mesons consisting of one heavy quark and one light quark and baryons containing two heavy quarks transform according to the representations $\underline{3}$ or $\underline{3}^*$. For baryons with one heavy and two light quarks both representations $\underline{3}^*$ and $\underline{6}$ are possible. The matrix elements of the operator $\bar{q}q$ between states transforming according to the representations $\underline{3}$, $\underline{3}^*$, $\underline{6}$ or $\underline{10}$ are determined by two constants.

In the following we exhibit the mass formulae for the multiplets $\underline{3}$, $\underline{3}^*$, $\underline{6}$ and $\underline{10}$ in a somewhat condensed form. We shall assume that the unperturbed multiplets $(m_u = m_d = m_s = 0)$ are well separated from one another, such that the mixing phenomenon discussed in appendix B is absent. We denote the matrix elements of the quark bilinears in the state with the largest value of isospin T, $T_3 = +T$ by B^{u}, B^{d}, B^{s}:

$$B^{\mathrm{u}} = \langle T_{\max}, T_3 = T_{\max} | \bar{u}u | T_3 = T_{\max}, T_{\max} \rangle \qquad \text{etc}\,.$$

Then we find the following first order quark mass expansions:

Triplet

Baryons	Mesons
$\left\{\begin{matrix} \mathrm{uhh, dhh, shh} \\ \Xi_{\mathrm{cc}}^{++}, \Xi_{\mathrm{cc}}^{+}, \Omega_{\mathrm{cc}}^{+} \end{matrix}\right\}$	$\left\{\begin{matrix} \mathrm{u\bar{h}, d\bar{h}, s\bar{h}} \\ \bar{\mathrm{D}}^0, \mathrm{D}^-, \mathrm{F}^- \end{matrix}\right\}$

$$M^2 = A_3 + \tfrac{1}{3}(m_{\mathrm{u}} + m_{\mathrm{d}} + m_{\mathrm{s}})(B_3^{\mathrm{u}} + 2B_3^{\mathrm{d}}) + 2(B_3^{\mathrm{u}} - B_3^{\mathrm{d}})[\tfrac{1}{2}(m_{\mathrm{u}} - m_{\mathrm{d}})T_3 + (\hat{m} - m_{\mathrm{s}})(T - \tfrac{1}{3})]$$

where we used

$$B_3^{\mathrm{s}} = B_3^{\mathrm{d}}\,.$$

Antitriplet

Baryons	Mesons
$\left\{\begin{matrix} \mathrm{ush, dsh, udh} \\ \Xi_{\mathrm{c}}^{\mathrm{A}+}, \Xi_{\mathrm{c}}^{\mathrm{A}0}, \Lambda_{\mathrm{c}}^{+} \end{matrix}\right\}$	$\left\{\begin{matrix} \mathrm{h\bar{d}, h\bar{u}, h\bar{s}} \\ \mathrm{D}^+, \mathrm{D}^0, \mathrm{F}^+ \end{matrix}\right\}$

$$M^2 = A_{3^*} + \tfrac{1}{3}(m_{\mathrm{u}} + m_{\mathrm{d}} + m_{\mathrm{s}})(B_{3^*}^{\mathrm{d}} + 2B_{3^*}^{\mathrm{u}}) + 2(B_{3^*}^{\mathrm{u}} - B_{3^*}^{\mathrm{d}})[\tfrac{1}{2}(m_{\mathrm{u}} - m_{\mathrm{d}})T_3 - (\hat{m} - m_{\mathrm{s}})(T - \tfrac{1}{3})]$$

with

$$B_{3^*}^{\mathrm{s}} = B_{3^*}^{\mathrm{u}}\,.$$

Sextet

$$\left\{\begin{matrix} \mathrm{uuh, udh, ddh, ush, dsh, ssh} \\ \Sigma_{\mathrm{c}}^{++}, \Sigma_{\mathrm{c}}^{+}, \Sigma_{\mathrm{c}}^{0}, \Xi_{\mathrm{c}}^{+}, \Xi_{\mathrm{c}}^{0}, \Omega_{\mathrm{c}} \end{matrix}\right\}$$

$$M^2 = A_6 + \tfrac{1}{3}(m_{\mathrm{u}} + m_{\mathrm{d}} + m_{\mathrm{s}})(B_6^{\mathrm{u}} + 2B_6^{\mathrm{d}}) + (B_6^{\mathrm{u}} - B_6^{\mathrm{d}})[\tfrac{1}{2}(m_{\mathrm{u}} - m_{\mathrm{d}})T_3 + (\hat{m} - m_{\mathrm{s}})(T - \tfrac{2}{3})]$$

with

$$B_6^{\mathrm{s}} = B_6^{\mathrm{d}}\,.$$

Decuplet

$$\begin{Bmatrix} \text{uuu, uud, udd, ddd, uus, uds, dds, uss, dss, sss} \\ \Delta^{++}, \Delta^{+}, \Delta^{0}, \Delta^{-}, \Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}, \Xi^{*0}, \Xi^{*-}, \Omega^{-} \end{Bmatrix}$$

$$M^2 = A_{10} + \tfrac{1}{3}(m_u + m_d + m_s)(B^u_{10} + 2B^d_{10}) + \tfrac{2}{3}(B^u_{10} - B^d_{10})[\tfrac{1}{2}(m_u - m_d)T_3 + (\hat{m} - m_s)(T-1)]$$

with

$$B^s_{10} = B^d_{10}\,. \tag{A.1}$$

For the multiplets $\underline{3}$, $\underline{3}^*$, $\underline{6}$ and $\underline{10}$ SU(3)-symmetry amounts to quark counting (equal spacing rule): for these multiplets the SU(3) mass formula may be written as

$$M^2 = a + b(N_u m_u + N_d m_d + N_s m_s)$$

where N_u is the number of up quarks in the state in question.

A.2. *SU*(4) *symmetry*

In the limit $m_u = m_d = m_s = m_c = 0$ the ground state baryons constitute a 20-dimensional representation of SU(4) and the meson multiplets transform according to $\underline{15} \oplus \underline{1}$. If the quark mass expansion of the proton mass in powers of m_u, m_d, m_s and m_c is known, one can calculate the expansion coefficients for all members of the $\underline{20} = \underline{8} \oplus \underline{6} \oplus \underline{3} \oplus \underline{3}^*$. The coefficients which occur in the quark mass expansion of $\underline{6}$, $\underline{3}$ and $\underline{3}^*$ with respect to m_u, m_d and m_s may then be given in terms of the quark mass expansion of the proton.

In fact let

$$M^2_p = \hat{A} + \hat{B}^u m_u + \hat{B}^d m_d + \hat{B}^s m_s + \hat{B}^c m_c$$

$$\hat{B}^u = \langle p|\bar{u}u|p\rangle \qquad \text{etc}\,.$$

$$\hat{B}^c = \hat{B}^s\,.$$

Then one has the following relations (valid to leading order in an expansion in m_c):

$$\begin{array}{lll} A_3 = \hat{A} + m_c\hat{B}^u\,; & A_{3^*} = \hat{A} + \tfrac{1}{3}m_c(2\hat{B}^s - \hat{B}^d + 2\hat{B}^u)\,; & A_6 = \hat{A} + m_c\hat{B}^d \\ B^u_3 = \hat{B}^d\,; & B^u_{3^*} = B^s_{3^*} = \tfrac{1}{6}(\hat{B}^s + 4\hat{B}^d + \hat{B}^u)\,; & B^u_6 = \hat{B}^u \\ B^d_3 = B^s_3 = \hat{B}^s\,; & B^d_{3^*} = \hat{B}^s\,; & B^d_6 = B^s_6 = \hat{B}^s\,. \end{array}$$

Equivalently, one may express the masses of the states in $\underline{3}$, $\underline{3}^*$, $\underline{6}$ and $\underline{8}$ in terms of the mass of the $T = 0$ state in the octet, M^2_Λ, and the F- and D-couplings of the octet

$$F = \tfrac{1}{3}(\hat{m} - m_s)(\hat{B}^u - \hat{B}^s) = \tfrac{1}{2}(M^2_N - M^2_\Xi)$$

$$D = \tfrac{1}{2}(\hat{m} - m_s)(\hat{B}^u + \hat{B}^s - 2\hat{B}^d) = \tfrac{3}{4}(M^2_\Sigma - M^2_\Lambda)\,.$$

We summarize the resulting expansion below [278]:

Triplet

$$M_3^2 = M_\Lambda^2 - 2\frac{m_c - \hat{m}}{m_s - \hat{m}} F + \tfrac{4}{3}D + (F - D)\left\{\frac{m_d - m_u}{m_s - \hat{m}} T_3 + 2T\right\}.$$

Antitriplet

$$M_{3^*}^2 = M_\Lambda^2 + \left(1 - \frac{m_c - \hat{m}}{m_s - \hat{m}}\right)(F + \tfrac{1}{3}D) + (F - \tfrac{2}{3}D)\left(\frac{m_d - m_u}{m_s - \hat{m}} T_3 - 2T\right)$$

Sextet

$$M_6^2 = M_\Lambda^2 - \left(1 + \frac{m_c - \hat{m}}{m_s - \hat{m}}\right)F + \frac{1}{3}\left(1 + 3\frac{m_c - \hat{m}}{m_s - \hat{m}}\right)D + F\left\{\frac{m_d - m_u}{m_s - \hat{m}} T_3 + 2T\right\}$$

Octet

$$M_8^2 = M_\Lambda^2 + FY + \tfrac{2}{3}D\left[T(T+1) - \frac{Y^2}{4}\right] + \frac{m_d - m_u}{m_s - \hat{m}}\{F + DY\}T_3 . \tag{A.2}$$

The quark mass differences $\hat{m} - m_s$ and $m_u - m_d$ induce mixing in zeroth order between states carrying the same electric charge, strangeness and charm quantum numbers. (i) The mass difference $\hat{m} - m_s$ mixes the isospin doublets (usc, dsc) that occur in the representations $\underline{3}^*$ and $\underline{6}$. The mixing angle is of order $(m_s - \hat{m}):(m_c - \hat{m})$. (ii) The isospin violating mass difference $m_u - m_d$ generates mixing of the two octet states Σ^0, Λ. The mixing angle is of order $(m_d - m_u):(m_s - \hat{m})$ [see section 9]. (iii) The mass difference $m_u - m_d$ also mixes the isosinglet udc in the representation $\underline{3}^*$ with the isovector state udc in the representation $\underline{6}$. The mixing angle is of order $(m_d - m_u):(m_c - \hat{m})$. [$m_u - m_d$ also slightly modifies the mixing among the isospin doublets mentioned above.]

For these 8 states the mass formula (A.2) therefore only holds up to terms of order $(m_s - \hat{m})^2:(m_c - \hat{m})$, $(m_d - m_u)^2:(m_s - \hat{m})$, $(m_d - m_u)^2:(m_c - \hat{m})$ and $(m_d - m_u)(m_s - \hat{m}):(m_c - \hat{m})$ respectively. [For a more detailed analysis of the mixing problem in the case $m_u = m_d$ see Borchardt, Mathur and Okubo [279].]

The mesons occur in 16-plets: on account of the OZI rule the representations $\underline{15}$ and $\underline{1}$ are nearly degenerate. We refer the reader to the literature [279, 280] for a detailed account of the general mass formulae. For the unmixed states the first order SU(4) mass formulae are equivalent to the quark counting rule:

$$M^2 = a + b(N_u m_u + N_d m_d + N_s m_s + N_c m_c).$$

This formula is also valid for the mixed states, provided the mixing is ideal (OZI rule). As discussed in appendix B (in the context of nonet degeneracy in SU(3)) the mixing is nearly ideal for all multiplets except for the pseudoscalars for which η_c is nearly pure $\bar{c}c$, the η is close to the SU(3) octet state and the η' is close to the SU(3) singlet state $(\bar{u}u + \bar{d}d + \bar{s}s)/\sqrt{3}$ [with a sizeable contamination from $F\tilde{F}$].

Appendix B. Meson nonets

As mentioned in section 7 the OZI rule implies that the singlet and octet mesons containing light quarks are nearly degenerate. For definiteness we use the notation appropriate for the vector meson nonet. Since the width of some of these mesons is not small in comparison with the mass splittings ($\Gamma_\rho = 158\,\mathrm{MeV}$, $M_{\mathrm{K}^*} - M_\rho = 116\,\mathrm{MeV}$) we have to consider both the real and the imaginary part of the position $E = M - \frac{1}{2}\mathrm{i}\Gamma$ of the pole in the propagator. Both M and Γ are functions of Λ and the quark masses. The width is very sensitive to the positions of the threshold for decay (phase-space) – it is a highly nonanalytic function of the quark masses. If we would e.g. put the masses of the u and d quarks equal to the physical value of $m_\mathrm{s}(m_\mathrm{u} = m_\mathrm{d} = m_\mathrm{s} = m_0)$ then the entire nonet would be stable; both octet and singlet would be close to 1020 MeV whereas the pseudoscalar mesons would occur at the mass $(m_\mathrm{s}/\hat{m})^{1/2} M_\pi = 700\,\mathrm{MeV}$, too large for the vector mesons to decay. As the masses of u and d are decreased at fixed m_s the multiplet splits, the states containing predominantly u and d quarks becoming lighter. When the mean mass $\hat{m} = \frac{1}{2}(m_\mathrm{u} + m_\mathrm{d})$ has reached a value of the order of $\frac{1}{2}m_\mathrm{s}$ the decay channel for the ρ is opened and the pole starts walking off the real axis. At the physical value of $\hat{m}$ the real part reaches the value 776 MeV and the imaginary part is -79 MeV. The poles of the other states stay closer to the real axis: the imaginary parts of M_ω, M_{K^*} and M_φ are -5 MeV, -25 MeV and -2 MeV respectively. First order perturbation theory only shifts the real parts, by an amount given by the expectation value of the perturbation $\bar{q}(m - m_0)q$ in the unperturbed states. It is difficult to control the higher order terms; the main effect of these higher order terms is to shift the poles in the direction of the negative imaginary axis.

To be on safe grounds we first analyze the mass formulae for unphysical values of m_u, m_d that are large enough for the vector mesons to be stable (say $\hat{m} \sim \frac{1}{2}m_\mathrm{s}$) and show that in this region of quark masses the φ state is almost ideally mixed. Since the decay channels respect the OZI rule we then conclude that the φ remains ideally mixed when the masses of m_u and m_d are lowered to their physical values. Finally we will analyze the mixing of ρ^0 and ω for which it does not make sense to ignore the imaginary part of the masses, but for which only the small isospin breaking piece of the quark mass perturbation counts, such that a first order calculation with respect to that perturbation is appropriate.

We first put $m_\mathrm{u} = m_\mathrm{d} = \hat{m}$ and suppose that $\hat{m}$ is large enough for the nonet to be stable. The first order mass formulae for the unmixed states are

$$M_\rho^2 = A_8 + 2\hat{m}B^\mathrm{u} + m_\mathrm{s}B^\mathrm{d}$$
$$M_{\mathrm{K}^*}^2 = A_8 + \hat{m}(B^\mathrm{u} + B^\mathrm{d}) + m_\mathrm{s}B^\mathrm{u}\,. \tag{B.1}$$

(The octet matrix elements of the operators $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$ involve only two unknowns because the F-coupling vanishes; in the notation used in section 9 for the baryon octet, the equality $M_{\rho^+} = M_{\rho^-}$ requires $B^\mathrm{s} = B^\mathrm{u}$.) We denote the mass eigenstates in the limit $m_\mathrm{u} = m_\mathrm{d}$ by $|\bar{\omega}\rangle$, $|\bar{\rho}\rangle$, $|\bar{\mathrm{K}}^*\rangle$, $|\bar{\varphi}\rangle$. The states $|\bar{\omega}\rangle$ and $|\bar{\varphi}\rangle$ are linear combinations of the singlet and of the octet member with $T = 0$:

$$|\bar{\omega}\rangle = \cos\varphi\,|1\rangle + \sin\varphi\,|8, T = 0\rangle$$
$$|\bar{\varphi}\rangle = \sin\varphi\,|1\rangle - \cos\varphi\,|8, T = 0\rangle \tag{B.2}$$

where we have introduced the mixing angle in such a manner that for $\mathrm{tg}\,\varphi = 1/\sqrt{2}$ the state $|\bar{\omega}\rangle$ becomes $(\bar{u}u + \bar{d}d)/\sqrt{2}$ whereas $|\bar{\varphi}\rangle$ becomes pure $\bar{s}s$.

To work out the masses of these states we need the matrix elements of the perturbation

$$\begin{aligned}&\langle 8, T=0|\bar{q}mq|8, T=0\rangle = \tfrac{2}{3}(\hat{m}+2m_s)B^u + \tfrac{1}{3}(4\hat{m}-m_s)B^d\\&\langle 8, T=0|\bar{q}mq|1\rangle = \tfrac{2}{3}\sqrt{2}(\hat{m}-m_s)B_{18}\\&\langle 1|\bar{q}mq|1\rangle = \tfrac{2}{3}(2\hat{m}+m_s)B_1\,.\end{aligned} \tag{B.3}$$

In terms of these matrix elements the masses are

$$\begin{aligned}&M_\omega^2 = a - \sqrt{b^2+c^2}\\&M_\varphi^2 = a + \sqrt{b^2+c^2}\\&a = M_8^2 + \tfrac{2}{3}(\hat{m}+2m_s)B^u + \tfrac{1}{3}(4\hat{m}-m_s)B^d - c\\&b = \tfrac{2}{3}\sqrt{2}(m_s-\hat{m})B_{18}\\&c = \tfrac{1}{2}(A_8-A_1) + \tfrac{1}{3}(\hat{m}+2m_s)B^u + \tfrac{1}{6}(4\hat{m}-m_s)B^d - \tfrac{1}{3}(2\hat{m}+m_s)B_1\end{aligned} \tag{B.4}$$

and the mixing angle is given by

$$\sin^2\varphi = \frac{3M_\varphi^2 - 4M_{K^*}^2 + M_\rho^2}{3(M_\varphi^2 - M_\omega^2)}\,. \tag{B.5}$$

The OZI rule requires $A_8 - A_1$ as well as the matrix elements $\langle\bar{u}u|\bar{q}mq|\bar{d}d\rangle$, $\langle\bar{u}u|\bar{q}mq|\bar{s}s\rangle$ and $\langle\bar{d}d|\bar{q}mq|\bar{s}s\rangle$ to be small. In terms of the invariant matrix elements this implies

$$\begin{aligned}&A_8 \simeq A_1\\&B_{18} \simeq (B^u - B^d) \qquad [\text{OZI}]\\&B_1 \simeq (B^u + \tfrac{1}{2}B^d)\,.\end{aligned} \tag{B.6}$$

It is easy to check that if these relations hold, then

$$\begin{aligned}&M_\rho^2 = M_\omega^2\\&M_\varphi^2 - M_{K^*}^2 = M_{K^*}^2 - M_\rho^2 = (m_s - \hat{m})(B^u - B^d) \qquad [\text{OZI}]\\&\mathrm{tg}\,\varphi = 1/\sqrt{2}\,; \qquad \varphi = 35.3^\circ\,.\end{aligned} \tag{B.7}$$

The deviations from ideal mixing are controlled by the ratio of the OZI violating amplitudes such as $A_8 - A_1$ in comparison to the mass splitting produced by the quark mass term, which is of order $(m_s - \hat{m})(B^u - B^d)$. The smallness of the ratio $(M_\omega - M_\rho):(M_{K^*} - M_\rho) \simeq 5\%$ shows that the OZI breaking amplitudes are very small. To the accuracy of first order mass formulae the mixing angle is ideal, the $|\bar{\varphi}\rangle$ is pure $\bar{s}s$, the $|\bar{\omega}\rangle$ is pure $(\bar{u}u + \bar{d}d)/\sqrt{2}$.

As discussed above it is not easy to predict exactly how the poles will move when the decay channels open. The real parts of the physical masses of φ, K^* and ρ satisfy the equal spacing rule (B.7) remarkably well for the linear mass formula. The differences showing up in the quadratic mass formula are well accounted for by the simple recipe for estimating terms of order $(m_{\mathrm{quark}})^2$ given in section 10. The asymmetries caused by the decay channels do therefore not seem to have sizeable effects on the real parts (this conclusion is confirmed by the success of the equal spacing rule in the baryon decuplet).

We now consider the perturbation caused by the isospin violating piece of the quark mass term. The unperturbed states are the isospin multiplets $|\bar{\omega}\rangle$, $|\bar{\rho}\rangle$, $|\bar{\mathrm{K}}^*\rangle$, $|\bar{\varphi}\rangle$. The unperturbed Hamiltonian includes the physical quark mass term except for the piece $\frac{1}{2}(m_{\mathrm{u}}-m_{\mathrm{d}})(\bar{u}u-\bar{d}d)$. To first order in $m_{\mathrm{u}}-m_{\mathrm{d}}$ the splittings are given by the expectation value in the unperturbed states:

$$
\begin{aligned}
&M^2_{\mathrm{K}^{*+}}-M^2_{\mathrm{K}^{*0}}=-\frac{m_{\mathrm{d}}-m_{\mathrm{u}}}{m_{\mathrm{s}}-\hat{m}}(M^2_{\mathrm{K}^*}-M^2_\rho)\\
&M^2_{\rho^+}-M^2_{\rho^0}=0\,.
\end{aligned}
\tag{B.8}
$$

As was discussed in section 9 the OZI selection rule inhibits the state $|\bar{\varphi}\rangle$ to mix with $|\bar{\omega}\rangle$ and $|\bar{\rho}^0\rangle$. To analyze the mixing of $|\bar{\omega}\rangle$ and $|\bar{\rho}^0\rangle$ produced by the perturbation it is important to account for the fact that the imaginary parts of their masses are very different. To first order in $m_{\mathrm{u}}-m_{\mathrm{d}}$ the effective Hamiltonian that reproduces the proper time evolution of the states (see appendix F) has the form

$$
H=\begin{pmatrix} M_\rho-\frac{1}{2}\mathrm{i}\Gamma_\rho & M_{\rho\omega}\\ M_{\omega\rho} & M_\omega-\frac{1}{2}\mathrm{i}\Gamma_\omega\end{pmatrix}
\tag{B.9}
$$

with

$$
M_{\rho\omega}=\frac{1}{4M_\rho}(m_{\mathrm{u}}-m_{\mathrm{d}})\langle\bar{\rho}^0|\bar{u}u-\bar{d}d|\bar{\omega}\rangle\,.
\tag{B.10}
$$

The eigenstates are

$$
\begin{aligned}
&|\rho^0\rangle=|\bar{\rho}^0\rangle-\frac{M_{\omega\rho}}{M_\omega-M_\rho-\frac{1}{2}\mathrm{i}(\Gamma_\omega-\Gamma_\rho)}|\bar{\omega}\rangle\\
&|\omega\rangle=|\bar{\omega}\rangle+\frac{M_{\rho\omega}}{M_\omega-M_\rho-\frac{1}{2}\mathrm{i}(\Gamma_\omega-\Gamma_\rho)}|\bar{\rho}^0\rangle\,.
\end{aligned}
\tag{B.11}
$$

There is no first order shift in the masses. Note that the states $|\rho^0\rangle$ and $|\omega\rangle$ are not orthogonal on account of $\Gamma_\rho\neq\Gamma_\omega$. The rate for ω to decay into $\pi^+\pi^-$ is given by

$$
\Gamma_{\omega\to\pi^+\pi^-}=\frac{|M_{\rho\omega}|^2}{(M_\rho-M_\omega)^2+\frac{1}{4}(\Gamma_\omega-\Gamma_\rho)^2}\Gamma_{\rho^0\to\pi^+\pi^-}\,.
\tag{B.12}
$$

The transition matrix element $M_{\rho\omega}$ may be expressed in terms of the invariants B^{u}, B^{d}, B_{18}:

$$
M_{\rho\omega}=M_{\omega\rho}=\frac{m_{\mathrm{u}}-m_{\mathrm{d}}}{2\sqrt{3}M_\rho}\{\sin\varphi\,(B^{\mathrm{u}}-B^{\mathrm{d}})+\cos\varphi\,\sqrt{2}\,B_{18}\}\,.
\tag{B.13}
$$

Using the OZI relations (B.6) this becomes

$$M_{\rho\omega} = -\frac{m_d - m_u}{m_s - \hat{m}}(M_{K^*} - M_\rho)\,. \tag{B.14}$$

The experimental information on $M_{\rho\omega}$ and the resulting value for the ratio $R = (m_s - \hat{m}):(m_d - m_u)$ are discussed in section 14.

Appendix C. Improved chiral perturbation theory (ICPT)

We split the Hamiltonian of QCD into the chirally symmetric part H_0 and the quark mass term H_1 and discuss the perturbation expansion in powers of H_1. The formal perturbation series for the mass of a physical state reads

$$M_n^2 = M_0^2 + \langle p|\bar{q}mq|p\rangle - \tfrac{1}{2}\mathrm{i}\int \mathrm{d}^4x\, \langle p|T\bar{q}(x)\, mq(x)\, \bar{q}(0)\, mq(0)|p\rangle + \cdots \tag{C.1}$$

where M_0 denotes the mass of the particle in the chiral limit and $|p\rangle$ stands for the unperturbed eigenstate of H_0. Since the energies are measured with respect to the physical ground state the time-ordered product appearing in the second order term only includes the connected part of the matrix element. The contribution of the state $|p\rangle$ to the sum over all intermediate states implicit in this matrix element has to be handled with care – in the corresponding second order formula for the shift of energy levels in quantum mechanics the sum only extends over the intermediate states orthogonal to $|p\rangle$. To eliminate the corresponding contribution in the covariant time-ordered product, one may look at the amplitude

$$T(p,q) = \mathrm{i}\int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}qx}\, \langle p|T\bar{q}(x)\ mq(x)\, \bar{q}(0)\, mq(0)|p\rangle_{\mathrm{conn.}} \tag{C.2}$$

where an average over the spin directions of the state $|p\rangle$ is understood. We are interested in the limit of $T(p,q)$ as $q \to 0$. The contribution of the intermediate state $|p\rangle$ shows up in the form of pole terms $\sim[M_0^2 - (p \pm q)^2]^{-1}$ which explode near $q = 0$. The proper second order contribution in the mass formula (C.1) is obtained by first taking the limit $q^2 \to 0$ and then taking the limit $pq \to 0$ [282].

Alternatively, one may consider the one-particle-irreducible piece of $T(p,q)$. This notion is however only defined within a framework that identifies the state $|p\rangle$ as the asymptotic state of an interpolating field with specified propagation properties. We want to avoid the use of such a formalism here, because it obscures the fact that we are only interested in physical matrix elements of the operator $\bar{q}mq$. Instead we may define the amplitude $T'(p,q)$ by

$$T'(p,q) = T(p,q) - \{[M_0^2 - (p-q)^2]^{-1} + [M_0^2 - (p+q)^2]^{-1}\}\, G(q^2) \tag{C.3}$$

where $G(q^2)$ is the residue of the pole in $T(pq, q^2)$ at $pq = \frac{1}{2}q^2$. The prescription given above amounts to the statement that the second order term in the series (C.1) is the value of $-\frac{1}{2}T'(p,q)$ at $q = 0$. For scalar particles the standard one-particle-irreducible piece coincides with T'; for particles of spin $\frac{1}{2}$ this is not the case, because the propagator leads to a polynomial in pq – the one-particle-reducible matrix element does contribute to the limit specified above and hence contributes to the mass shift [282]. This difference in the one-particle-reducible matrix elements reflects the fact that the Lagrangian for scalar fields is quadratic in the mass, while the Lagrangian for spin $\frac{1}{2}$ fields is linear in the mass.

After these kinematical preliminaries we now turn to the dynamical problems generated by the fact that the unperturbed system contains $N_f^2 - 1$ massless Goldstone bosons (N_f is the number of flavours whose mass terms are included in H_1). We first consider the perturbation series for bound states that are massive in the chiral limit, $M_0 \neq 0$. Since these states occur in degenerate multiplets of SU(N_f) it is important that the unperturbed basis vectors are chosen such that the perturbation is diagonal (Λ–Σ^0-mixing). The expectation value of the perturbation then determines the mass shift in first order of m_{quark}. The second order term in (C.1) is infrared divergent. The divergence arises from the integration over x: the integrand exists, but does not decrease sufficiently fast as $x \to \infty$. The divergence has to do with the fact that a chiral proton is surrounded by a cloud of massless virtual pions. (For definiteness we discuss $N_f = 2$ and consider the expansion of the proton mass in powers of m_u, m_d at fixed m_s. In this case only the pions produce an infrared divergence.) The probability for finding a virtual pion in the vicinity of the proton only falls off with a power of the distance. Because the matrix element $\langle\pi|\bar{q}mq|\pi\rangle$ is different from zero [to lowest order this matrix element is the (mass)^2 of the pion] the quantity $\langle p|T\bar{q}mq\,\bar{q}mq|p\rangle$ contains contributions that only decrease like a power of x. The crucial point here is that these contributions arise from configurations in which the pion is far away from the proton. The coefficient of the leading infrared divergence is determined by physical properties of the chiral theory: by the π–p scattering amplitude for pions of zero momentum and by the one particle matrix elements of the perturbation. The low energy theorems of current algebra furthermore show that the scattering amplitude is determined by the one particle matrix element g_A of the axial current (Goldberger–Treiman relation). Any model Hamiltonian of the type $H_0 + H_1$ for which

(i) H_0 has the same one particle spectrum as the chiral Hamiltonian of QCD,

(ii) H_0 conserves an axial vector current with the correct matrix elements between the vacuum and the pseudoscalar one particle states and the correct value g_A for the matrix element between one particle states at zero momentum transfer,

(iii) H_1 has the same one particle matrix elements as the quark mass term in QCD,

produces the same leading infrared divergences in the perturbation series. [Note that only the linear infrared divergence in the second order term is determined by g_A. If g_A vanishes for the particle in question, the second order term in the quark mass expansion does not contain a linear infrared divergence, but may contain a logarithmic one. The conditions given above have to be sharpened if they are to guarantee that the model Hamiltonian reproduces the logarithmic singularity correctly.] Effective chiral Lagrangians which describe the pseudoscalar mesons and the nucleons as elementary fields have the above properties, provided one restricts oneself to tree diagrams. Fig. 3a shows the diagram responsible for the leading infrared divergence in M_p in the language of such a chiral Lagrangian. Note that we are not using this Lagrangian beyond the tree graph approximation here: we are not carrying out the loop integral, but only look at the behaviour of the integrand at small values of the pion momentum k – there is no significance to this loop integral for large values of k. The behaviour of the integrand near $k = 0$ produces a linear infrared divergence – the second order mass formula (C.1) does therefore not make sense.

What one has to do to carry the quark mass expansion beyond first order is to reorder the series by summing up dangerous diagrams [281]. In fact the diagrams depicted in fig. 3b produce factors of the type $[\langle\pi|\bar{q}mq|\pi\rangle/k^2]^n$ which are more and more infrared singular as the order n increases. The diagrams involve the same chiral one particle matrix elements and it is easy to sum them up: one merely has to replace the massless pion propagator in the chiral effective Lagrangian by a massive propagator with

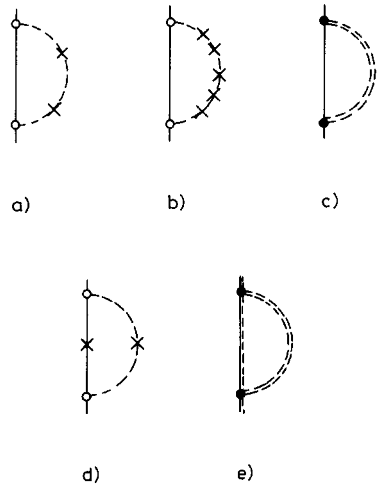

Fig. 3. Summing up infrared singularities. Single lines represent baryons (—) and mesons (---) in the chiral limit, the crosses denote mass insertions generated by the chiral symmetry breaking term $\bar{q}mq$. Open circles indicate derivative coupling proportional to g_A/f_π. Double lines denote propagators with physical masses, and the full circles represent the corresponding vertex for physical particles, proportional to the axial vector form factor.

$M_\pi^2 = \langle\pi|\bar{q}mq|\pi\rangle$ (fig. 3c). In addition to the sum of the contributions with two or more mass insertions this diagram also contains a contribution with no mass insertion and a contribution with one insertion. We could subtract these parts which are not infrared divergent. This is not necessary, however, for the following reason: the diagram with no mass insertion only changes the value of M_0; the diagram with one mass insertion is linear in the quark masses with coefficients that respect the SU(N_f) symmetry properties of the chiral limit. This contribution may therefore be absorbed in the first order term – it merely changes the values of the coefficients B^u, B^d, B^s without affecting their symmetry properties. As long as we only extract quark mass ratios from the spectrum of the physical particles on the basis of these symmetry properties and are not interested in the value of the nucleon mass in the chiral limit or in the coefficients B^u, B^d, B^s we do not have to subtract these contributions. (To extract the σ-terms one does have to take these modifications into account; see Gasser [282].)

The sum of the leading infrared singularities regularizes the divergence in the second order term: the diagram 3c, evaluated with a massive pion propagator is convergent. The divergence $\sim\int d^4k\, k_\mu\, k_\nu\, (k^2)^{-3}\, (pk)^{-1}$ is converted into a factor $1/M_\pi$. The entire contribution from graph 3c has an expansion of the form $a + bM_\pi^2 + cM_\pi^3 + \cdots$. The third term in this expansion is proportional to $(m_u + m_d)^{3/2}$. The quark mass expansion fails, because this function is not analytic at $m_u = m_d = 0$.

To reproduce the leading nonanalytic term we may in fact use the physical pion mass rather than the first order expression $M_\pi^2 = \langle\pi|\bar{q}mq|\pi\rangle$, because the difference between this expression and the physical pion (mass)^2 is of order $M_\pi^4 \ln M_\pi^2$ and hence only shows up in the terms of order $m_q^2 \ln m_q$.

The result is very simple: to get rid of the leading nonanalytic terms in the expansion of the proton mass $M_p(\Lambda, m_u, m_d, m_s, \ldots)$ in powers of m_u and m_d one subtracts the corresponding lowest order diagram, fig. 3c, using the physical mass in the pion propagator. The nucleon propagator and the pion–nucleon vertex are taken in the symmetry limit. The behaviour of the vertex at nonzero values of k is irrelevant as long as it is symmetric under SU(N_f) and reduces to g_A/f_π for $k \to 0$. The integral over k may be cut off at some k_{max} or, alternatively, one may suppress large values of k by using a form factor at the vertices:

$$\bar{M}_{\mathrm{p}}^2 = M_{\mathrm{p}}^2 - \frac{3}{32\pi^2}\frac{g_{\mathrm{A}}^2}{f_\pi^2} J(M_0, M_0, M_\pi)$$

$$J(M_1, M_2, m) = \frac{\mathrm{i}}{\pi^2}\int \frac{\mathrm{d}^4k}{m^2-k^2} G^2(k^2)\, \bar{u}(p)\not{k}\gamma_5 \frac{(M_2+\not{p}-\not{k})}{M_2^2-(p-k)^2}\not{k}\gamma_5\, u(p)$$

$$p^2 = M_1^2 \tag{C.4}$$

where $G(k^2)$ is the axial vector form factor with $G(0)=1$. In contrast to the proton mass itself, the quantity $\bar{M}_{\mathrm{p}}^2$ is analytic in the quark masses m_{u}, m_{d} up to terms of order $m_{\mathrm{q}}^2 \ln m_{\mathrm{q}}$.

The next-to leading infrared singularities ($\ln k_{\min}$ rather than $1/k_{\min}$) are less easy to deal with. One contribution arises from the diagram in fig. 3d: one mass insertion is attached to the nucleon line, the other to the pion line. The diagrams containing any number of nucleon mass insertions (internal as well as external lines) are again summed up if one replaces the chiral nucleon propagator (mass M_0) by the corresponding propagator with the physical nucleon mass. A prescription that cures the infrared singularities also on the first nonleading level must involve propagators with physical masses both for the pion and for the nucleon lines (fig. 3e). It is however not correct to simply use the symmetric couplings given by the effective chiral Lagrangian and to only consider the chiral asymmetries due to mass terms in this effective Lagrangian (symmetric vertices, propagators with physical masses). On the next-to-leading level there is also a contribution due to the σ-term in π N-scattering. In the quark mass expansion of the nucleon mass the σ-term shows up in a nonanalytic piece proportional to $\langle \mathrm{p}|\hat{m}(\bar{u}u+\bar{d}d)|\mathrm{p}\rangle M_\pi^2 \ln M_\pi^2$. In the language of an effective Lagrangian the σ-term amounts to a coupling of the type $\bar{\psi}\psi\pi^2$ with a coupling constant that vanishes in the chiral limit. If one wants to include all nonanalytic terms of order $m_{\mathrm{q}}^2 \ln m_{\mathrm{q}}$ one has to strengthen the conditions (i), (ii) and (iii) on the effective Lagrangian and insure that it reproduces the low energy scattering amplitudes not only in the chiral limit, but up to and including the terms of order m. [Note that the nonanalytic term proportional to $M_\pi^4 \ln M_\pi^2$ in the quark mass expansion (C.6) of M_η^2 is due to the analogous vertex correction.]

We did not take these next-to-leading nonanalytic terms in the quark mass expansion of the baryon masses into account. The quantity $\bar{M}_n$ defined by

$$\bar{M}_n^2 = M_n^2 - \frac{1}{32\pi^2}\frac{g_{\mathrm{A}}^2}{f_\pi^2}\sum_m \sum_{i=\pi,k,\eta} C_{nm}^i\, J(M_n, M_m, M_i) \tag{C.5}$$

involves only physical masses. [The constants C_{nm}^i are the Clebsch–Gordan coefficients involving the ratio $\alpha = d:(f+d)$ of the axial vector coupling constants; the integral J is defined in (C.4).] In the above we have shown that in contrast to $M_n(\Lambda, m_{\mathrm{u}}, m_{\mathrm{d}}, m_{\mathrm{s}}, \ldots)$ the quantity $\bar{M}_n(\Lambda, m_{\mathrm{u}}, m_{\mathrm{d}}, m_{\mathrm{s}}, \ldots)$ is analytic in the quark masses m_{u}, m_{d}, m_{s} up to terms of order $m_{\mathrm{q}}^2 \ln m_{\mathrm{q}}$.

Numerical values for the purified masses $\bar{M}_n$ are given in table 6. In column 1 we list the masses of the particles in pure QCD, obtained by subtracting the electromagnetic Born terms from the physical masses (we do not indicate the error bars associated with these numbers, but work with the mean values throughout). In the top half of the table we give the mean masses of the isospin multiplets, i.e. ignore the difference $m_{\mathrm{u}} - m_{\mathrm{d}}$. Column 2 contains the values for $\bar{M}_n$ according to equation (C.5) with $\alpha = 0.62$ (SU(3) symmetry for the axial vector couplings g_{A}/f_π). We have used the dipole formula for the axial vector form factor: $G(t) = (1 - t/m^2)^{-2}$ with $m^2 = 0.71\ \mathrm{GeV}^2$. The main effect is an overall shift of the

Table 6
Masses of baryons stripped of their meson clouds. Column 1 contains the physical masses corrected for electromagnetic self-energy. Columns 2 and 3 give the purified mass $\bar{M}_n$ for SU(3) symmetric axial vector couplings and SU(3) symmetric meson–baryon coupling constants respectively. The quantity Δ measures the deviations from the Gell-Mann–Okubo formula.

	QCD 1	g_A/f_π 2	renormalized 2a	$\theta = 0$ 2b	$g_{M\bar{B}B}$ 3
N	939	1082	939	1082	1080
Λ	1116	1262	1119	1262	1230
Σ	1193	1331	1188	1331	1289
Ξ	1318	1460	1317	1460	1406
Δ	−0.070	−0.084	−0.084	−0.084	−0.018
p − n	−2.05	−2.52	−2.52	−2.74	−2.49
$\Sigma^+ - \Sigma^-$	−7.81	−8.49	−8.49	−9.11	−7.97
$\Xi^0 - \Xi^-$	−5.56	−5.52	−5.52	−5.80	−5.09

masses by about 140 MeV. One may renormalize the purified masses by subtracting an SU(3) singlet, e.g. in such a manner that the nucleon stays where it is (column 2a): the renormalized masses differ from the physical masses by at most 5 MeV. Alternatively, one may look at the Gell-Mann–Okubo combination

$$\Delta = \frac{2N + 2\Xi - 3\Lambda - \Sigma}{\Xi - N}\,.$$

The nonanalytic corrections modify this combination only very little. In column 3 we show the results obtained if the meson–baryon coupling constants $g_{M\bar{B}B} = (g_A/\sqrt{2}f_\pi)(M_{\bar{B}} + M_B)$ rather than the constants g_A/f_π are taken SU(3) symmetric. Since the corresponding leading nonanalytic terms are the same we have no reason to prefer one prescription to the other. According to the coupling constant analysis provided by Nagels et al. [283] the ratio $d:(f+d)$ has the value $\alpha = 0.76$ in this case. As can be seen by comparing columns 2 and 3 the individual shifts are substantially different for the two prescriptions. What matters, however, are not the overall shifts, but only the parts that are not octet or singlet. These parts differ only very little as can be seen e.g. from the Gell-Mann–Okubo formula which is again almost unaffected.

The lower half of the table quotes the corresponding mass differences within the isospin multiplets. The mass difference $m_u - m_d$ induces mixing between π^0 and η as well as between Σ^0 and Λ. The SU(3) symmetric coupling constants determine the probability amplitudes for emission and absorption of the mesonic octet states: the meson mass matrix that determines the propagation of these states is however not diagonal in the octet basis. In order to be able to work with diagonal meson propagators one has to transform the vertices accordingly. To see the effect of $\pi^0 - \eta$ mixing on the mass differences one may compare columns 2 and 2b: in column 2 we give the numbers that one obtains if the baryon–meson coupling constants are subject to the mixing transformation (with the mixing angle $\theta = 0.57°$ that follows from the meson spectrum) and in column 2b we give the corresponding values if the mixing angle is set equal to zero. As discussed in section 10 the nonanalytic terms increase the mass difference between proton and neutron in comparison to the splittings due to $m_s - \hat{m}$. The effect is not sensitive to the manner in which one determines the baryon-meson coupling constants (compare columns 2 and 3).

The leading nonanalytic terms are independent of the spin of the particle in question. What counts is the analog of g_A, the matrix element of the axial charge at infinite momentum. For the Goldstone bosons, the constant g_A vanishes. The leading infrared divergence in the second order term of the quark mass expansion is therefore only logarithmic [284]. To reproduce the leading nonanalytic terms in the quark mass expansion of the Goldstone boson masses with an effective Lagrangian, this Lagrangian must include the relevant σ-terms. The sum of the leading infrared divergences is again given by the lowest order self-energy diagrams with physical masses in the propagators and one ends up with an expression analogous to (C.5). The corresponding leading nonanalytic contributions are proportional to $M_\pi^4 \ln M_\pi^2$, $M_K^4 \ln M_K^2$ and $M_\eta^4 \ln M_\eta^2$. In fact, the approximation obtained by expanding the result of ICPT and retaining only the leading nonanalytic terms is indeed reliable here [282]. This is by no means the case for the baryon mass formulae, which become completely unphysical if one only retains the leading nonanalytic terms proportional to M_π^3, M_K^3 and M_η^3 (see section 10). For the Goldstone bosons the net result is that the infrared logarithm is replaced by a logarithm of the meson masses and the formulae analogous to (C.5) may be written as

$$\begin{aligned}
\bar{M}_\pi^2 &= M_\pi^2\{1 - 3L_\pi + L_\eta\} \\
\bar{M}_K^2 &= M_K^2\{1 - 2L_\eta\} \\
\bar{M}_\eta^2 &= M_\eta^2\{1 - 6L_K + 4L_\eta\} + M_\pi^2\{3L_\pi - 2L_K - L_\eta\} \\
\bar{M}_{K^0}^2 - \bar{M}_{K^+}^2 &= (M_{K^0}^2 - M_{K^+}^2)\left\{1 - 2L_\eta + \frac{4M_K^2}{M_\eta^2 - M_\pi^2}(L_\pi - L_\eta)\right\} \\
L_n &= (48\pi^2 f_\pi^2)^{-1} M_n^2 \ln(M_n^2/\mu^2)\,.
\end{aligned} \tag{C.6}$$

In these approximate expressions the form factor only enters through the value of μ in the argument of the logarithm. For a value of μ in the range $0.5\,\mathrm{GeV} < \mu < 1\,\mathrm{GeV}$ the mass shifts due to these corrections amount to less than 5%.

We have mentioned in section 8 that the mass difference $M_{\pi^+} - M_{\pi^0}$ vanishes to first order in $m_u - m_d$. The same is true of the corresponding chiral corrections. Retaining terms of order $(m_u - m_d)^2/(m_s - \hat{m})$ and taking π^0–η-mixing into account the expression for the mass difference becomes

$$\begin{aligned}
M_{\pi^+}^2 - M_{\pi^0}^2 = \frac{1}{4}\frac{(m_d - m_u)^2}{m_s - \hat{m}} B\Big\{&1 - 18L_\pi + 18L_K + 8L_\eta \\
&+ 6\frac{M_\pi^2}{M_K^2 - M_\pi^2}(L_\pi - L_K) + \frac{4M_\pi^2}{M_\eta^2 - M_\pi^2}(L_\eta - L_\pi)\Big\}\,.
\end{aligned} \tag{C.7}$$

Appendix D. The pion–nucleon σ-term

The pion–nucleon σ-term in $\pi^\pm p \to \pi^\pm p$ scattering is defined by

$$\sigma = \frac{\hat{m}}{2M_p}\langle p|\bar{u}u + \bar{d}d|p\rangle \tag{D.1}$$

where $|p\rangle$ denotes the physical one-proton state, and M_p is the proton mass. In a first stage of our discussion we shall disregard isospin violating effects due to $m_u \neq m_d$ (isospin violation due to photon exchange must be taken into account by correcting the scattering data before extracting a value for σ. These effects will not be considered here). At the end of this section we shall come back to the question of isospin asymmetries in σ-terms in connection with other channels of πN-scattering.

The interest in the quantity σ derives from its connection with πN-scattering [285–296]: Define the form factor

$$\bar{u}(p')\, u(p)\, \sigma(t) = \hat{m}\, \langle p'|\bar{u}u + \bar{d}d|p\rangle\, ; \qquad t = (p' - p)^2\, .$$

As shown by Brown, Pardee and Peccei [297] the on-shell pion–nucleon scattering amplitude at the unphysical Cheng–Dashen point $s = M_N^2$, $t = 2M_\pi^2$ differs from the form factor $\sigma(t = 2M_\pi^2)$ only through terms that are formally of order M_π^4. Furthermore Pagels and Pardee [298] have shown that the relation between $\sigma(2M_\pi^2)$ and the scattering amplitude at the Cheng–Dashen point does not contain nonanalytic terms on the level of M_π^3, but does contain corrections of order $M_\pi^4 \ln M_\pi^2$. On the basis of their analysis we conclude that the on-shell amplitude at the Cheng–Dashen point should differ only very little from $\sigma(2M_\pi^2)$. To connect the data on π–N scattering to the forward matrix element, i.e. to the proper σ-term, $\sigma = \sigma(0)$ defined in (D.1) one then only needs to study the t-dependence of $\sigma(t)$. From an analysis that is identical to the one given by Pagels and Pardee [298] we obtain

$$\sigma(2M_\pi^2) = \sigma(0) + \frac{3g_A^2 M_\pi^3}{32\pi f_\pi^2} + O(M_\pi^4 \ln M_\pi^2) \tag{D.2}$$

with $f_\pi = 132$ MeV. The correction amounts to 7 MeV (the analogous formula given by Pagels and Pardee differs from eq. (D.2) by a factor of 2). Improving the calculation with the methods described in appendix C we find that the correction is reduced by a factor of two; the final value, including contributions from the kaon and eta clouds is

$$\sigma(2M_\pi^2) - \sigma(0) = 3.5 \pm 1.2\,\text{MeV} + O(M_\pi^4 \ln M_\pi^2)\, . \tag{D.3}$$

The remaining corrections of order $m_{\text{quark}}^2 \ln(m_{\text{quark}})$ have not been given in the literature. We expect them also to be very small.

Recent evaluations (refs. [293, 299]; for a compilation of recent data, see Nagels et al. [291b]) of scattering data give values around $\sigma(2M_\pi^2) = 65$ MeV with remarkably small error bars. In view of the above discussion this value implies

$$\sigma \simeq 60\,\text{MeV}\, . \tag{D.4}$$

A popular estimate for the size of the σ-term is obtained if one assumes that the matrix elements of the operators $\bar{u}u$, $\bar{d}d$ and $\bar{s}s$ are approximately equal to the matrix elements of the corresponding charge densities u^+u, d^+d and s^+s at rest (additivity rule). In this picture the quark mass difference $m_s - \hat{m}$ is given by $\frac{1}{2}(M_\Xi - M_N) \simeq 190$ MeV whereas the σ-term is given by $3\hat{m}$. Using the ratio $m_s : \hat{m} = 25$ one thus gets $\hat{m} \simeq 8$ MeV, $\sigma \simeq 24$ MeV.

A more precise determination follows from the mass formulae for the baryon octet, discussed in sections 9 and 10. To get a value for σ one however needs one additional piece of information concerning the SU(3) singlet component of the quark mass operator. It suffices to know the value of the nucleon mass in the chiral limit or the size of the nucleon matrix element of the operator $\bar{s}s$. We first leave the value of $\langle \mathrm{p}|\bar{s}s|\mathrm{p}\rangle$ open and parametrize it in terms of

$$y = \frac{2\langle \mathrm{p}|\bar{s}s|\mathrm{p}\rangle}{\langle \mathrm{p}|\bar{u}u + \bar{d}d|\mathrm{p}\rangle}\,.$$

The σ-term may then be worked out either by using the observed SU(3)-breaking mass differences between N, Λ, Σ and Ξ or the observed isospin breaking mass differences p − n, $\Sigma^+ - \Sigma^-$, $\Xi^0 - \Xi^-$. The linear *first order* mass formulae give

$$\sigma = \frac{\hat{m}}{(m_s - \hat{m})}\frac{1}{(1-y)}(M_\Xi + M_\Sigma - 2M_\mathrm{N})$$

$$\sigma = \frac{m_\mathrm{u} + m_\mathrm{d}}{(m_\mathrm{d} - m_\mathrm{u})(1-y)}[M_{\Sigma^-} - M_{\Sigma^+} - \tfrac{1}{2}(M_\mathrm{n} - M_\mathrm{p})]\,. \tag{D.5}$$

With $y = 0$, $m_s : \hat{m} = 25$, $(m_s - \hat{m}) : (m_\mathrm{d} - m_\mathrm{u}) = 43.5$ one finds $\sigma = 26\,\mathrm{MeV}$ and $\sigma = 25\,\mathrm{MeV}$ from these two determinations respectively (the contribution of the electromagnetic interaction to the isospin splittings has been corrected for according to table 3).

The order of magnitude of the corrections due to higher order terms in the quark mass expansion can be seen by using first order mass formulae for M^2 rather than for M: this gives $\sigma = 31\,\mathrm{MeV}$ and $\sigma = 32\,\mathrm{MeV}$ respectively. One of us (Gasser [282]) has analyzed the higher order terms in the quark mass expansion on the basis of the method described in section 10 and has shown that the asymmetries of the meson clouds increase the value of the σ-term by about 10 MeV. Taking the higher order terms in the quark mass expansion into account one obtains the value

$$\sigma = \frac{1}{1-y}\sigma_0\,; \qquad \sigma_0 = 35 \pm 5\,\mathrm{MeV}\,. \tag{D.6}$$

For $\langle \mathrm{p}|\bar{s}s|\mathrm{p}\rangle = 0$ this result is smaller than the values extracted from πN-scattering by about a factor two. One may of course blame the difference on the expectation value of $\bar{s}s$ in the nucleon which is indeed not known. Since the vacuum expectation value $\langle 0|\bar{s}s|0\rangle$ is substantial one should not dismiss the possibility that $\langle \mathrm{p}|\bar{s}s|\mathrm{p}\rangle$ is also rather large without further thought. Note however, that y would have to be of order 0.4 if this effect is made responsible for the discrepancy with the observed value. As noted above the value of y determines the value of the nucleon mass M_0 in the chiral limit. For $y = 0$ one has $M_0 \simeq 870\,\mathrm{MeV}$ and $y = 0.2$ corresponds to $M_0 \simeq 670\,\mathrm{MeV}$. If y should turn out to be as large as 0.4 then the value of the nucleon mass in the chiral limit would be completely different from what it is in this world. We conclude that $\sigma(2M_\pi^2) \simeq 65\,\mathrm{MeV}$ is not consistent with QCD unless very strange things happen in the chiral limit (see also appendix E).

Let us now come back to the question of isospin asymmetries. The discussion applies also to the σ-terms which occur in other channels of πN-scattering. Isospin-breaking effects (from $m_\mathrm{u} \neq m_\mathrm{d}$; the data are assumed to be corrected for electromagnetic interactions) in σ-terms occur in 3 places [300]. *First* in the (theoretical) connection between physical pion–nucleon scattering amplitudes and σ-terms. To the best of our knowledge an analysis similar to the one done by Brown, Pardee and Peccei [297]

Table 7
σ-terms in πN-scattering. The quantity δ measures the isospin asymmetries due to $m_u \neq m_d$.

Channel	σ-term
$\pi^{\pm}p \to \pi^{\pm}p$	σ
$\pi^{\pm}n \to \pi^{\pm}n$	σ
$\pi^0 p \to \pi^0 p$	$\sigma - \frac{1}{2}\delta$
$\pi^0 n \to \pi^0 n$	$\sigma + \frac{1}{2}\delta$
$\pi^0 p \to \pi^+ n$	$\frac{1}{4}\sqrt{2}\,\delta$
$\pi^- p \to \pi^0 n$	$-\frac{1}{4}\sqrt{2}\,\delta$

and Pagels and Pardee [298] has not yet been performed for the case $m_u \neq m_d$. We have nothing to add to this problem. *Second*, isospin asymmetries must be taken into account in the data analysis which is usually done with isospin symmetric amplitudes. [See Höhler [299] and Gensini [301] for a discussion of these problems.] In the *third* place, $m_u \neq m_d$ produces isospin asymmetries in the current algebra expression for the σ-terms. This point is easy to discuss. The σ-terms for the various channels are given in table 7, where we have made use of isospin symmetry to express the relevant matrix elements of the operators $\bar{u}u$, $\bar{u}d$, $\bar{d}u$ and $\bar{d}d$ between protons and neutrons in terms of the isospin symmetric part σ defined in (D.1) and the asymmetry δ given by

$$\delta = \frac{m_d - m_u}{2M_N} \langle p|\bar{u}u - \bar{d}d|p\rangle . \tag{D.7}$$

The asymmetry is related to the proton–neutron mass difference. In fact, to lowest order in the quark mass expansion in powers of m_u and m_d the mass difference $M_n - M_p$ is given by δ:

$$\delta = M_n - M_p = (2.05 \pm 0.30)\,\text{MeV} \tag{D.8}$$

(compare table 3). This shows that the isospin asymmetries are not very large. The σ-term in $\pi^0 p$ scattering is smaller than the σ-term in $\pi^0 n$ scattering by about 2 MeV.

This isospin violation is substantially smaller than what one may obtain from rough estimates based on the additivity rule. In fact, this rule ($\langle p|\bar{u}u|p\rangle \simeq \langle p|u^+u|p\rangle$ etc.) is too crude an approximation if one wants to obtain a good estimate of both the size and the asymmetries of the σ-terms (picking a value like $m_s = 150$ MeV, e.g., and using the quark mass ratios obtained from the meson spectrum one finds $\sigma \simeq 18$ MeV and $\delta \simeq 3.3$ MeV rather than the values $\sigma \simeq 35$ MeV and $\delta \simeq 2$ MeV given above). The additivity rule does give an acceptable estimate for the size of the F coupling in the mass operator, but it does not reproduce the observed isospin breakings because D coupling plays an essential role here.

Appendix E. What if $\langle 0|\bar{q}q|0\rangle$ vanishes?

The discussion of symmetry breaking reviewed in this paper is based on the assumption that the expectation value of $\bar{q}q$ in the ground state does not vanish in the chiral limit. From the consistency of the various determinations of the chiral symmetry breaking quark mass ratios one might be tempted to conclude that this analysis at the same time also checks the correctness of the underlying assumption. As has been demonstrated by a number of authors (see e.g. refs. [302–307] and the references quoted in these papers) phenomenology is however equally consistent with an alternative picture based on the assumption that the quantity $\langle 0|\bar{q}q|0\rangle$ is of order m_{quark}, i.e. vanishes in the chiral limit. Theoretically, this alternative is not "natural" because a quantity that has no reason to vanish in general is indeed different from zero. This argument for ruling the alternative out may however fail for one of the following reasons:

(i) For the analysis given in sections 13, 14 and 15 to make sense it is not sufficient to know that the order parameter $\langle 0|\bar{q}q|0\rangle$ is different from zero in the chiral limit. It is important that this quantity is not small in units of the scale of QCD. The theoretical understanding of the interrelation between the chiral order parameter and the renormalization group invariant scale Λ is still rather bleak. The only direct evidence for a sizeable value of $\langle 0|\bar{q}q|0\rangle$ comes from lattice calculations which break chiral symmetry through terms that are theoretically inessential, but might jeopardize the numerical values found for the order parameter.

(ii) It is conceivable that a discrete subgroup of the U(1) group generated by the axial current is a symmetry of the ground state, implying that $\langle 0|\bar{q}q|0\rangle$ vanishes in the chiral limit [308, 307].

Before these issues are clarified one should not take the standard framework for granted, even if the alternative based on $\langle 0|\bar{q}q|0\rangle = \mathrm{O}(m_{\text{quark}})$ is not very appealing. One deficiency of this alternative is that the Gell-Mann–Okubo formula for the η meson is lost. The meson masses are of order m_{quark} rather than of order $(m_{\text{quark}})^2$; hence the ratio $m_s : \hat{m}$ assumes a smaller value such as 5 or 10 instead of 24. The masses of the quarks u and d are larger than in the standard framework and one therefore obtains violations of chiral $\mathrm{SU}(2)\times\mathrm{SU}(2)$ that are substantially larger. Indications for relatively large deviations from chiral $\mathrm{SU}(2)\times\mathrm{SU}(2)$ have long been noticed in pion–nucleon scattering where the Goldberger–Treiman prediction for the pion–nucleon coupling constant appears to be off by as much as 7%. Also values given for the σ-term both in pion–nucleon scattering (see appendix D) and in pion–pion scattering appear to be too large to be consistent with the standard framework, in which $\mathrm{SU}(2)\times\mathrm{SU}(2)$ is an almost perfect symmetry of the Hamiltonian because m_u and m_d are tiny in comparison to the scale of QCD. One may of course try to understand these deviations within the alternative scheme based on the assumption $\langle 0|\bar{q}q|0\rangle = \mathrm{O}(m_{\text{quark}})$. It is however clear that the question cannot be resolved by looking at small deviations in the phenomenology; what one has to do is to calculate $\langle 0|\bar{q}q|0\rangle$ in terms of Λ. We will therefore not comment on ratios and absolute values of quark masses found on the basis of the assumption $\langle 0|\bar{q}q|0\rangle = \mathrm{O}(m_{\text{quark}})$.

Appendix F. A quantum mechanical model for $\rho-\omega$ mixing

We set up a quantum mechanical model which describes $\rho-\omega$ mixing; especially we shall show that the effect of $\rho-\omega$ mixing in the process $e^+e^- \to \pi^+\pi^-$ near the $\rho-\omega$ resonances is to replace the ρ-propagator by

$$\frac{1}{s-M_\rho^2+\mathrm{i}M_\rho\Gamma_\rho}+\frac{F_\omega}{F_\rho}\frac{2M_\rho M_{\rho\omega}}{(s-M_\rho^2+\mathrm{i}M_\rho\Gamma_\rho)(s-M_\omega^2+\mathrm{i}M_\omega\Gamma_\omega)} \tag{F.1}$$

where F_ω and F_ρ are the photon–vector meson coupling strengths $\langle 0|j_\mu^{\text{e.m.}}|\mathbf{p},\mathrm{h},i\rangle = \varepsilon_\mu F_i M_i$, $i=\rho,\omega$; $M_{\rho\omega}$ denotes the nondiagonal piece in the mass matrix of the $\rho-\omega$ system (eq. (B.10)).

To set up notation, we start with the exponential decay law for an unstable particle.

F.1. Decay of an unstable particle

We split the Hamiltonian into two pieces $H = H_0 + V$. The spectrum of H_0 is assumed to consist of a continuous and discrete part,

$$H_0|E) = E|E)\,, \qquad (E'|E) = \delta(E'-E)\,, \qquad E \geq \mu$$

$$H_0|1) = E_1|1)\,, \qquad (1|1) = 1\,, \qquad E_1 > \mu$$

$$|1)(1| + \int \mathrm{d}E|E)(E| = \mathbb{1}\,.$$

V causes transitions from the state $|1)$ to the continuum, $(1|V|E) \neq 0$. We shall use the notation $(1|V|E) = V(1,E)$, $(E|V|1) = V(E,1)$, $(E|V|E') = V(E,E')$, $(1|V|1) = V(1,1)$.

Production of the state $|1)$ at $t = 0$ and subsequent decay is described by the initial value problem

$$\mathrm{i}\partial_t|\psi(t)) = H|\psi(t))$$

$$|\psi(t)) = b(t)\,\mathrm{e}^{-\mathrm{i}E_1 t}|1) + \int \mathrm{d}E\, C(E,t)\,\mathrm{e}^{-\mathrm{i}Et}|E)$$

$$b(0) = 1\,, \qquad C(E,0) = 0\,.$$

The Fourier transform of $b(t)$ (with $b(t) = 0$, $t < 0$)

$$b(t) = \frac{\mathrm{i}}{2\pi}\int \mathrm{d}E'\exp\{\mathrm{i}(E_1 - E')t\}\, G(E_1, E')$$

is found to be

$$G(E_1, E)^{-1} = E - M(E) + \mathrm{i}\varepsilon$$

$$M(E) = E_1 + V(1,1) + \int_\mu^\infty \frac{\mathrm{d}E'\, V(1,E')\, U(E',E)}{E - E' + \mathrm{i}\varepsilon}\,.$$

The function $U(E',E)$ is the solution of the integral equation

$$U(E',E) = V(E',1) + \int_\mu^\infty \frac{\mathrm{d}E''\, V(E',E'')\, U(E'',E)}{E - E'' + \mathrm{i}\varepsilon}\,.$$

Exponential decay of the state $|1)$ amounts in this picture to the approximation $M(E) \simeq M(E_1)$. Then one has

$$(1|\mathrm{e}^{-\mathrm{i}Ht}|1) = \mathrm{e}^{-\mathrm{i}M(E_1)\cdot t}$$

$$M(E_1) = E_R - \tfrac{1}{2}\mathrm{i}\Gamma$$

where

$$E_{\mathrm{R}} = E_1 + V(1,1) + \mathcal{P}\!\!\int_{\mu}^{\infty} \frac{\mathrm{d}E'\,|V(1,E')|^2}{E_1 - E'} + \mathrm{O}(V^3)$$

$$\Gamma/2 = \pi |V(1, E = E_1)|^2 + \mathrm{O}(V^3)\,. \tag{F.2}$$

Next we consider production of the state $|1)$ in a scattering experiment.

F.2. Production of the unstable state $|1)$ *in* $\mathrm{e}^+\mathrm{e}^- \to \pi^+\pi^-$

Consider the process $\mathrm{e}^+\mathrm{e}^- \to \pi^+\pi^-$ near the $\rho - \omega$ complex. The transition amplitude is proportional to the matrix element

$$\langle \pi^+\pi^- \text{ out} \,|\, j_\mu^{\text{e.m.}} |0\rangle\,.$$

We mimick this matrix element in our quantum mechanical model by

$${}_{-}(E|\phi|\psi)$$

where $|E)_-$ is the solution of the Lippmann–Schwinger equation

$$|E)_- = |E) + \frac{1}{E - H_0 - \mathrm{i}\varepsilon} V|E)_-\,.$$

$|\psi)$ is an arbitrary state, ϕ may be any operator for which the matrix element $(E|\phi|\psi)$ is a smooth function of E.

We expand $|E)_-$ in terms of the complete system $|1)$, $|E)$, viz.,

$$|E)_- = |E) + Y(E)\,|1) + \int \mathrm{d}E'\,\chi(E,E')\,|E')$$

and solve for $Y(E)$, $\chi(E,E')$. (We shall set $(E'|V|E) = 0$ for simplicity.) Inserting the functions $Y(E)$ and $\chi(E,E')$ in the matrix element ${}_{-}(E|\phi|\psi)$ one finds

$${}_{-}(E|\phi|\psi) = (E|\phi|\psi) + \frac{V(E,1)\,(1|\phi|\psi)}{E - M(E)} \left\{ 1 + \frac{1}{(1|\phi|\psi)} \int \frac{\mathrm{d}E'\,V(1,E')\,(E'|\phi|\psi)}{E - E' + \mathrm{i}\varepsilon} \right\}.$$

In the approximation $M(E) = M(E_1)$ (which leads to the exponential decay law) we find

$${}_{-}(E|\phi|\psi) = (E|\phi|\psi) + \frac{V(E,1)\,(1|\phi|\psi)}{E - M(E_1)} \{1 + \mathrm{O}(V)\}\,. \tag{F.3}$$

This shows that ${}_{-}(E|\phi|\psi)$ has a Breit–Wigner pole at $E = E_{\mathrm{R}} - \frac{1}{2}\mathrm{i}\Gamma$.

Mixing is considered in the next paragraph.

F.3. Production of two resonances which mix

Assume now that the spectrum of H_0 has the form

$$H_0|E, i) = E|E, i)\,, \qquad E \geq \mu_i$$

$$H_0|i) = E_i|i)\,, \qquad E_i > \mu_i$$

$$(E', i|E, j) = \delta(E' - E)\,\delta_{ij}$$

$$(i|j) = \delta_{ij} \qquad\qquad i = 1, 2\,; \qquad E_1 \sim E_2\,.$$

The interaction V induces transitions between the bound states $|i)$ and the continuum,

$$(j|V|E, i) = \delta_{ij}\,V(j, E)\,, \qquad (E, i|V|j) = \delta_{ij}\,V(E, j)$$

and mixes the states $|1)$ and $|2)$,

$$(1|V|2) = (2|V|1)^*\,.$$

We shall again assume $(E, i|V|E', j) = 0$ to simplify the calculation and use the notation $(1|V|2) = V(1, 2)$, $(2|V|1) = V(2, 1)$.

Let ϕ, $|\psi)$ be arbitrary as before, with $(E, i|\phi|\psi)$ a smooth function of E. Now consider the matrix element

$$_-(E, 1|\phi|\psi)\,; \qquad |E, 1)_- = |E, 1) + \frac{1}{E - H_0 - \mathrm{i}\epsilon}\,V|E, 1)_-\,.$$

We expect two types of dominating contributions in the vicinity of $E \sim E_1 \sim E_2$:

(i) a peak at $E \sim E_1$ due to the transitions

$$|\psi) \xrightarrow[\phi]{} |1) \xrightarrow[V]{} |E, 1)\,,$$

(ii) another peak at $E \sim E_2$ due to the transitions

$$|\psi) \xrightarrow[\phi]{} |2) \xrightarrow[V]{} |1) \xrightarrow[V]{} |E, 1)\,.$$

The following calculation unravels these two effects. Let

$$|E, 1)_- = |E, 1) + Y_1(E)\,|1) + Y_2(E)\,|2) + \int \mathrm{d}E'\,\chi(E, E')\,|E', 1) + \int \mathrm{d}E'\,D(E, E')\,|E', 2)\,.$$

In the approximation $(E', i|V|E, j) = 0$ we find

$$(E - M_1^*(E))\, Y_1(E) = V(1, E) + V(1,2)\, Y_2(E)$$
$$(E - M_2^*(E))\, Y_2(E) = V(2, 1)\, Y_1(E)$$
$$\chi(E, E') = (E - E' - \mathrm{i}\varepsilon)^{-1}\, V(E', 1)\, Y_1(E)$$
$$D(E, E') = (E - E' - \mathrm{i}\varepsilon)^{-1}\, V(E', 2)\, Y_2(E)$$

with

$$M_i(E) = E_i + V(i, i) + \int_\mu^\infty \frac{\mathrm{d}E'\, |V(i, E')|^2}{E - E' + \mathrm{i}\varepsilon}\,.$$

Insertion of these expressions into the matrix element $_-(E|\phi|\psi)$ leads to

$$_-(E, 1|\phi|\psi) = (E, 1|\phi|\psi) + \frac{V(E, 1)\,(1|\phi|\psi)}{E - M_1(E)} + \frac{V(E, 1)\, V(1, 2)\,(2|\phi|\psi)}{(E - M_1(E))\,(E - M_2(E))} + \text{corrections}\,. \tag{F.4}$$

Now set $M_i(E) = M_i(E_i)$, $i = 1, 2$ and identify $|1) \leftrightarrow |\rho^0)$, $|2) \leftrightarrow |\omega)$, $V(1, 2) \leftrightarrow M_{\rho\omega}$, $(1|\phi|\psi) = F_\rho M_\rho$, $(2|\phi|\psi) = F_\omega M_\omega$. Then it is evident from eq. (F.4) that the dominant contributions to the matrix element $_-(E|\phi|\psi)$ at $E \sim E_1 \sim E_2$ are described by terms of the structure given in eq. (F.1). In fact the relativistic treatment is an immediate generalization of this quantum mechanical system and leads directly to this modified form of the ρ-propagator.

References

Section 1

[1] Gell-Mann, Caltech preprint 1961, unpublished; reprinted in: The eightfold way: a theory of strong interaction symmetry, eds. M. Gell-Mann and Y. Ne'eman (W.A. Benjamin, New York, Inc., 1964).
[2] Y. Nambu, Phys. Rev. Lett. 4 (1960) 380; Phys. Rev. 117 (1960) 648.
[3] Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345; Phys. Rev. 124 (1961) 246.
[4] H. Koyama, Progr. Theor. Phys. 38 (1967) 1369.
[5] S.L. Glashow and S. Weinberg, Phys. Rev. Lett. 20 (1968) 224.
[6] M. Gell-Mann, R.J. Oakes and B. Renner, Phys. Rev. 175 (1968) 2195.
[7] S. Coleman and S.L. Glashow, Phys. Rev. Lett. 6 (1961) 423; Phys. Rev. 134B (1964) 671.
[8] K.G. Wilson, Phys. Rev. 179 (1969) 1499.
[9] H.A. Kastrup, Nucl. Phys. B58 (1964) 561.
[10] G. Mack, Nucl. Phys. B5 (1968) 499.
[11] Broken scale invariance and the light cone, Lectures from Coral Gables Conf. on Fundamental Interactions at High Energy, eds. M. Gell-Mann and K. Wilson, Tracts in Mathematics and the Natural Sciences, Vol. 2 (Gordon and Breach, 1971).
[12] R. Crewther, Phys. Rev. Lett. 28 (1972) 1421.
[13] M.S. Chanowitz and J. Ellis, Phys. Lett. 40B (1972) 397.
[14] P. Minkowski, On the anomalous divergence of the dilatation current in gauge theories, Berne University preprint 1976.
[15] S.L. Adler, J.C. Collins and A. Duncan, Phys. Rev. D15 (1977) 1712.
[16] J.C. Collins, A. Duncan and S.D. Joglekar, Phys. Rev. D16 (1977) 438.
[17] Z. Maki and I. Umemura, Progr. Theor. Phys. 38 (1967) 1392.
[18] Y. Tomozawa, Phys. Rev. 186 (1969) 1504.
[19] S. Okubo, Phys. Rev. 188 (1969) 2293, 2300.
[20] C. Bernard, A. Duncan and S. Weinberg, Phys. Rev. D12 (1975) 792.

[21] A.I. Vainshtein, M.B. Voloshin, V.I. Zakharov, V.A. Novikov, L.B. Okun and M.A. Shifman, Sov. J. Nucl. Phys. 27 (1978) 274.
[22] H. Leutwyler, Phys. Lett. 48B (1974) 431; Nucl. Phys. B76 (1974) 413 and in: Light Cone Physics and PCAC, Proc. Adriatic Summer Meeting on Particle Physics, Rovinj, in: Particle Physics, eds. M. Martinis, S. Pallua and N. Zovko (North-Holland, 1974).
[23] Y. Okumura, Progr. Theor. Phys. 48 (1972) 331, 490.
[24] T. Das, L.K. Pandit and P. Roy, Nucl. Phys. B53 (1973) 567.
[25] R.L. Jaffe and C.H. Llewellyn Smith, Phys. Rev. D7 (1973) 2506.
[26] J.F. Gunion, Phys. Rev. D8 (1973) 517.
[27] M. Testa, Phys. Lett. 56B (1975) 53.
[28] J. Gasser and H. Leutwyler, Nucl. Phys. B94 (1975) 269.
[29] D. Gross, S.B. Treiman and F. Wilczek, Phys. Rev. D19 (1979) 2188.
[30] S. Weinberg, in: A Festschrift for I.I. Rabi, ed. L. Motz (New York Academy of Sciences, New York, 1977) p. 185.
[31] G. Furlan, N. Paver and C. Verzegnassi, Springer Tracts in Modern Physics 62 (1972) 118.
[32] G. Cicogna, F. Strocchi and R. Vergara Caffarelli, Phys. Res. D6 (1972) 301; Phys. Lett. 46B (1973) 217.
[33] G. Furlan and N. Paver, Acta Physica Austriaca 40 (1974) 119.
[34] E. Gava, F. Legovini and N. Paver, Nuovo Cim. Lett. 14 (1975) 41.
[35] H. Sazdjian and J. Stern, Nucl. Phys. B94 (1975) 163.
[36] H. Sazdjian, Nucl. Phys. B129 (1977) 319.
[37] N.H. Fuchs, Phys. Rev. D16 (1977) 1535.
[38] S. Mallik, Helv. Phys. Acta 50 (1978) 825.
[39] M.D. Scadron, Rep. Prog. Phys. 44 (1981) 213; J. Phys. G7 (1981) 1325.
[40] H. Sazdjian and J. Stern, Orsay preprint 1982, to appear.
[41] R. Dashen, Lectures on chiral symmetry breaking, in: Proc. Intern. School of Physics Enrico Fermi Course 54, Aug. 2–14, 1971 (Academic Press, New York, 1972).
[42] C.A. Dominguez, Chiral-symmetry breaking corrections to the Goldberger–Treiman relation, Texas A+M University preprint, 1981, DOE/ER/05223-45.
[43] H. Georgi and H.D. Politzer, Phys. Rev. D14 (1976) 1829.
[44] A.J. Buras, 1981 Intern. Symp. on Lepton and Photon Interactions at High Energies, Bonn, Aug. 24–29, 1981, ed. W. Pfeil (Phys. Inst., University of Bonn, Bonn).
[45] J. Drees, 1981 Intern. Symp. on Lepton and Photon Interactions at High Energies, Bonn, Aug. 24–29, 1981, ed. W. Pfeil (Phys. Inst., University of Bonn, Bonn).
[46] R. Marshall, Report given at the EPS Intern. Conf. on High Energy Physics, Lisbon 1981.
[47] P. Franzini, Report given at the EPS Intern. Conf. on High Energy Physics, Lisbon 1981.
[48] H. Wahl, Report given at the EPS Intern. Conf. on High Energy Physics, Lisbon 1981.
[49] H.D. Politzer, Nucl. Phys. B117 (1976) 397.

Section 3

For a general review of QCD containing references to the literature see

[50] W. Marciano and H. Pagels, Phys. Reports 36C (1978) 137.

A review of the renormalization group is given by

[51] A. Peterman, Phys. Reports 53C (1979) 157.

For the relation among the different conventions defining the renormalization group invariant scale in QCD without fermions see

[52] R. Dashen and D. Gross, Phys. Rev. D23 (1981) 2340.

[53] E.C.G. Stueckelberg and A. Peterman, Helv. Phys. Acta 26 (1953) 499.
[54] M. Gell-Mann and F.E. Low, Phys. Rev. 95 (1954) 1300.
[55] C.G. Callan, Phys. Rev. D2 (1970) 1541.
[56] K. Symanzik, Commun. Math. Phys. 18 (1970) 227.
[57] H.D. Politzer, Phys. Rev. Lett. 30 (1973) 1346; Phys. Reports 14C (1974) 129.
[58] D. Gross and F. Wilczek, Phys. Rev. Lett. 30 (1973) 1343; Phys. Rev. D8 (1973) 3622; Phys. Rev. D9 (1974) 980.
[59] D.R.T. Jones, Nucl. Phys. B75 (1974) 531.
[60] W.E. Caswell, Phys. Rev. Lett. 33 (1974) 244.
[61] G. 't Hooft, Nucl. Phys. B61 (1973) 455.
[62] J.C. Collins and A.J. Macfarlane, Phys. Rev. D10 (1974) 1201.

[63] W.A. Bardeen, A.J. Buras, D.W. Duke and T. Muta, Phys. Rev. D18 (1978) 3998.
[64] E.G. Floratos, S. Narison and E. de Rafael, Nucl. Phys. B155 (1979) 115.
[65] D.V. Nanopoulos and D.A. Ross, Nucl. Phys. B157 (1979) 273.
[66] R. Tarrach, Nucl. Phys. B183 (1981) 384.
[67] O. Nachtmann and W. Wetzel, The β-function for effective quark masses to two loops in QCD, Heidelberg preprint 1981.
[68] K. Symanzik, Commun. Math. Phys. 34 (1973) 7.
[69] T. Appelquist and J. Carazzone, Phys. Rev. D11 (1975) 2856.
[70] B. Ovrut and H. Schnitzer, Nucl. Phys. B179 (1981) 381.
[71] W. Bernreuther and W. Wetzel, Decoupling of heavy quarks in the $\overline{\mathrm{MS}}$ scheme, preprint Cornell, July 1981, CLNS 81/500.

Section 4

[72] R. Crewther, Riv. Nuovo Cim. 2, No. 8 (1979) 63 and in: Field theoretical methods in particle physics, Kaiserslautern 1979, ed. W. Rühl, Vol. 55 B, Nato Advanced Study Institutes Series (Plenum, 1980) p. 529.

Section 5

[73] Y. Nambu, Phys. Rev. Lett. 4 (1960) 380; Phys. Rev. 117 (1960) 648.
[74] Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345; Phys. Rev. 124 (1961) 246.
[75] J. Goldstone, Nuovo Cim. 19 (1961) 154.

Section 6

[76] M. Gell-Mann, R.J. Oakes and B. Renner, Phys. Rev. 175 (1968) 2195.
[77] V.A. Novikov, M.A. Shifman, A.I. Vainshtein and V.I. Zakharov, Nucl. Phys. B191 (1981) 301.

Section 7

[78] R. Dashen, Phys. Rev. 183 (1969) 1245.
[79] R. Dashen and M. Weinstein, Phys. Rev. 183 (1969) 1291.
[80] S. Okubo, Phys. Lett. 5 (1963) 165.
[81] G. Zweig, CERN reports 1964, reprinted in: Developments in the quark theory of hadrons, a reprint collection, Vol. I, 1964–1978, eds. D.B. Lichtenberg and S.P. Rosen (Hadronic Press, Nonantum, Mass., 1980).
[82] J. Iizuka, Progr. Theor. Phys. Suppl. 37/38 (1966) 21.
[83] S. Okubo, Suppl. Progr. Theor. Phys. 63 (1978) 1.
[84] L.-F. Li and H. Pagels, Phys. Rev. Lett. 26 (1971) 1204; Phys. Rev. Lett. 27 (1971) 1089.
[85] D.T. Cornwell, Nucl. Phys. B55 (1973) 436.
[86] P. Langacker and H. Pagels, Phys. Rev. D10 (1974) 2904.

Section 10

For a review of chiral perturbation theory see
[87] H. Pagels, Phys. Reports 16C (1975) 219.

[88] L.-F. Li and H. Pagels, Phys. Rev. Lett. 26 (1971) 1204; Phys. Rev. Lett. 27 (1971) 1089; Phys. Rev. D5 (1972) 1509.
[89] P. Langacker and H. Pagels, Phys. Rev. D8 (1973) 4595; Phys. Rev. D10 (1974) 2904.
[90] J. Gasser and A. Zepeda, Nucl. Phys. B174 (1980) 445.
[91] J. Gasser, Ann. of Phys. 136 (1981) 62.
[92] E.M. Henley and W. Thirring, Elementary quantum field theory (McGraw-Hill Book Company Inc., New York, 1962).
[93] D.T. Cornwell, Nucl. Phys. B55 (1973) 436.

Section 12

[94] W.N. Cottingham, Ann. of Phys. 25 (1963) 424.
[95] H. Leutwyler and P. Otterson, in: Scale and conformal symmetry in hadron physics, ed. R. Gatto (Wiley, New York, 1973) p. 3.
[96] J.C. Collins, Nucl. Phys. B149 (1979) 90.
[97] J. Gasser and H. Leutwyler, Nucl. Phys. B94 (1975) 269.
[98] A. Zee, Phys. Reports 3C (1972) 129.

[99] M. Cini and P. Stichel, Text book on elementary particle physics, electromagnetic interactions, Sept. 1972, unpublished.
[100] J.F. Gunion, Phys. Rev. D8 (1973) 517.
[101] S. Coleman and S.L. Glashow, Phys. Rev. Lett. 6 (1961) 423.
[102] S. Coleman and H. Schnitzer, Phys. Rev. 136B (1964) 223.
[103] Y. Miyamoto, Progr. Theor. Phys. 35 (1966) 175.
[104] A. De Rújula, H. Georgi and S.L. Glashow, Phys. Rev. D12 (1975) 147.
[105] K. Lane and S. Weinberg, Phys. Rev. Lett. 37 (1976) 717.
[106] H. Fritzsch, (a) Phys. Lett. 63B (1976) 419; (b) Phys. Lett. 71B (1977) 429.
[107] S. Ono, Phys. Rev. Lett. 37 (1976) 655.
[108] A.D. Dolgov et al., Phys. Lett. 15 (1965) 84.
[109] W. Thirring, Acta Physica Austriaca, Suppl. II (1965) 205.
[110] J. Arafune and Y. Iwasaki, Progr. Theor. Phys. 35 (1966) 339.
[111] G. Barton and D. Dare, Nuovo Cim. 46A (1966) 433.
[112] S. Ishida, K. Konno and H. Shimodaira, Nuovo Cim. 46A (1966) 194.
[113] T. Minamikawa et al., Suppl. Progr. Theor. Phys. 37/38 (1966) 56.
[114] H.R. Rubinstein, Phys. Rev. Lett. 17 (1966) 41.
[115] A. Gal and F. Scheck, Nucl. Phys. B2 (1967) 110.
[116] S. Ishida et al., Nucl. Phys. B2 (1967) 307.
[117] H.J. Lipkin, Nucl. Phys. B1 (1967) 597.
[118] H.R. Rubinstein, F. Scheck and R.H. Socolow, Phys. Rev. 154 (1967) 1608.
[119] R.P. Feynman, Photon hadron interactions (W.A. Benjamin, Inc., Reading, Mass., 1972).
[120] A. De Rújula, H. Georgi and S.L. Glashow, Phys. Rev. D12 (1975) 147.
[121] D.B. Lichtenberg, Phys. Rev. D12 (1975) 3760.
[122] D.B. Lichtenberg, Phys. Rev. D14 (1976) 1412.
[123] W. Celmaster, Phys. Rev. Lett. 37 (1976) 1042.
[124] C. Itoh et al., Progr. Theor. Phys. 61 (1979) 548.
[125] N. Isgur, Phys. Rev. D21 (1980) 779.
[126] T. Das, G. Guralnik, V. Mathur, F. Low and J. Young, Phys. Rev. Lett. 18 (1967) 759.
[127] D. Gross, S. Treiman and F. Wilczek, Phys. Rev. D19 (1979) 2188.
[128] R. Socolow, Phys. Rev. 137B (1965) 1221.
[129] C.J. Bebek et al., Phys. Rev. D17 (1978) 1693.
[130] R. Dashen, Phys. Rev. 183 (1969) 1245.
[131] P. Langacker and H. Pagels, Phys. Rev. D8 (1973) 4620.
[132] D.H. Boal and A.C.D. Wright, Phys. Rev. D16 (1977) 1505.
[133] V.S. Zidell, R.A. Arndt and L.D. Roper, Phys. Rev. D21 (1980) 1255, 1289.

Section 13

The material presented in this section is essentially taken from
[134] J. Gasser, Ann. of Phys. 136 (1981) 62.

[135] H. Leutwyler, in: Proc. Topical Meeting on Intermediate Energy Physics, April 2–12, Zuoz 1975 (SIN-Documentation Group, CH-5234 Villigen, Switzerland).
[136] A. Halprin, B.W. Lee and P. Sorba, Phys. Rev. D14 (1976) 2343.
[137] S. Weinberg, in: A Festschrift for I.I. Rabi, ed. L. Motz (New York Academy of Sciences, New York, 1977) p. 185.
[138] C.A. Dominguez and A. Zepeda, Phys. Rev. D18 (1978) 884.
[139] C.A. Dominguez, Phys. Lett. 86B (1979) 171.
[140] P. Langacker and H. Pagels, Phys. Rev. D19 (1979) 2020.
[141] P. Langacker, Phys. Rev. D20 (1979) 2983.
[142] M.A. Shifman, A.I. Vainshtein and V.I. Zakharov, Nucl. Phys. B147 (1979) 385, 519.
[143] P. Minkowski and A. Zepeda, Nucl. Phys. B164 (1980) 25.
[144] H. Fritzsch and P. Minkowski, Phys. Reports 73C (1981) 67.

Section 14

[145] S.L. Glashow, Phys. Rev. Lett. 7 (1961) 469.
[146] S. Coleman and S.L. Glashow, Phys. Rev. 134B (1964) 671.
[147] M. Gourdin, L. Stodolsky and F.M. Renard, Phys. Lett. 30B (1969) 347.

[148] F.M. Renard, Nucl. Phys. B15 (1970) 118.
[149] R.P. Feynman, Photon hadron interactions (W.A. Benjamin, Inc., Reading, Mass., 1972).
[150] A.S. Goldhaber et al., Phys. Lett. 30B (1969) 249;
S.A. Coon et al., Nucl. Phys. A287 (1977) 381.
[151] M.A. Shifman, A.I. Vainshtein and V.I. Zakharov, Nucl. Phys. B147 (1979) 519.
[152] P. Langacker, Phys. Rev. D20 (1979) 2983.
[153] D. Benaksas et al., Phys. Lett. 39B (1972) 289; Phys. Lett. 42B (1972) 507.
[154] A. Quenzer et al., Phys. Lett. 76B (1978) 512.
[155] G. Segré and J. Weyers, Phys. Lett. 62B (1976) 91.
[156] H. Genz, Lett. Nuovo Cim. 21 (1978) 270.
[157] P. Langacker, Phys. Lett. 90B (1980) 447.
[158] B.L. Ioffe, Yad. Fiz. 29 (1979) 1611.
[159] M. Voloshin, Proc. Workshop on physical program of e^+e^- storage rings (Novosibirsk, 1979).
[160] B.L. Ioffe and M.A. Shifman, Phys. Lett. 95B (1980) 99.
[161] M. Oreglia et al., Phys. Rev. Lett. 45 (1980) 959.
[162] T.M. Himel et al., Phys. Rev. Lett. 44 (1980) 920.
[163] D.G. Sutherland, Phys. Lett. 23 (1966) 384.
[164] S.K. Bose and A.H. Zimmermann, Nuovo Cim. 43A (1966) 1165.
[165] P. Dittner, P.H. Dondi and S. Eliezer, Phys. Rev. D8 (1973) 2253.
[166] A. Browman et al., Phys. Rev. Lett. 32 (1974) 1067.
[167] C. Baglin et al., BAPS 12 (1967) 567; Phys. Lett. 29B (1969) 445.
[168] F.W. Bullock et al., Phys. Lett. 27B (1968) 402.
[169] A. Neveu and J. Scherk, Ann. of Phys. 57 (1970) 39.
[170] H. Osborn and D.J. Wallace, Nucl. Phys. B20 (1970) 23.
[171] N.G. Deshpande and T.N. Truong, Phys. Rev. Lett. 41 (1978) 1579.
[172] P. Langacker and H. Pagels, Phys. Rev. D10 (1974) 2904; Phys. Rev. D19 (1979) 2070.
[173] S. Weinberg, Phys. Rev. D11 (1975) 3583.
[174] R. Crewther, Cargèse Lectures (Plenum Press, 1975).
[175] S. Raby, Phys. Rev. D13 (1976) 2594.
[176] C.A. Dominguez and A. Zepeda, Phys. Rev. D18 (1978) 884.
[177] C. Roiesnel and T.N. Truong, Nucl. Phys. B187 (1981) 293.
[178] W.I. Brandelik et al., Phys. Lett. 70B (1977) 132; Phys. Lett. 80B (1979) 412; Phys. Lett. 76B (1978) 361.

Section 15

[179] A.I. Vainshtein et al., Sov. J. Nucl. Phys. 27 (1978) 274.
[180] A.I. Vainshtein, V.I. Zakharov and M.A. Shifman, JETP Lett. 27 (1977) 55.
[181] M.A. Shifman, A.I. Vainshtein and V.I. Zakharov, Nucl. Phys. B147 (1979) 385, 448, 519.
[182] R. Aron and R.S. Longacre, Phys. Rev. D24 (1981) 1207.
[183] M. Bonesini et al., Phys. Lett. 103B (1981) 75.
[184] C. Becchi, S. Narison, E. de Rafael and F.J. Yndurain, Z. Physik C8 (1981) 335.
[185] S. Narison and E. de Rafael, Phys. Lett. 103B (1981) 57.
[186] J. Schwinger, Particles, sources and fields, Vol. II. (Addison-Wesley, 1973).
[187] M.A. Shifman, A.I. Vainshtein, M.B. Voloshin and V.I. Zakharov, Phys. Lett. 77B (1978) 80.
[188] E.G. Floratos, S. Narison and E. de Rafael, Nucl. Phys. B155 (1979) 115.
[189] L.J. Reinders, H.R. Rubinstein and S. Yazaki, Nucl. Phys. B186 (1981) 109.
[190] K. Schilcher, Minh D. Tran and N.F. Nasrallah, Nucl. Phys. B181 (1981) 91.
[191] D.J. Broadhurst, Phys. Lett. 101B (1981) 423.
[192] W. Hubschmid and S. Mallik, Nucl. Phys. B193 (1981) 368.
[193] S.M. Eidelman, L.M. Kurdadze and A.I. Vainshtein, Phys. Lett. 82B (1979) 278.
[194] H. Sazdjian, Nucl. Phys. B129 (1977) 319.
[195] C.A. Dominguez, Phys. Rev. D15 (1977) 1350; Phys. Rev. D16 (1977) 2313.
[196] C.A. Dominguez, Chiral-symmetry breaking corrections to the Goldberger-Treiman relation, Texas A + M University preprint. 1981. DOE/ER/05223-45.
[197] S. Mallik, preprint Karlsruhe 1982, to be published.
[198] N. Craigie and J. Stern, Trieste preprint 1981, to be published.
[199] B.L. Ioffe, Nucl. Phys. B188 (1981) 317.
[200] Y. Chung, H.G. Dosch, M. Kremer and D. Schall, Nucl. Phys. B197 (1982) 55.

Section 16

[201] V.A. Novikov, L.B. Okun, M.A. Shifman, A.I. Vainshtein, M.B. Voloshin and V.I. Zakharov, Phys. Rev. Lett. 38 (1977) 626; Phys. Lett. 67B (1977) 409; Phys. Reports 41C (1978) 1.
[202] J.S. Bell and R.A. Bertlmann, Z. Physik C Particles and Fields 4 (1980) 11; Nucl. Phys. B187 (1981) 285; Nucl. Phys. B177 (1981) 218.
[203] R.A. Bertlmann, Duality between resonances and asymptotia, Report UWThPh-80-38, Vienna University, Austria (1980).
[204] L.J. Reinders, H.R. Rubinstein and S. Yazaki, Nucl. Phys. B186 (1981) 109.
[205] L.J. Reinders, H.R. Rubinstein and S. Yazaki, Phys. Lett. 94B (1980) 203; Phys. Lett. 95B (1980) 103.
[206] B. Guberina, R. Meckbach, R.D. Peccei and R. Rückl, Nucl. Phys. B184 (1981) 476.
[207] L.J. Reinders, Lectures given at the Intern. School of Elementary Particle Physics, Kupari-Dubrovnik (Yugoslavia) 1981, preprint Rutherford Lab. 81-078.
[208] M.A. Shifman, Report Intern. Symp. on Lepton and Photon Interactions at High Energies, Bonn, Aug. 24–29, 1981, ed. W. Pfeil (Phys. Inst., University of Bonn, Bonn).
[209] H. Georgi and D.H. Politzer, Phys. Rev. D14 (1976) 1829.
[210] R. Coquereaux, Ann. of Phys. 125 (1980) 401.
[211] M.B. Voloshin, Yad. Fiz. 29 (1979) 1368 [Sov. J. Nucl. Phys. 29 (1979) 703]; Nucl. Phys. B154 (1979) 365; ITEP preprint ITEP-21, 1980.
[212] S. Iwao, Lett. Nuovo Cim. 29 (1981) 343.
[213] A. Martin, Talk at the EPS Intern. Conf. on High Energy Physics, Lisbon 1981.
[214] R. Tarrach, Nucl. Phys. B183 (1981) 384.
[215] H. Leutwyler, Phys. Lett. 98B (1981) 447.
[216] M.B. Voloshin, Nonperturbative effects in pre-Coulombic levels of a heavy quarkonium, preprint ITEP-30, 1981.
[217] A.I. Vainshtein et al., Sov. J. Nucl. Phys. 27 (1978) 274.
[218] R.A. Bertlmann, Heavy quark-antiquark system from exponential moments in QCD, CERN preprint TH-3192, 1981.
[219] B. Ovrut and H. Schnitzer, Nucl. Phys. B179 (1981) 381.
[220] W. Bernreuther and W. Wetzel, Decoupling of heavy quarks in the $\overline{\text{MS}}$ scheme, preprint Cornell, July 1981, CLNS 81/500.

Section 17

[221] M.K. Gaillard, B.W. Lee and J.L. Rosner, Rev. Mod. Phys. 47 (1975) 277.
[222] S. Borchardt, V.S. Mathur and S. Okubo, Phys. Rev. Lett. 34 (1975) 38, 236.
[223] R. Vergara Caffarelli, Phys. Lett. 55B (1975) 481; Phys. Lett. 58B (1975) 100.
[224] A. Kazi, G. Kramer and D.H. Schiller, Acta Physica Austriaca 45 (1976) 65.
[225] S. Iwao, Ann. Sci. (Kanazawa Univ.) 13 (1976); Nuovo Cim. Lett. 15 (1976) 331, 569; Progr. Theor. Phys. 55 (1976) 943.
[226] H. Hayashi, I. Ishiwata, S. Iwao, M. Shako and S. Takeshita, Ann. of Phys. 101 (1976) 394; Progr. Theor. Phys. 55 (1976) 1912.
[227] Z. Maki and I. Umemura, Progr. Theor. Phys. 59 (1978) 507.
[228] Z. Maki T. Teshima and I. Umemura, Progr. Theor. Phys. 60 (1978) 1127.
[229] J. Kaudaswamy, J. Schechter and M. Singer, Phys. Rev. D17 (1978) 1430.
[230] K. Yamawaki, Nuovo Cim. 51A (1979) 159.

For an analysis of the effective spin–spin interaction see
[231] A.D. Sakharov, JETP Lett. 21 (1975) 9; JETP Lett. 78 (1980) 2115.

[232] H. Leutwyler, in: Proc. Topical Meeting on Intermediate Energy Physics, March 29–April 8, Zuoz 1978 (SIN-Documentation Group, CH-5234 Villigen, Switzerland).

Section 18

For a review of grand unified theories see
[233] P. Langacker, Phys. Reports 72C (1981) 185.

[234] J.C. Pati and A. Salam, Phys. Rev. D8 (1973) 1240; Phys. Rev. Lett. 31 (1973) 661; Phys. Rev. D10 (1974) 275.
[235] H. Georgi and S.L. Glashow, Phys. Rev. Lett. 32 (1974) 438.
[236] H. Georgi, H. Quinn and S. Weinberg, Phys. Rev. Lett. 32 (1974) 451.
[237] A.J. Buras, J. Ellis, M.K. Gaillard and D.V. Nanopoulos, Nucl. Phys. B135 (1978) 66.
[238] I. Antoniadis, C. Kounnas and C. Roiesnel, Symmetry breaking effects in grand unified theories, preprint Centre de physique théorique, Ecole polytechnique, Palaiseau, 1981.
[239] E. Gildener, Phys. Rev. D14 (1976) 1667.
[240] J. Ellis, Lectures given at the 21st Scottish Universities Summer School in Physics, St Andrews, Scotland 1980.
[241] M.S. Chanowitz, J. Ellis and M.K. Gaillard, Nucl. Phys. B128 (1977) 506.

[242] D.V. Nanopoulos and D.A. Ross, Nucl. Phys. B157 (1979) 273.
[243] P. Binétruy and T. Schücker, Nucl. Phys. B178 (1981) 293, 307.
[244] J. Ellis and M.K. Gaillard, Phys. Lett. 88B (1979) 315.
[245] H. Georgi and C. Jarlskog, Phys. Lett. 86B (1979) 297.
[246] R. Gatto, G. Sartori and M. Tonin, Phys. Lett. 28B (1968) 128.
[247] N. Gabibbo and L. Maiani, Phys. Lett. 28B (1968) 131.
[248] R. Gatto, G. Morchio and F. Strocchi, Phys. Lett. 80B (1979) 265; Phys. Lett. 83B (1979) 348.
[249] G. Sartori, Phys. Lett. 82B (1979) 255.
[250] H. Georgi and D.V. Nanopoulos, Nucl. Phys. B155 (1979) 52; Nucl. Phys. B159 (1979) 16; Phys. Lett. 82B (1979) 329.
[251] R.N. Mohapatra and D. Wyler, Phys. Lett. 89B (1980) 81.
[252] R.N. Mohapatra and B. Sakita, Phys. Rev. D21 (1980) 1062.
[253] R. Barbieri, D.V. Nanopoulos and D. Wyler, Phys. Lett. 106B (1981) 303.
[254] G. 't Hooft, Nucl. Phys. B79 (1974) 276.
[255] A.M. Polyakov, JETP Lett. 20 (1974) 194.
[256] M.A.B. Bég and A. Sirlin, Ann. Rev. Nucl. Sci. 24 (1974) 379.
[257] S. Weinberg, Phys. Rev. D13 (1976) 974; Phys. Rev. D19 (1979) 1277.
[258] L. Susskind, Phys. Rev. D20 (1979) 2619.
[259] M. Gell-Mann, P. Ramond and R. Slansky, in: Supergravity, eds. P. van Nieuwenhuizen and D.Z. Freedman (North-Holland, 1979) p. 315.
[260] S. Dimopoulos and L. Susskind, Nucl. Phys. B155 (1979) 237.
[261] E. Eichten and K. Lane, Phys. Lett. 90B (1980) 125.
[262] S. Coleman and E. Weinberg, Phys. Rev. D7 (1973) 1888.
[263] S. Weinberg, in: A Festschrift for I.I. Rabi, ed. L. Motz (New York Academy of Sciences, New York, 1977) p. 185.
[264] S. Weinberg, Phys. Lett. 82B (1979) 387.

Section 19

[265] P. Hasenfratz, Talk given at the EPS Intern. Conf. on High Energy Physics, Lisbon, 1981.
[266] P. Hasenfratz, 1981 Intern. Symp. on Lepton and Photon Interactions at High Energies, Bonn, Aug. 24–29, 1981, ed. W. Pfeil (Phys. Inst., University of Bonn, Bonn).
[267] E. Marinari, G. Parisi and C. Rebbi, Phys. Rev. Lett. 47 (1981) 1795.
[268] H. Hamber and G. Parisi, Phys. Rev. Lett. 47 (1981) 1792.
[269] D.H. Weingarten, Phys. Lett. 109B (1982) 57.
[270] H. Hamber, E. Marinari, G. Parisi and C. Rebbi, Phys. Lett. 108B (1982) 314.
[271] A. Hasenfratz, Z. Kunszt, P. Hasenfratz and C.B. Lang, Hopping parameter expansion for the meson spectrum in SU(3) lattice QCD, preprint CERN TH-3220, 1981.
[272] A. Casher, Phys. Lett. 83B (1979) 395.
[273] T. Banks and A. Casher, Nucl. Phys. B169 (1980) 103.
[274] J. Smit, Nucl. Phys. B175 (1980) 307.
[275] H. Pagels and S. Stokar, Phys. Rev. D20 (1979) 2947; Phys. Rev. D22 (1980) 2876.
[276] J.F. Donoghue and K. Johnson, Phys. Rev. D21 (1980) 1975.
[277] M. Ida and J. Okada, Spontaneous breaking of chiral flavour symmetry, preprint Kobe University 81-4 (1981).

Appendices

[278] S. Okubo, Phys. Rev. D11 (1975) 3261; J. Math. Phys. 16 (1975) 528.
[279] S. Borchardt, V.S. Mathur and S. Okubo, Phys. Rev. D11 (1975) 2572; Phys. Rev. Lett. 34 (1975) 236.
[280] A. Kazi, G. Kramer and D.H. Schiller, Acta Physica Austriaca 45 (1976) 65.
[281] S. Weinberg, Physica 96A (1979) 327.
[282] J. Gasser and A. Zepeda, Nucl. Phys. B174 (1980) 445;
J. Gasser, Ann. of Phys. 136 (1981) 62.
[283] M.M. Nagels et al., Nucl. Phys. B147 (1979) 189.
[284] D.T. Cornwell, Nucl. Phys. B55 (1973) 436.

For references to early work on the σ-term see
[285] E. Reya, Rev. Mod. Phys. 46 (1974) 545.

For more recent papers see
[286] T.P. Cheng, Phys. Rev. D13 (1976) 2161.
[287] R.L. Jaffe, Phys. Rev. D21 (1980) 3215.
[288] C.A. Dominguez and P. Langacker, Phys. Rev. D24 (1981) 1905 and the references quoted therein.

Numerical values are tabulated in
[289] R.J. Oakes, Acta Physica Austriaca Suppl. 9 (1972) 518.
[290] H. Pilkuhn et al., Nucl. Phys. B65 (1973) 460.
[291] M.M. Nagels et al., (a) Nucl. Phys. B109 (1976) 1; (b) B147 (1979) 189.
[292] M.K. Banerjee and J.B. Cammarata, Phys. Rev. D16 (1977) 1334; Phys. Rev. C17 (1978) 1125; Phys. Rev. D18 (1978) 4078; Phys. Rev. D19 (1979) 145.
[293] G. Höhler, F. Kaiser, R. Koch and E. Pietarinen, Handbook of pion–nucleon scattering, 1979, Fachinformationszentrum, Karlsruhe.
[294] W. Langbein, Nuovo Cimento 51A (1979) 219.
[295] M.G. Olsson and E.T. Osypowski, J. Phys. G6 (1980) 423.
[296] M.G. Olsson, J. Phys. G6 (1980) 431.
[297] L.S. Brown, W.J. Pardee and R.D. Peccei, Phys. Rev. D4 (1971) 2801.
[298] H. Pagels and W.J. Pardee, Phys. Rev. D4 (1971) 3335.
[299] G. Höhler, Pion–nucleon scattering and quark models, Lecture given at the 3rd Adriatic Meeting on Particle Physics, Dubrovnik, September 1980.
[300] S. Weinberg, in: A Festschrift for I.I. Rabi, ed. L. Motz (New York Academy of Sciences, New York, 1977) p. 185.
[301] P.M. Gensini, Lectures given at the Workshop on Low- and Intermediate Energy Kaon Physics, Rome, March 24–28, 1980, Report UL/IF-81-79/80, Università di Lecce.
[302] J.F. Gunion, P.C. McNamee and M.D. Scadron, Nucl. Phys. B123 (1977) 445.
[303] H. Sazdjian, Nucl. Phys. B129 (1977) 319.
[304] N.H. Fuchs and H. Sazdjian, Phys. Rev. D18 (1978) 889.
[305] N.H. Fuchs and M.D. Scadron, Phys. Rev. D20 (1979) 2421.
[306] M.D. Scadron, Rep. Progr. Phys. 44 (1981) 213; J. Phys. G7 (1981) 1325.
[307] H. Sazdjian and J. Stern, Orsay Preprint 1982, to appear.
[308] R. Dashen, Phys. Rev. 183 (1969) 1245.